THE ULTIMATE DIY GUIDE ON HOW TO ACHIEVE THE ANAMORPHIC LOOK ON THE CHEAP!

- LENS FLARES!
- OVAL BOKEH!
- DISTORTION!
- ASPECT RATIO!

ANAMORFAKE IT UNTIL YOU MAKE IT!

BY TITO FERRADANS

I did not invent the anamorfake method/process. I did come up with the name though and have pushed for it more than anyone.

The first time I saw an anamorfake lens was on Amir Safari's "Is that thing anamorphic?" video, way before anamorphic became mainstream and anamorfakes were sold on eBay. Amir had modified a Helios 44M and posted the results to Vimeo. Between October of 2012 and June of 2013 we exchanged messages and Amir pointed me to some of the lenses in this guide as well as taught me the steps to open them up.

I was living in Brazil at the time and could not find a place to get laser cuts done. Amir sent me a few discs all the way from Austria which allowed me to get started. He is one of the biggest influences in my DIY anamorphic/anamorfake journey, and I dedicate this guide to him.

Thank you!

To Amir Safari.

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01. INTRODUCTION

The anamorphic format has a lot of ups and downs and if you are reading this guide you have already done some research on the subject. One of the biggest barriers for entering the anamorphic world and start shooting is the price of admission. Due to low availability and high demand many anamorphic adapters now cost well over $1000, which brings up the question: "are they really worth it?".

If you are a hardcore tinkerer, if you love taking expensive things apart and fiddling with their insides, sometimes succeeding sometimes failing; or if you are more focused on the learning process than the resulting images for a few

The vintage Isco-Göttingen anamorphic family:
Iscomorphot 8/1.5x - $750, Isco Widescreen 2000 MC - $1300, Iscorama pre-36 - $3500, and Iscorama 54 - $4000.

weeks or months - and if you have money to spare -, shooting with anamorphic adapters is for you. If you are not passionate about any (or all) of these things, there are different paths to choose from and, most importantly, they all lead to the anamorphic look.

The first path is an easy one: rent proper anamorphic glass whenever you have a project that matches the look. There are plenty of options, from vintage LOMOs, to SLR Magic primes, to Atlas' Orion series, P+S Technik and all the way up to Hawk, Zeiss, Cooke and Panavision. These are all-in-one units that do not display all the quirks of shooting with adapters. They have good minimum focus, standard mounts, regular focus and aperture rings, no play on the mount or in the middle of the lens, and they output excellent images. They are too expensive to own though, so you will be renting every time and that can still run pricy.

The second path involves some tinkering - on a lighter level - and it will give you much faster results. It is like warming up to shooting with adapters. Going down this path does not require you to change your shooting style to accomodate lenses' limitations. As a matter of fact, you might not even have to change your regular lenses! This is anamorfaking: a combination of tweaks to spherical lenses and post-processing that aims to achieve a convincing anamorphic look.

Frame grabs from Train Station, by Amir Safari. Full anamorfake set.

02. WHAT IS ANAMORFAKE?

I have been working on anamorfake projects and tutorials for more than five years - check out this playlist - and, when it comes to the resulting images, most of the audience is unable to tell the difference. Depending on the case, even the experts will have a hard time telling them apart.

I call this process **anamorfake** because it delivers some of the most pronounced features of the anamorphic look but it does not involve anamorphic glass, hence "faking it". Two things that scream anamorphic on the screen are streak flares and oval bokeh. Anamorfaking consists in forcing these traits on spherical lenses through light modification.

Contax Zeiss 28mm f/2.8 @ f/4 with anamorfake mod on Sony a7s II

In short: we are going to add an oval cutout at the aperture mechanism of spherical lenses to shape bokeh and out-of-focus areas. We are also adding a fishing line to create the streak flares from strong light sources. In this guide I will go over the process of opening multiple lenses, provide templates for the oval inserts so you can get them cut and explain how to make custom cutouts for lenses not covered in this book. I also expand on the streak flares, how to control them and optimal scenarios, as well as a little bit of post-production.

The first step is to cover what are the limitations of anamorfaking. I am a strong believer that you need to know the bad before you know the good. If you are unable to work with these limitations, it is not even worth looking at the

amazing things anamorfakes are capable of rendering. Then we will move on to the good stuff: what are the advantages of anamorfakes over anamorphic adapters - there are several - and how to use them effectively.

If you are absolutely not into tinkering, there is a chapter for ready-to-use products on the market and brief reviews about them, like the Cinemorph filter from Vid-Atlantic, Richard Gale's optics (former DogSchidtOptiks), IronGlass, Motion Six and some other independent manufacturers.

From there we have a chapter on how to design oval inserts for any lens you are able to open besides the ones I am explaining here, then streak flares customization. Lastly we move to the tutorials themselves, covering how to modify six famous (and cheap) Soviet M42 lenses, a full set of six Rokinon/ Samyang Cine primes and a budget set of five Contax Zeiss lenses, with useful advice from Ron Sim at **SIMMOD Lens**.

The last chapter is about post-production, getting the right aspect ratio, how to add a little bit of distortion to your shots, a hardcore method for oval bokeh and digital flares too.

The additional files that come with this guide include oval cutout templates for different apertures and stretch factors for all the lenses shown here. I also included a few templates for post and masks for different aspect ratios that can be easily imported into any editing software. Just for fun I added some tutorials, samples and other maker files on top. You know, a production value boost.

ARE YOU READY?

Helios 44-2 58mm f/2 @ f/2 with anamorfake mod and amber tint on Sony a7s II - Amber Helios Test Shoot

03. THE WORST PARTS OF FAKING IT

Let's start with the downsides of anamorfaking. Some of these things are just me being picky, others are downsides when compared to shooting anamorphic, and others are downsides when compared to shooting in general.

Shooting with anamorphic glass widens your field of view due to the horizontal compression. Anamorfaking, however, will not give you any extra field of view. If you are using a 50mm, you will still get the field of view of a 50mm: there is no squeeze.

This is a bit of misconception that happens every once in a while. I have seen folks believe that because the lens has an oval shape inside it has been turned into anamorphic and the footage is going to be squeezed, field of view extended and so on. That does not happen. At all. No gain in field of view, but no loss either.

Since there is no squeeze, that means that your shooting aspect ratio is your base aspect ratio for post. If you are shooting 16:9 (most of us are), you will have to crop or mask parts of the frame in order to output your final product as 2.4:1 or any other wide aspect ratio (this process is covered in **Chapter 10**). Ultimately this means that you are losing some resolution by anamorfaking (roughly 25% of your total pixels when going from 16:9 to 2.4:1).

The striped areas represent parts of the image that would be discarded from 16:9 to 2.4:1 crop. Roughly 25%.

By using the oval inserts as new aperture shapes you will give up the ability to stop down the lens to control exposure. This is a little confusing so bear with me: by following the tutorials in this guide you are still going to be able to stop down the lenses, but when doing so you will lose the oval shaped bokeh. Because of this limitation you will have to think thoroughly what is your most used aperture and prepare to be stuck with it. Yes, you can always change the disc inside each lens but that can be a time consuming and precise process not fit to be done on set.

Having a locked aperture will require you to use good neutral density (ND) filters to control exposure without changing ISO and/or shutter speed. SIMMOD Lens offers excellent variable NDs if you want to check that out.

The streak filter also introduces some artifacts. The most noticeable of them is a vertical line cutting through the middle of your bokeh. This is something very few people notice when watching a finished film, but I promised I was going to tell you all about the negative aspects of anamorfaking before we got to the good parts.

Canon 85mm f/1.8 @ f/2 + Cinemorph 58mm from Vid-Atlantic on Canon EOS 5D MkIII

A brief review of the downsides: no extra field of view, cropping is required to achieve widescreen aspect ratios, oval bokeh requires fixed aperture plus ND filters, and streak flares introduce a line through bokeh. It is not that bad, right? This list is twice as long when shooting with anamorphic adapters, trust me.

04. COULD I GET SOME GOOD NEWS?

Now let's have a look at the advantages of anamorfaking over real anamorphic adapters. The advantages over spherical are pretty obvious: a neat look that feels way more expensive than it actually costs.

Low cost is one of the key points for anamorfakes. All you need is the lens you want to modify, an oval aperture disc (easy to make and cheap to buy) and tools that are easily accessible. No "months of stalking eBay for the perfect adapter". No need to pay $500+ for an entry level adapter. Just pick your spherical lens and you are good to go. If you start with $25 lenses you can build an entire set for a fraction of the price of an anamorphic adapter.

Simplicity is another good point. When using an anamorfake lens all you need to do is attach it to your camera and press record. You avoid all the step-rings, clamps, rails and lens supports that most anamorphic adapters require. You also avoid the extra weight!

Focusing is one of anamorphic adapters' worst quirks. If you buy a cheap adapter you are doomed to double focusing - setting both your taking lens and anamorphic adapter at the same distance to produce a sharp image - or buying a single focus solution - are you ready to fork out another $500+ to buy rack-focusing capabilities? Focusing with anamorfakes is the same as focusing any standard lens. If you modify an auto-focus lens you can even have your camera do all the work for you.

Anamorfake Canon EF 50mm f/1.4 USM - autofocus good and ready to go!

Besides all of these amazing feats, every anamorfake you make teaches you a little more about the mechanics of lenses, what parts usually connect together, how the glass is held in place and how to come up with creative solutions for unexpected problems. It also makes you appreciate your gear a lot more: you see the amount of detail that goes into building even a simple lens and you will treat it with proper respect.

Another thing about anamorfaking: never tell anyone you are faking it. Most people will never figure out how your footage looks so special. Plus, if you start using anamorphic adapters you can always add them on top of your anamorfakes for even more pronounced oval bokeh!

Three disassembled USSR lenses in the process of anamorfaking. You will get there too!

05. BUY IT NOW!

If DIY is not really your thing, there are at least a handful of options to start crafting your anamorfake look without the need to open a single lens. In this chapter we will take a look at the Cinemorph filters by Vid-Atlantic, Streak filters, IronGlass lenses, cheaper modifications on eBay, the Helios 44-2 Cat Eye mod and a quick word about the folks in the UK (Richard Gale and Motion Six).

CINEMORPH FILTER (VID-ATLANTIC) - $70-90

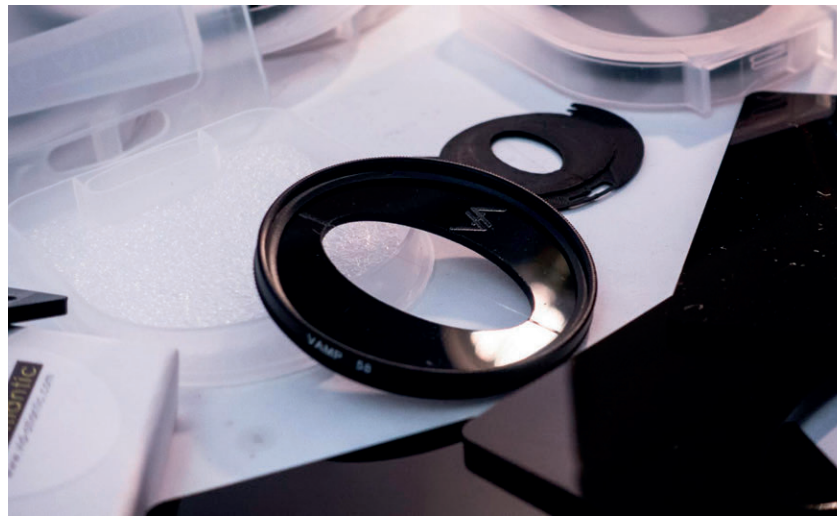
BUY IT NOW!

Use code TITO10 for 10% discount

The Cinemorph filter was created by the folks at Vid-Atlantic while trying to work around the hassle of double focus for anamorphic adapters and pick-up days without much budget for the gear used in the main shoot. Their goal was to make a filter that would allow them to intercut anamorphic and spherical footage without jarring differences and they originally only used it in their own projects.

The thing grew when others started to chase the anamorphic look on the cheap and Vid-Atlantic decided to sell the filter online. After several iterations, the current Cinemorph is a 3d-printed insert that mounts into a rotating filter frame.

All variations of the Cinemorph filter. Rear mount, 4x4 and 77mm.



The rotating frame allows you to reposition the oval however you prefer - the downside of the rotating frame is you can easily knock the oval out of alignment, so once you have it in position use a little piece of tape to lock it.

Canon EF 135mm f/2L @ f/2 + Cinemorph 77mm on Canon EOS 5D MkIII - proper and 90 degrees orientation.

As of late 2017, the Cinemorph comes without any glass in order to cut down ghosting artifacts. You can also pick your preference of streak filter for horizontal flares.

The Cinemorph comes in two sizes (**58mm** and **77mm**) and you should get the one that is closest to your lens' filter thread. Get a filter that is too small you will have heavy vignette. Get a filter that is too big and the oval bokeh will not show in the footage. Using a full frame camera I got my best results around 85mm, but was able to go down to 35mm without seeing vignette. None of my zoom lenses loved it, though.

These work really well with modern lenses and cost next to nothing to give you strong anamorphic vibes on a Canon L series prime set. The downside is the Cinemorph will cut around two stops of light, so Canon L glass is a really good call for it!

Vid-Atlantic's website provides plenty of information on the filters as well as a ton of material shot with them. They are also amazing in customer service so do not hesitate in contacting them if you have questions. You can also watch my review of the Cinemorph filter on YouTube for footage samples and a more in-depth analysis.

Besides the original thread-mounted filters Vid-Atlantic also offers **4x4"** and **4x5.65"** versions for mattebox mounted ovals with different sizes so you can find the one that works best with your lens. These have little notches so you can easily make and place your own streak filters.

CINEMORPH MOD (VID-ATLANTIC) - $35-50

BUY IT NOW!

Use code TITO10 for 10% discount

This version of the Cinemorph is aimed at the Sigma 18-35mm f/1.8 ART in EF mount and a few modern Zeiss Distagons that come in EF mount. I have only used it with the Sigma and it is my failsafe for wide shots when anamorphic adapters fail. I always have a cinemorphed 18-35mm in my lens kit when shooting anamorphic or anamorfake. It allows for super wide shots, super shallow depth of field and incredible close-ups. This kind of versatility can turn the scales on a tough schedule.

Cinemorphed Sigma 18-35mm f/1.8 ART in EF mount

Please do not worry about the “mod” word in the name of this Cinemorph. The modification process here is super simple and straightforward. All you need to do is loosen three screws on the back of the lens, slide in the filter, rotate it into position and tighten the screws back. The process has a more detailed explanation in the leaflet that comes with the mod, but that is the bulk of it.

This Cinemorph is a thin 3d-printed rear filter that will work over the entire zoom range of the Sigma. This is a fragile piece of gear and it can get tangled up with EF adapters, so be careful. On the thought that it can get caught on adapters the filter is made in an easily breakable fashion so it does not damage either the adapter or the lens.

When fiddling with aperture, stay between wide open (f/1.8) and f/2.8, as you will start to see vignetting on the sides of the frame past f/3.5 at 18mm. This means the best way to control your exposure is again through the use of ND filters.

For more detailed information and samples of what this filter delivers you can check out Vid-Atlantic’s website as well as watch my in-depth review.

Cinemorphed Sigma 18-35mm f/1.8 ART on Sony a7s II (S35 mode) - Multiverse Dating for Beginners

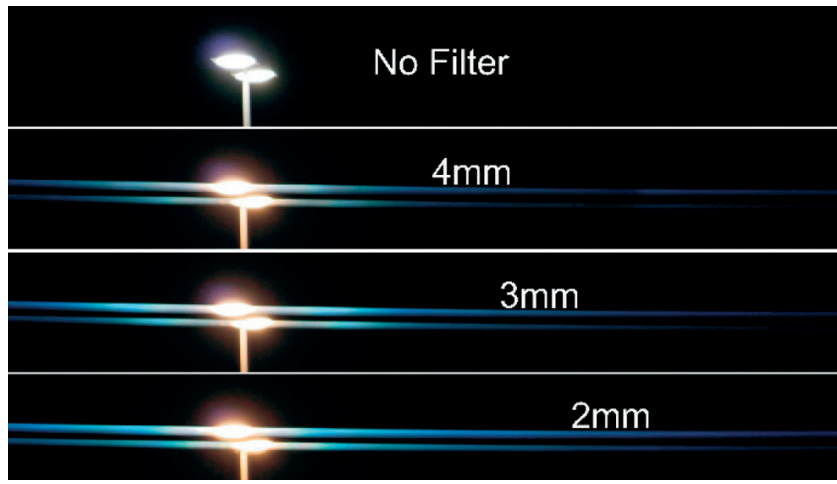
STREAK FILTERS (TIFFEN/SCHNEIDER) - $250-500
BUY IT NOW!

Streak filters is how the fancy folks cope with anamorfaking. These are way pricier than the Cinemorph and deliver only, well, streak flares. They do so through a grid of thin, evenly spaced lines. The space between the lines is how you rate them, going from 1mm (strongest flares) to 4mm (subtler effect).

The lines can be colored, tinting the flare, or neutral, and flaring the same color as the light source.

Coming from Tiffen and Schneider, these filters are aimed at bigger gear and usually come in 4x4" or 4x5.65" sizes. It was only recently I started seeing options in 77mm or 82mm with a rotating tray (just like the Cinemorph) so you can align the flares.

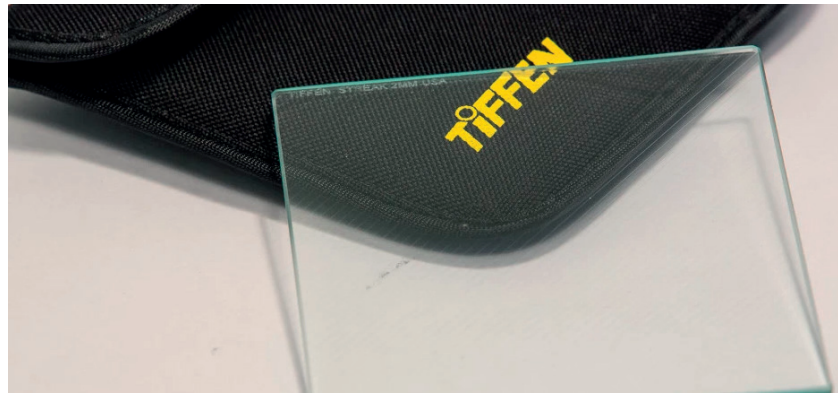
Blue streak filter comparison. Image from B&H Photo Video



The grid of lines guarantees you will get flares from lights all across the frame and the ability to pick the spacing between the lines allows you to control the intensity of the effect.

2mm Clear Streak + Helios 44-2 58mm f/2 with anamorfake mod on Canon EOS 5D MkIII

The picture above is part of my tests using a Tiffen 2mm Clear Streak filter. As you can see, the flare is pretty nice and strong. The other thing you can also notice straight away is that the lines from the filter show up very noticeably in bokeh (right side of the frame). I noticed the line pattern to show up no only at bokeh but also on any light halos or even on faint flares coming from big light sources like a china ball.



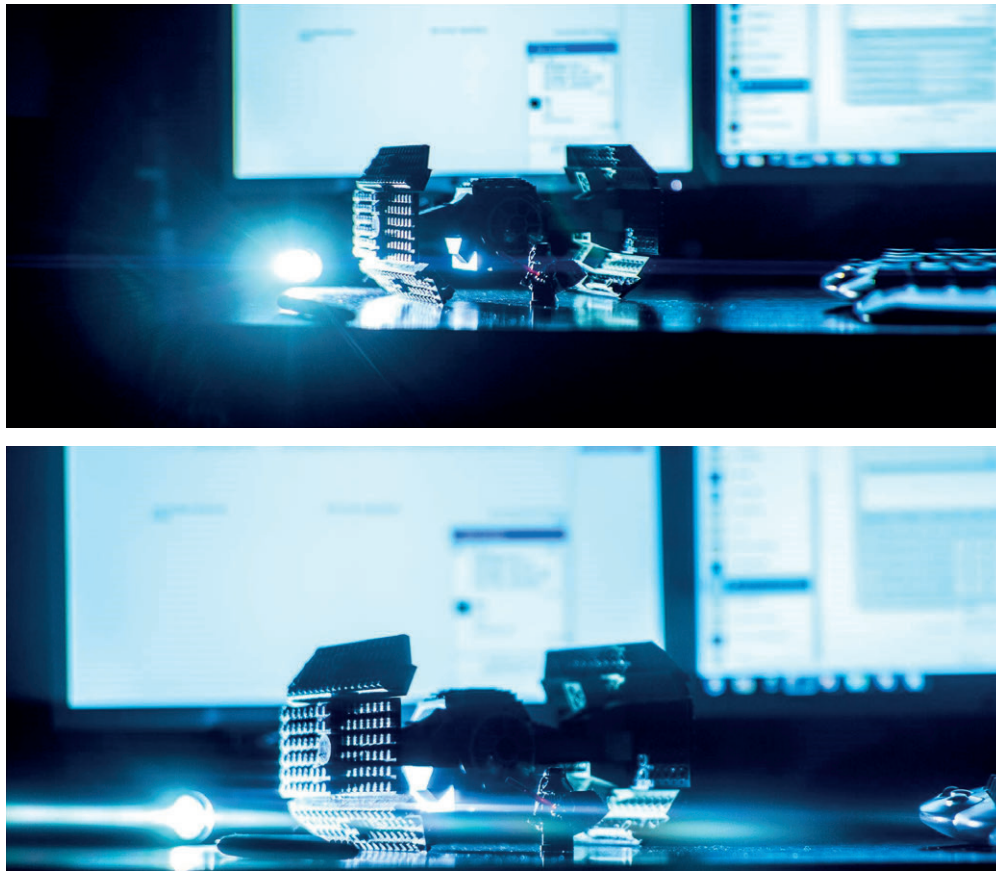
2mm Tiffen Clear Streak 4x4" Filter - it is hard to see the lines, but they will show up in bokeh.

One of the biggest advantages of the Streak filter over the Cinemorph and other anamorfake techniques is that Streak filters still work well with wide angle lenses, but the flares start to get "bendy" as they move towards the edges of the frame. I like the effect as it makes me feel the image is not flat and these are not just "slapped-in-post" flares.

I was lucky enough to find my 4x4" used on eBay for relatively cheap, but if you want to play with the effect before committing to it, Cokin offers much cheaper options. Its A and P lines sells between $10 and $50 and the filter you are looking for is the Cokin P 058 2pt Star. You will also have to commit to Cokin's filter holder, but it is not a tough experience.

Another thing to watch out for is the naming. For strange reasons, sometimes these are also called Star filters, except Star filters have many intersecting lines and produce a cheesy, non-anamorphic effect. This applies to both the Cokin and the Tiffen/Schneider filters. When buying one, watch out for the line's pattern and avoid crossed lines.

Below is a comparison between a real anamorphic flare and the one generated by the 2mm Clear Streak filter. The filter version is much stronger than the anamorphic, and it does not blend with the footage as organically - in fairness, we are comparing a $4500 adapter with a $200 filter.



Top: Iscorama 54 1.5x anamorphic adapter flare. Bottom: 2mm Clear Streak Filter.
Both paired with a Helios 44-2 58mm f/2 on Canon EOS 5D MkIII

The anamorphic's flare is also more saturated and visibly more blue. That is a noteworthy quality: most anamorphics flare towards blue/purple or orange. Because of that, if you want to stay as close as possible to the anamorphic look, I would say the Blue or Orange streak filters are your strongest bet.

IRONGLASS ANAMORFAKE LENSES - $120-2400
BUY IT NOW!

IronGlass became a player in the anamorfake market after I started posting tutorials on how to modify USSR lenses on my channel. IronGlass has taken it further than I did, modifying lenses I have never done before. Located in Ukraine, they have quick and easy access to all the base lenses plus proper tools for intense modifications.

Soviet Lens Army: Mir 20M 20mm f/3.5, Mir 10A 28mm f/3.5, Mir 24N 35mm f/2, Mir 1B 37mm f/2.8, Helios 44-2 58mm f/2, Jupiter 9 85mm f/2, Tair 11A 135mm f/2.8, Tair 3 300mm f/4.5

IronGlass currently offers anamorfakes in EF, E-mount and PL mounts. You can buy single lenses or various combinations ranging from 3-lens sets all the way up to 7-lens sets, covering every imaginable focal length. Orders can be placed either on their eBay store or on their website. You also get some level of customization by choosing flare and aperture colors. The lenses are all USSR glass, which I am a big fan of, but some people do not love their strong character that transpires low contrast and whacky flares.

The lenses come with solid focus gears - some come with iris gears as well -, declicked apertures and standard front sizes/threads, which makes them ready for production. Some of the larger sets even come in their own hard case. I found their oval shapes to be wider than a 2x oval would be, more like 1.33x, so watch for that.

5-lens anamorfake set from Iron Glass: Mir 1, Helios 44-2, Jupiter 9, Kaleinar 5n and Jupiter 37A

These lenses are not the cheapest solution around but if you need something right away and would prefer to avoid the DIY path and still save money compared to buying an anamorphic adapter, IronGlass is a good option.

OTHER EBAY SELLERS - $70-150
BUY IT NOW!

A bunch of other sellers started making anamorfake lenses. These are mostly Helios 44's and all fall in the same price range. They usually do not include focus gears or any other fancy mods - even the streak flare is uncommon in this group.

I noticed in some of them that the original iris of the lens is taken out and replaced with the oval disc. To me that makes the lens less versatile and you have to rely on ND filters to control exposure. On the plus side, you can adjust the rotation of the oval disc through the aperture ring. Just like the Cinemorph, I recommend you use a small piece of tape to lock rotation once you find your optimal setting.

Helios 44-2 58mm f/2 with anamorfake mod on Canon EOS 5D MkIII. Simple, yet super effective.

These are good options if you just want to play and explore the look without committing either a lot of money or a lot of effort in doing your own modifications.

HELIOS 44-2 CAT EYE MOD - $150-200
BUY IT, MAYBE!

The Cat Eye mod is a Helios 44-2 with a modified iris. The biggest difference from regular anamorfakes here is that you can open and close the iris somewhat normally. The fastest iris is around f/5.6 and the slowest is f/11. The shape is not a perfect oval, it has a pinch on the top and bottom. It also gets thinner as you stop it down, instead of just becoming smaller. You can see some test shots in this review.

I first came across this mod on this video by RetroFotoHouse. From there I started talking to Roman (who runs the channel and this eBay store) and I ordered a few units. These are time-consuming to make and require special equipment and skill beyond my level.

Cat Eye Helios 44-2 58mm f/2 - Wide open at f/5.6-ish and stopped down to f/11-ish

The result of the pinch and not-so perfect oval is a stronger waterfall effect on bokeh, even when at f/5.6. I highly recommend using a rotating M42 adapter so you can align the cat eye shape as vertically as possible.

Helios 44-2 58mm f/2 @ f/5.6 due to Cat Eye mod on Sony a7s II

The greatest challenge with this mod is actually getting one. Roman only makes them in batches, but then what are you going to do with six or ten of these? They are not usually available at his eBay store, but try sending him a message!

A NOTE ABOUT OTHER ANAMORFAKE MAKERS

Unless you have been living under a rock, you should be aware that there are other known anamorfake manufacturers. Namely DogSchidt Optiks (DSO), now **Richard Gale Optics**, and **Motion Six**.

Gale's products are delivered with astonishing quality since the beginning. I followed their origin at the EOSHD Anamorphic forum with the FlareFactory58 then growing into the TRUMP series, which continues to evolve and upgrade. You can find detailed information about some of Gale's gear in my three-episode series on YouTube.

Originally designed to enhance the anamorphic look of 1.33x and 1.5x adapters through tailored flares and oval bokeh, DSO's modification of the Helios 44-2 quickly became something completely new.

DogSchidt Optiks' FlareFactory58, TRUMP, Optical Attachments and various accessories.

DSO's lenses offered an unprecedented level of customization and I still miss the original FlareFactory58. It had a much friendlier price point while delivering plenty of customization options. I would not think twice before pulling the trigger on a FlareFactory58 with the right modifications if I saw one on eBay. On current products, Gale's prices are steep, but I stand by its quality and craftsmanship.

FlareFactory58 @ f/4 on Canon EOS 5D MkIII
Custom build: 2x oval bokeh, f/4, blue flares, medium contrast, high glow and low cleaning marks

Gale shut me down after I made the extreme Helios 44-2 modification tutorial and accused me of stealing his ideas from playing with the FlareFactory58 and TRUMP lenses. Since then I have tried to re-establish contact multiple times without success. Because of this I am aprehensive of adding more information about his lenses in this guide since I have no channel for checking the data with him.

Motion Six is another anamorfake maker in the U.K. While I had some exchanges with Gale and his products, Motion Six is an entirely different story. I never owned any Motion Six lenses and I do not plan on buying them. I have strong reasons to believe they would not sell me one either.

I only learned about them after receiving a threatening message on eBay - also after I published the Helios 44-2 extreme mod tutorial and sold some of the lenses I had modified. I have no positive experience with these guys.

06. PREP YOUR WORKSTATION

Time to get started on your own anamorfakes! Before that, let's go over the list of tools you are likely to need and then have a quick word about what lenses are best fit for this process.

This list of ingredients is quite comprehensive, so I am underscoring the essential items. You are good to start fiddling if you have these. The rest of the list just increases your possibilities.

TOOLS AND MATERIALS SHOPPING LIST

LENS WRENCH. This is your most important tool. You are going to use it to take many things apart.

SANDPAPER (COARSER GRITS). Sandpaper is useful for removing the original paint of many metal parts in case you want to repaint them and reinforce flares. It is also key when thinning out thick oval aperture discs.

SPRAY PAINT (BLACK, ORANGE, BLUE, ETC). Spray paint goes hand in hand with the sandpaper. Pick the colors you want to reinforce in your flares and internal reflections

SHARPIES (BLACK, ORANGE, BLUE, AS MANY COLORS AS YOU CAN FIND). These are what allow you to tint your aperture and flares as well as some element sides. I bought a 36-color pack and never looked back. You can get away with less colors if you know what you want. Blue and orange cover 95% of my modifications.

DOUBLE SIDED TAPE. Double sided tape will be useful to secure ovals in place as well as hold flare lines down while you close your anamorfakes.

MASKING TAPE. Masking tape is key when spray painting anything. You do not want paint in any moving parts!

SCISSORS. It is needless to try and explain why scissors are important. You will be cutting many things. Fishing line, tapes, ovals...

FISHING LINE (MULTIPLE DIAMETERS). These are the flare-makers! Each diameter gives you a different intensity of flare, and Sharpies tint the flares.

OVAL CUTOUTS (MULTIPLE SIZES). You can either make your own from the provided templates or buy them right here.

ELECTRICAL TAPE. More resistant than masking tape and useful to hold things in place and patch holes from imprecise scissor cuts.

CALIPER. A caliper will be useful for measuring and making your own oval aperture discs and anything related to 3d printing (lens caps, focus gears)

VECTOR DRAWING SOFTWARE. I use Adobe Illustrator but there are plenty of free options out there (Inkscape). All you need is to be able to output accurate sizes for laser cutting your oval apertures.

ND FILTERS (OR VARIABLE NDS). Many of these mods will result in a scenario where you do not want to change your aperture. Because of that, NDs become key in controlling exposure.

CHEAP CPL FILTERS. You are going to get rid of the glass (or plastic) in these filters and just make use of the rotating frame. Pick sizes that match the lenses you are working with.

CHEAP UV FILTERS. Same thing as the CPL. You want the cheapest ones you can find, as the glass is going to be discarded. Sometimes these are useful as spacers.

A GOOD SET OF SMALL SCREWDRIVERS. These are just as important as your lens wrench, except you can find them in any hardware store. Get something that has #00 and #000 sizes for Phillips and JIS. Watch repair screwdrivers also work well.

ACETONE. Acetone is a controversial item in this list. It can eat away at optical cement, but it is also a great tool for wiping out the blackening material on the sides of glass elements, allowing you to repaint them and customize your flares even further. Never handle acetone without gloves!

COTTON PADS. These are the best way to keep your acetone under control. Dose small drops and rub away at the elements until they are clean!

METAL POLISH. Another controversial item, this will introduce a huge amount of microscratches and lower the contrast of your lens by a ton. It also wipes away your coatings, making the lens flare a lot more (almost uncontrollably).

NITRILE GLOVES. Always do all your anamorfake work with gloves. You want to avoid greasy fingerprints on the inside of your lenses - as well as avoid lens grease, sharpie paint, spray paint, acetone burns and metal polish on your bare skin!

3D PRINTER. A 3d printer is a handy item and allows you to make focus gears and lens caps as well as many other useful things. Depending on your printer, you might even be able to make oval aperture discs with it instead of laser cutting.

ISOPROPYL ALCOHOL + LENS WIPES. These two come together. They are the regular stuff you would use to clean your lenses, but now you will be cleaning the inside elements too before closing it all.

CANNED AIR. It is always a good measure to blast off any dust from inside the lenses before you close them up.

ROTATING M42 ADAPTERS. All the USSR lenses in this guide are M42 and getting the oval + flare alignment right in the first try can be daunting. These adapters will save you a lot of time while also transforming the M42 mount into whatever you need to attach to your camera.

This should cover all the buying part besides the lenses themselves. Even if you get all of it, these items will not cost much.

A masking tape masterpiece for spray painting the Mir 1B.

CLEAR THE SPACE!

Do all of your work in a well-lit, clean and flat area, without dust and other types of debris. As good measure I like to photograph or record every step of modification so I can backtrack on it if I become unsure of where some part goes. I also learned through experience to not do any lens disassembling on the kitchen table or even my work table because I might get stuck in the middle of the process and all of the sudden I need that space for a different activity, so pieces get mixed together, dust falls everywhere, you lose track of what part went where and so on.

Make room - and time - for this endeavor, otherwise it can be quite unpleasant.

A far from ideal work area: cramped, dark and dusty. Spare yourself the stress.

THE IMPORTANCE OF THE HELIOS 44-2

It is not hard to notice the amount of mentions to the Helios 44-2 in this guide. Sixteen up to now, and going up. The Helios packs many qualities that make it a great candidate for

anamorfaking. It has a very simple build, not too many elements, it is super cheap and readily available.

This combination by itself already lowers the stress-bar for cracking it open. You will not be playing with some irreplaceable, ultra complex piece of gear. You will be experimenting on the AK-47 of lenses: it can take a beating, be assembled all wrong and it will still work to some extent. You will see in detail when we get to the proper tutorials, but the Helios is super easy to take apart and put back together.

On top of that the Helios 44-2 features one of the simplest and most effective optical designs: the double-gauss. It is used in several high-quality, cheap and fast lenses around the 50mm focal length. If you have a 50mm f/1.8, it is likely to be a double-gauss design.

A formula for success, right? These are some of the reasons the Helios 44-2 is recommended as an entry-lens for people who want to experiment with vintage optics and also why that is the first lens we are going to anamorfake.

IMPORTANT QUALITIES FOR ANAMORFAKING A LENS

While all lenses can be anamorfaked, I prefer to focus my efforts on the ones that show certain qualities that make the work easier.

The first one of these qualities is full manual control. No auto-focus. No electronics. While camera-controlled aperture and autofocus are great, they require the inside of the lenses to be crammed with chips, wires and motors. This adds tons of steps and raises the risks of damage to the lenses while disassembling them. Plus, there is no way to know how the lens' chip will react to you jamming something inside of it like a virus.

That leaves us with mostly vintage lenses, which is not bad. You have a great array to choose from and lower prices when compared to modern lenses - electronics cost more after all.

Cost is the next important quality. I try to avoid modifying expensive lenses. I know I will be torn and never forgive myself if I mess it up a $900 lens. There are still a million cheap vintage lenses to pick from. If you want to modify an expensive set, I recommend getting in touch with a professional tech - such as Ron at SIMMOD Lens.

When working on your own anamorfakes beyond this guide, I recommend going with the cheapest lenses you can find. Try Craigslist! I also do a lot of researching online to check if someone has already posted a disassembly guide for the lens I am trying to open. The first dozen or so are learning experience, aimed to make you more comfortable with the process of disassembling lenses and making your own adjustments.

Pentacon Electric 29mm f/2.8 with anamorfake mod. Cost: $45

Do not be afraid of modifying multiple copies of the same lens. There is always something that we can improve or play with. It took me more than ten Helios 44-2 mods before I settled on the full process described in **Chapter 09**.

A lens' construction is also a key quality when it comes to the modding process. As you will see later on, many lenses have a quick-access path to the aperture mechanism. If a lens allows you to easily take out all the optical groups behind or in front of the iris, this is a good candidate for anamorfaking. Some other lenses have each element on its own retaining ring and getting to the aperture grows into an endless list of steps, making it hard to remember the way back and keep track of what part goes where. Plus, if you need to go back in and make adjustments, the easier the process, the better.

The last quality I take into account is a lens' character. This one should only come into play after you get acquainted with modding in general.

Let me use the lenses featured in this guide as examples of good choices for anamorfaking. I love the look of USSR glass: the low contrast, the hazing, the easy flares and other artifacts, so I felt it would be interesting to build on top of that - plus they are dirt cheap. The Contax on the other hand are much cleaner but still have great character and can be found for modest prices. The Rokinon/Samyang are just everywhere these days and their popularity should encourage people to start moving towards the anamorphic look.

07. DESIGN YOUR OWN OVAL APERTURES

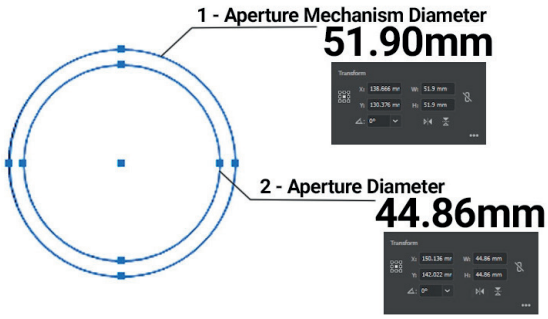
(a.k.a. Anamorfake Any Lens!)

This chapter is all about math. Here I am going to cover how to measure and make oval apertures for any lens you manage to crack open. It includes calculations for light loss, maximum use of the aperture area, anamorphic stretch simulation and even combining anamorfakes with real anamorphic adapters. I have a video version of this chapter, in case you prefer that instead of reading!

The tools we are going to need are a caliper, any lens you managed to open - I am going to use the Tair 11A as an example - and a vector-based program such as Adobe Illustrator.

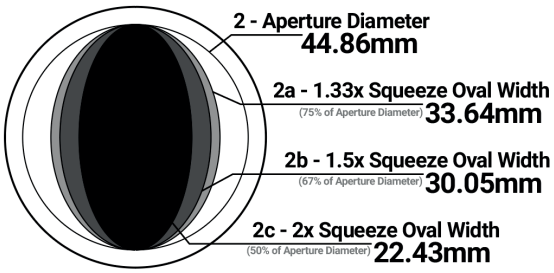
When you get to the aperture, there are two things for you to measure with the caliper and write down. The first one is the diameter of the aperture mechanism area, the second one is the diameter of the aperture itself, which is our base number for calculating the new f-stop of the lens after the modification. On the image below these are marked with 1 and 2 respectively, along with its sizes. Use at least two decimal cases for precision.

With these two numbers go to Illustrator (or your favorite vector software) and create two concentric circles. Remember to include the measurement unit (mm) to make the vectors match the real world. The outer circle is going to be the size of the Aperture Mechanism Diameter (1), 51.90mm, and the inner circle is scaled like the Aperture Diameter (2), 44.86mm. So far so good, but nothing anamorfake about it.



Select the inner circle and change its width without touching the height - on Illustrator you have to disable the little chainlink icon. The new width of the inner circle is a percentage of the Aperture Diameter (2). That percentage is given by how strong you want the ovals to look like: divide 1 by the stretch you are aiming to achieve.

1/1.33 = 0.75 or 75%
1/1.5 = 0.67 or 67%
1/2 = 0.50 or 50%



This design will give you a disc that fits over the aperture mechanism. It will not fall deeper into the lens or move anywhere. At the same time it takes full advantage of the aperture's height, minimizing light loss to as little as possible.

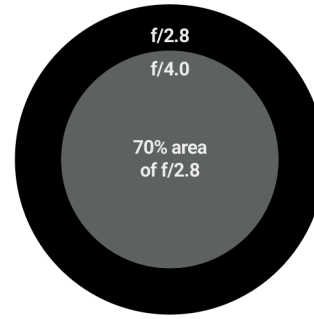
After this point we are going even deeper in the math rabbit hole of calculating light stops and anamorphic combinations. Get your pad and pencil.

First we are going to calculate how the oval insert affects the lens' maximum f-stop. It is known that every time the aperture area is divided by $\sqrt{2}$, light transmission is divided in half, cutting down one stop of light. If you want a more detailed explanation read this Wikipedia article.

The f-stop is directly related to the aperture area
Let's consider aperture area = 1
Dividing the aperture area by $\sqrt{2}$ cuts one stop of light
In an accurate approximation, $\sqrt{2}$ = ~1.41

1/1.41 = 0.7 or 70%
full aperture area /$\sqrt{2}$ = new aperture area or % of original aperture area

So every time the aperture area is rescaled to 70% of its original size, we cut one stop of light



This means that every time we shrink the area of a circle to 70% of its original size, we are cutting one stop of light. Now look at the percentages we are using: 75% and 67% are pretty close to 70%. Interesting, no?

With that in mind, let's continue with the Tair 11A as our example. We need a formula for the aperture area. The area of a circle is given by (πr^2), and we can increase its usefulness by remembering a circle is a specific type of ellipse (the mathematical name for ovals). Here is the formula for the area of an ellipse

Ellipse Area = π x radius H x radius W

Radius is an important measurement when dealing with circles and ellipses. Since a circle's height and width are identical, its radius is given by the formula (diameter/2). For the Tair, that is 44.86/2 = 22.43mm. Lastly, for practical reasons, I am rounding π to 3.14.

f/2.8 area = 3.14 x 22.43 x 22.43 = **1579.74 mm²**
f/4 area = 3.14 x 18.76 x 18.76 = **1105.08 mm²**

Now let's turn the full aperture into a 1.5x oval by reducing its width to 67% of the original size. That means the width radius will be 22.43*0.67 = 15.02mm, ultimately reducing the full aperture area to 67% of its original value.

f-stop area with 1.5x squeeze = 3.14 x 22.43 x 15.02 = **1057.86 mm²**

If we compare the area of the Tair 11A at f/4 (1105.08 mm²) and the aperture value at 1.5x squeeze (1057.86 mm²), one could say the difference is negligible. Because of this we can say that inserting a 1.5x oval into a f/2.8 lens will cut down one stop of light, turning it into a f/4 lens.

It makes sense when we compare the numbers, but what if this is a coincidence? Without the actual Tair numbers, the logic would be:

1.5x squeeze oval aperture = 67% of the original aperture width
67% of the original aperture width ≈ 70% of the aperture area
70% of the aperture area ≈ dividing the aperture area by √2
dividing the aperture area by √2 = **one less stop of light transmission**
1.5x squeeze oval aperture = one less stop of light transmission

This can be applied to any lens. If your base lens is f/4, it will turn into f/5.6, if it is f/1.4, the new maximum aperture is going to be f/2. While I demonstrated it for 1.5x, one could say 1.33x is close enough to have the same effects (75%). Math is magic.

The difference is when making it a 2x oval, because we reduce the aperture area to half of its original size (50%), and that means $\sqrt{2^2}$, translating to two stops of light loss.

These changes have effect in resolution and depth of field. Since your vertical aperture is at least one stop faster than your horizontal aperture, you will end up with better image resolution and sharpness horizontally. The oval shape acts as a narrower f-stop horizontally, while vertically we still use the full aperture f-stop. Same applies for depth of field: shallower vertically and deeper horizontally.

This is interesting because it is the exact opposite of what actual anamorphics do. When shooting with anamorphics you have increased vertical resolution and sharpness. This happens because anamorphics effectively squeeze more information in the horizontal axis - think of it as an unusual codec compression method.

This is the only time in this guide I will bring up something that has to do with shooting with anamorphics. Consider it a sample of the "Anamorphic Cookbook".

Keeping in mind that anamorfakes and anamorphics work in opposite ways when it comes to resolution and depth of field, but deliver similar looking results, you can combine both techniques. A positive aspect of doing so is that you "even out" the strengths and weaknesses of each technique, creating a setup that takes advantage of the best of both worlds.

Spherical and 16:9 on top, 37mm. Black letterbox indicates cropping. Anamorphic 2.66:1 below, 75mm

Many people like to use anamorfake lenses to enhance bokeh in anamorphic adapters with weaker squeezes - such as 1.33x and 1.5x. By doing so one can easily achieve unrealistic bokeh compression like 3x or even 4x. For the sake of consistency, let's define the main goal is to achieve 2x squeeze bokeh. If you understand the math, you will be able to figure out your own insane squeezes.

The only anamorphics we can enhance to achieve 2x bokeh are 1.33x and 1.5x. Here are the steps to design the right aperture, making sure you have the best light transmission and oval goodness.

Squeeze works in a similar way to everything we saw so far - minus the light loss. A 1.5x anamorphic adapter will make bokeh 1.5x thinner, which is 67% (1/1.5) of its circular width.

The real squeeze is multiplied by the simulated squeeze of the oval aperture. Continuing with our 1.5x oval aperture example from above, if we combine it with a 1.5x anamorphic adapter, bokeh squeeze will be:

0.67 (from the oval aperture) **x 0.67** (from the adapter) = **0.44 or 44%**

0.44 is less than 0.5 or 50%, which says that this bokeh is more squeezed than 2x bokeh and it is no good for our purposes. We agreed on that, remember? This makes for unrealistic bokeh and it is also cutting more light than we need to lose. Time to make a more subtle oval aperture.

The best way to tackle the problem is to start from the result we want to achieve. 0.5 or 50% has the be the product of two other numbers, the squeeze factor and the anamorfake aperture. At least one of these has to be known in advance.

50% bokeh = 1.5x anamorphic squeeze (1/1.5 = 0.67) x anamorfake squeeze (x)

$$0.5 = 0.67 \times x$$
$$x = 0.74 \text{ or } 74\%$$

50% bokeh = 1.33x anamorphic squeeze (1/1.33 = 0.75) x anamorfake squeeze (x)

$$0.5 = 0.75 \times x$$
$$x = 0.66 \text{ or } 66\%$$

The resulting number is how much we need to scale the width of the oval aperture from our original circular shape. Both cases are close enough to 70% to say only one stop of light will be lost. It is not too expensive to cheat 2x bokeh on an adapter without that much squeeze!

The interesting part is that when we pay attention to the numbers, on the first example a 1.5x anamorphic requires 1.33x anamorfake to match 2x bokeh, and in the second example a 1.33x anamorphic needs 1.5x anamorfake (no surprise here though, the order of the factors does not change the product in multiplication).

08. STREAK FLARE

Streak flares are the most adored trait of anamorphic lenses - more than oval bokeh, I dare to say. So in the anamorfaking process we must have options for achieving said flares.

Lens flares in general are an artifact created when strong rays of light bounce around the various glass elements of a lens, aperture blades and the insides of the lens' body. Why do only anamorphics have streak flares then? In an anamorphic lens the cylidrical glass elements responsible for the squeeze are the ones generating the elongated internal reflections - and the coatings on the glass cast the colors shown on those streaks.

If you tried simulating the anamorphic look at any point in life, you probably know fishing line across the front of a lens can give a streak flare on any lens. Since I could not figure out myself why that happens, I went to Brian Caldwell, creator of the Speedbooster and father of the Chameleon Anamorphic Lenses to ask why does a fishing create a streak flare. This is what I got:

> *"[The] streak flare from fishing line is similar to star pattern from an iris opening. [It is] caused by diffraction from either side of the wire, so it's really two overlapping horizontal lines assuming the fishing line is vertical. Similarly, a three-sided aperture will have six flare star points emanating in either direction perpendicular to each of the three flat faces. A perpendicular knife edge should also cause horizontal flare."*

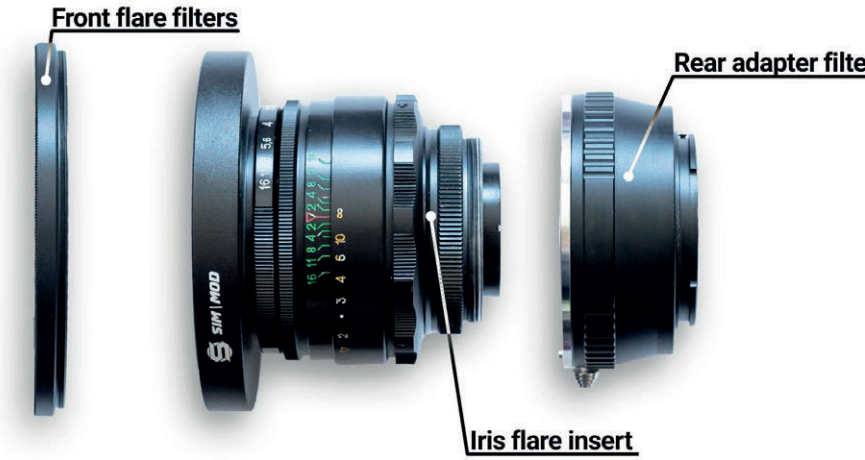
As I was still a little confused, this article about lens flares from Cambridge in Colour set things straight as I was able to count the star points and figure out the number of sides in a lens' aperture. Plus, I was able to get a streak flare from a knife in front of the lens, so it all checks out:

This is the camera setup for the flare right above. Sony A7s2 + Helios 40 + giant knife.

Anamorfake flares lack the depth and element separation of the real flares but they are the next best thing - and most of them are super cheap, so it is a worthy consideration.

Now that theory is out of the way, there are a few different ways to create streak flares. I am splitting these methods based on the placement of the strings in regards to the lens. Our three options are:

- **IN FRONT OF THE LENS**
- **IN THE MIDDLE OF THE LENS (WITH AN IRIS INSERT)**
- **BEHIND THE LENS**



The three options for placing flare filters

Regardless of where we place our flare-making filters, all optical solutions hurt bokeh. Some will be more noticeable, others less. We are effectively introducing obstacles to the light coming through the lens and these obstacles show up in the out-of-focus-areas. The only way to achieve streak flares and clean bokeh is through post-production, so if bokeh is untouchable for you, check **Chapter 10** and skip this chapter altogether.

Another unavoidable downside of creating flares with filters is that real anamorphic flares will not rotate when the camera rotates while anamorfake flares will follow the shot's orientation. I had never thought about this before I shot these tests, so unless your audience is very familiar with how real anamorphic flares work, this quirk will go unnoticed!

Anamorphic flares are made up by the reflections of multiple elements.

The colors shown in anamorfake flares are directly related to the color present in the diffractive surface (also known as the fishing line). If your streak filter features blue, you will get blue flares. If the lines are clear, the flares will take on the color of the light source creating the flare. Multiple light sources with different colors will create a shot that shows flares with all those different colors. My tests show various options. For the DIY solutions, Sharpie markers offer great adherence and are cheap enough to buy an arsenal of different colors.

A small glimpse of my Sharpie collection.

Starting with flare filters that go in front of the lens, I tried to cover as many options as possible: some entirely DIY, then the cheap variety, all the way up to expensive filters. Do not be mistaken though: there is always a more expensive option.

In order of cost, first up is a basic scratch across a UV filter, followed by a line across the front of the lens, then Vormax's Pseudoanamorphic filter, an old Cokin filter, Vid-Atlantic's streak flare and then Schneider's streak flares. Past that point, if you really want to spend money, you should look into Tiffen, Schneider and Formatt filters, or even Lindsey Optics' options.

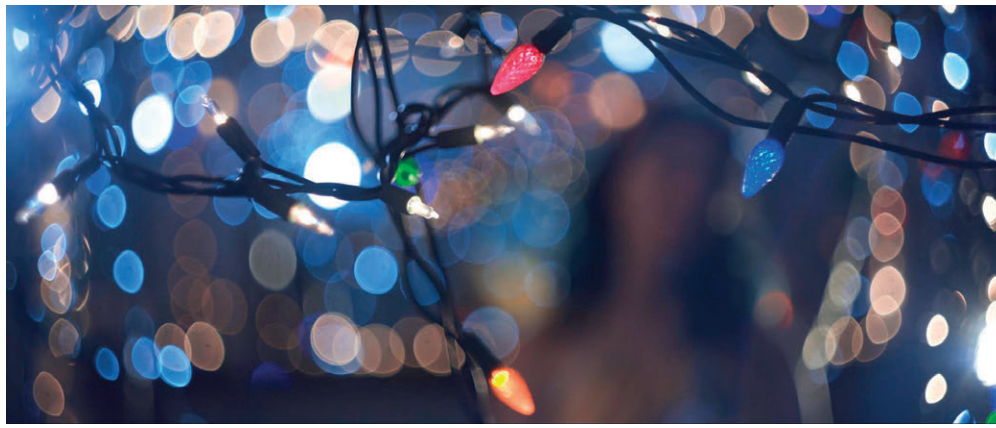
Lindsey Optics' awesome variety of streak filters. More information on them here!

I shot the comparisons for all the flare options in this chapter together, back to back, which made it really easy to notice the differences between them. The camera settings were kept constant unless I needed to adjust something drastic.

I used the Sony A7s2, shooting at 4K, 30fps, S-Log2, WB at 3200K and all lenses were wide open unless noted. Same for ISO, 1600 unless specified. Shutter varied to accommodate for light intensity, LED flickering and maximum aperture (I did not want to introduce a Vari ND into a filter test).

I have multiple light sources and colors, different depths to compare how the filter affects bokeh, and various focal lengths to consider if longer or wider lenses break the effect.

For reference, I shot a sample with no filters, so we can look at the original frame, flares and bokeh.



Clean reference frames shot with the Helios 44-2 and no flare filters. Aperture wide open.

You can find all these frames in full resolution in along with other flare samples in ./Appendix/08_StreakFlares/

SCRATCHED UV FILTER - $2

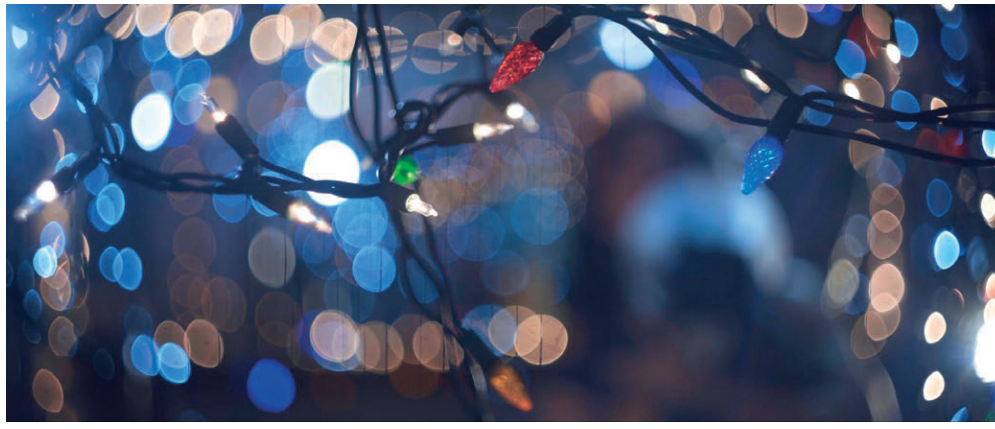
It is hard for me to say something is straight up bad. This is one of these cases and here is my take on it: not worth it. Still, here is the data.

Unbeatable simplicity: get a cheap UV filter, a ruler and an exacto knife and run a straight cut/scratch right down the center of the filter. The straighter the cut, the cleaner the flare.

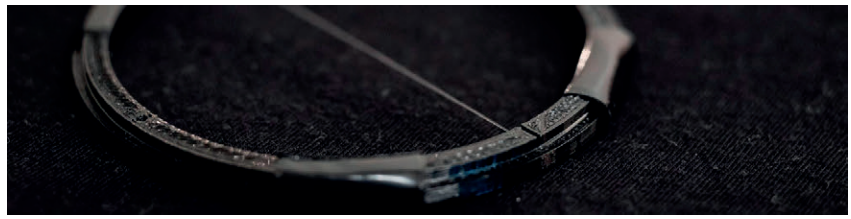
You can find a video tutorial for this filter in ./Videos/08_StreakFlares/ScratchFlare.mp4

While making mine I had to run the exacto knife multiple times over the same line in order to get a decent strike on the glass. It gets easier with every pass. Also, a UV filter is not the best mounting frame because it does not allow you to freely rotate the scratch for perfect alignment. I recommend screwing this on an empty polarizer frame so you can make sure the flare is perfectly horizontal every time. Tint is optional, use a fine point marker to give color to your flare.

I noticed loss of contrast, hardly visible flares and a giant line through bokeh. Since I did not tint the scratch, the flares take on the color of the light sources that create them.



FISHING LINE ACROSS THE FRONT OF THE LENS - $5

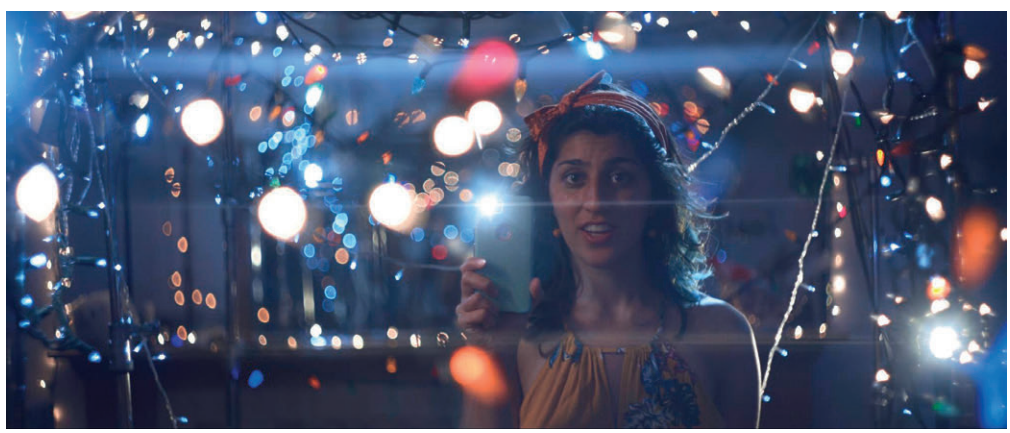


On my first try I just taped the line on both sides of the barrel. This ended up giving me skewed results so I made a 3d printed part that keeps the line straight. It can be inserted into a 77mm filter frame and rotated for alignment! It was a bit too complicated and I should have just improved my tape job. I learned from it so you could also learn from my mistake! You can also 3D print yours and disagree with me.

You can find the STL file for 3d printing in ./Modding Parts/Flare Filters/77_LineFlare.stl
You can also find a video tutorial of assembly in ./Videos/08_StreakFlares/LineFlare.mp4

Different diameters of fishing line give different flares. My personal take is "the thinner the better" and these samples used a 0.2mm line. Tint is optional, use a marker to give color to your flare.

I had a clear string across the front of the lens, which made stronger neutral flares. No tint means flares matching light sources. It also has somewhat of a high threshold as to which lights create flares. You will notice in the upcoming tests there are way "flarier" options.



The only thing that makes this costlier than wrecking a UV filter is that I was unable to buy just a small amount of fishing line, you have to buy the whole reel! This will give you a line down bokeh, unavoidable.

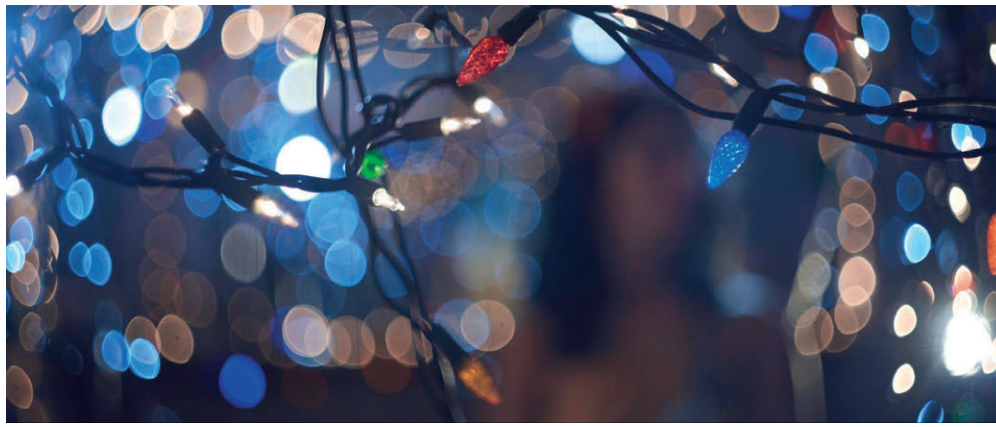
I noticed all the streak filters involving fishing line had splintery flares. They are thin around the light source, but as the flare stretches out, the ends widen and we get many multi-directional splinters. I did not love the effect but could not pinpoint if it was my particular type of fishing line.

VID-ATLANTIC FLARE/STREAK FILTER - $70

I find Vid-Atlantic's streak filters produce better results than the basic fishing line when they actually flare. There is something about the material they use for the flares, which gives off a much stronger and saturated streak. They come in two sizes (58 and 77mm), on a rotating frame that allows you to quickly realign your flares. I also find it useful to lock rotation using a small piece of tape, preventing accidentally skewed flares.

While their website shows beautiful shots and nice streaks, the ones I got on this test were quite muted. Maybe they need a very strong light source and I failed at that, although the one on the top left of the frame is a 300W tungsten light pointed directly at camera (but it barely made a line).

The string across the filter is strong red which should yield beautiful flares but they are far too subtle for my taste. On the bright side, damage to bokeh is almost imperceptible.



If your shoot uses natural light and you are getting flares straight from the Sun or other stupidly strong light source you should have better results than I did

SINGLE LINE VS MULTIPLE LINES

Now we step into a different category of streak filters. All options so far featured a single line right down the middle of the lens. The advantages of having multiple lines is that you get stronger flares overall and flares show up more

easily. The spacing between the lines determines the strength - for example, 2mm spaced lines give off much stronger flares than 4mm spaced lines. The downside is that now you will have multiple lines cutting through your bokeh.

Subtler flares and only one hit through bokeh versus stronger flares and multiple hits through bokeh.

I learned as I tested that 50mm is the ideal base focal length and you can go longer without problems while using any of the filters. The flares thicken as you switch for longer lenses and they always look nice.

Bokeh is affected the exact same way regardless of focal length, so longer lenses make bokeh artifacts more noticeable simply because bokeh is larger in the frame.

Contax 35, 50, 85 and 135mm top to bottom.

As you go wider, though, single line filters will start to fall apart and give you shorter and thicker flares while multi-line filters hold up pretty well even on wide angles.

Front streak filters will start to distort with the optics too, becoming bendy at the top and bottom of the frame, which also happens with wide anamorphics, so I find that to be a nice touch. The adapter-mounted filter keeps the lines straight.

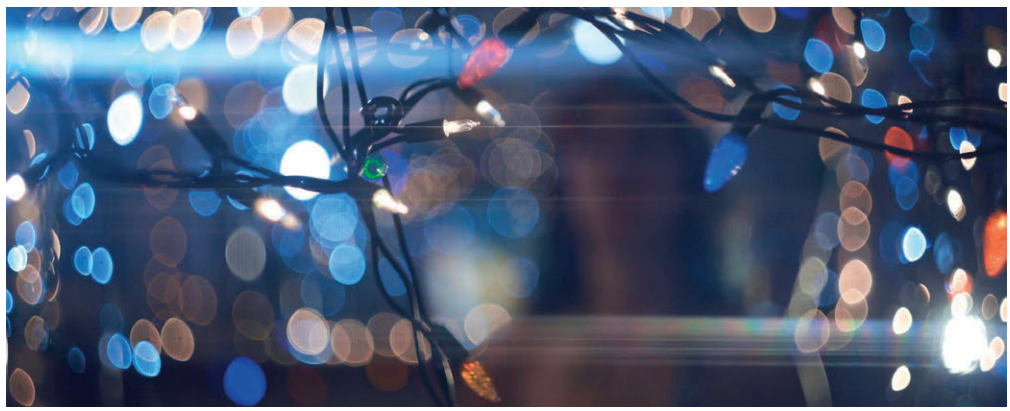
Top: Contax 21/2.8 with rear adapter. Bottom: Contax 28/2.8 with Cokin in front.

When it comes to focus, unlike real anamorphic flares, these are "focusable", which means that the streak will be sharpest and thinnest when its source is in focus (such as the phone in all these tests). As you rack away, it kind of expands vertically, which is not displeasing and works well for background and out-of-frame light sources.

VORMAX PSEUDOANAMORPHIC FILTER - $15

I was pleasantly surprised with Vormax's Pseudoanamorphic filter. It gives unique looking flares in the sense that all streaks are rainbows. It also flares quite a lot. It comes mounted on a rotating frame which facilitates the job of aligning the streaks. The frame is a bit thick though, considering today's slim filters and all. It is almost as tall as two full-sized filters stacked.

The Pseudoanamorphic filter is made in Russia and it is also dirt cheap. For its cost, this is a pretty good alternative. Expect long shipping times to North America. This is the first filter in this test that uses multiple well marked lines and it actually blends in pretty well with bokeh. These are subtle cuts in our highlights!

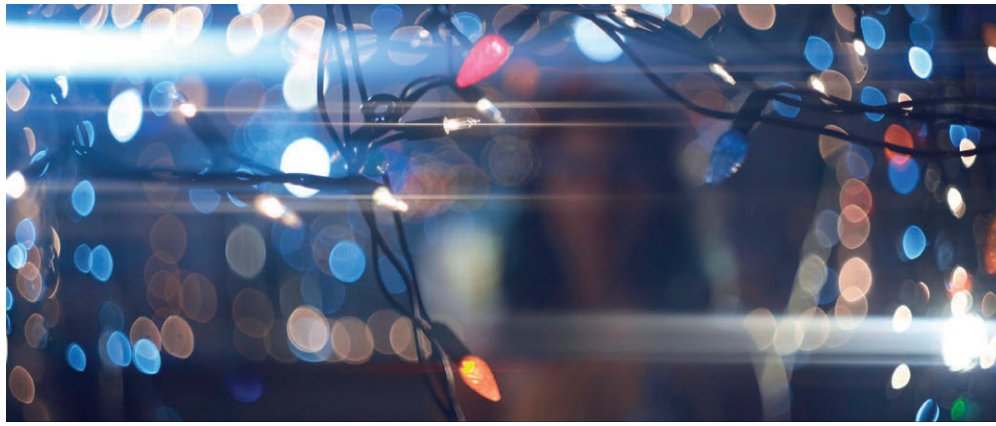


COKIN A OR P 058 STAR EFFECT 2 FILTERS - $30-60

To me this was the biggest surprise of the whole lot. I had to get creative to mount it on a regular 77mm thread, but I got it working. Take this creative mounting into account if you pursue the Cokins - mine was also dirt cheap. You can find some more info on Cokin's sizes here. This was an A series (glass is around 58mm wide), but the best ones are P series (77mm wide).

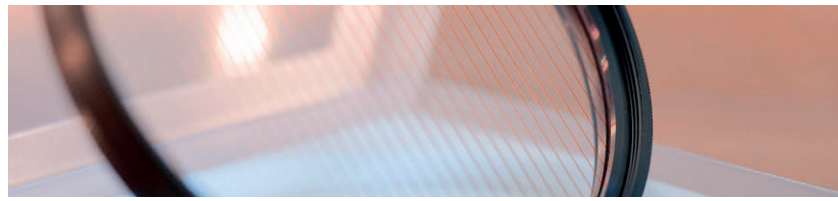
They follow Cokin's nomenclature and fall under star filters. This would be a two point star - which happens to be a line.

The streaks are neutral and everything flares. Even the tiniest twinkly lights had streaks. Flares are not splintery as the regular fishing line and have a nice falloff as they get further from their sources.



Bokeh gets busier if you know what you are looking for but if you are unaware this is actually the most discreet result I got for streaks in out-of-focus highlights - especially when considering how many streaks it makes!

TIFFEN OR SCHNEIDER STREAK FILTERS - $100-350

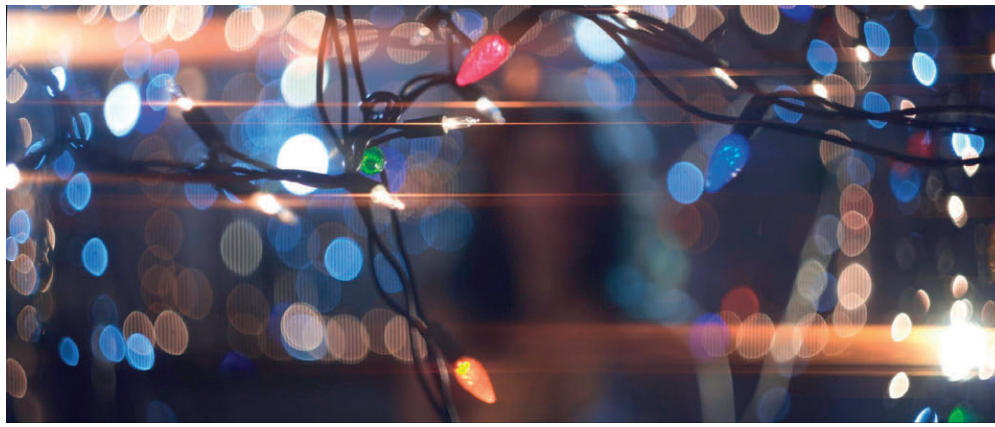


Tiffen and Schneider Streak filters can be found in different sizes and different mounting systems. If you use a compact camera rig you can get them on rotating 77 or 82mm threads, but if you have a mattebox with filter trays, there is also the option to go with 4x4" and 4x5.65" sizes - at a more expensive price.

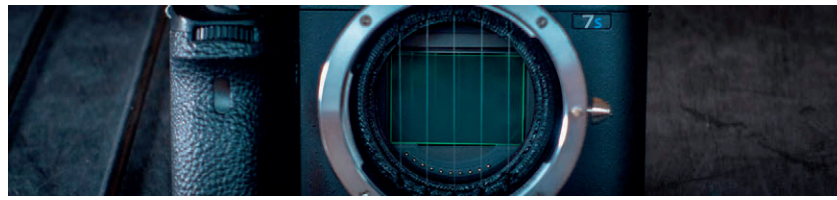
The one I used for this test is a Schneider 77mm Streak Orange 2mm. Again I used a little piece of tape to lock the rotating frame once I was happy with the flare alignment.

The color of the lines control the color of the flares, and these are undoubtedly orange regardless of the light source's color. I think they are a bit too intense, but the audience will definitely notice these flares!

The “2mm” number indicates the spacing between the flare lines. Smaller spacing induces stronger flares but it also means more lines crossing through your bokeh.



ADAPTER MOUNTED FISHING LINES - $25-50



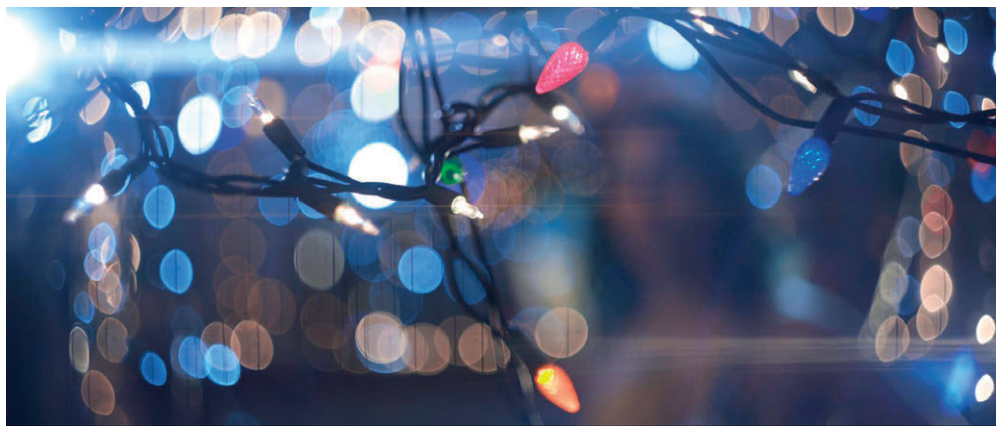
I had the idea for this contraption after seeing an ARRI Alexa modification by Clairmont Camera that put a bunch of streak lines inside the camera mount, behind the lens.

I bought a cheap EF to E mount adapter without any electronics, measured its insides and made a 3d model to hold five lines across the sensor. I assume the size would fit most adapters from EF mount to different sensor sizes.

You can find the STL file for 3d printing in ./Modding Parts/Flare Filters/EFtoE_FlareAdapter.stl
You can also find a video tutorial of assembly in ./Videos/08_StreakFlares/EFtoE_FlareAdapter.mp4

As much as I wanted to love this option, it has very pronounced quirks.
It is super convenient to have the flares made inside your lens adapter. Getting all the strings in place can take a few minutes, but once the setup is complete you are done forever with it. You only have to align the flares once regardless of which lens you are using.

As all previous string-based filters, the flares are quite splintery. There was no tint, so the flares keep the lights' colors and again we see a fairly high threshold for which lights flare up. Bokeh is cut up unevenly depending on its position in the frame (which also bothered me).



Strangest of all is performance when not shooting wide open. I had flares disappear and reappear through panning shots when the lens was stopped down (f/4-5.6). It has to do something with the path of the light and the space between the strings. By having them in front of the lens, light cannot miss the filter, but behind the whole optical system loopholes exist...

You can find a video of this behaviour at ./Appendix/08_StreakFlares/AdapterMounted_panFlares.mp4

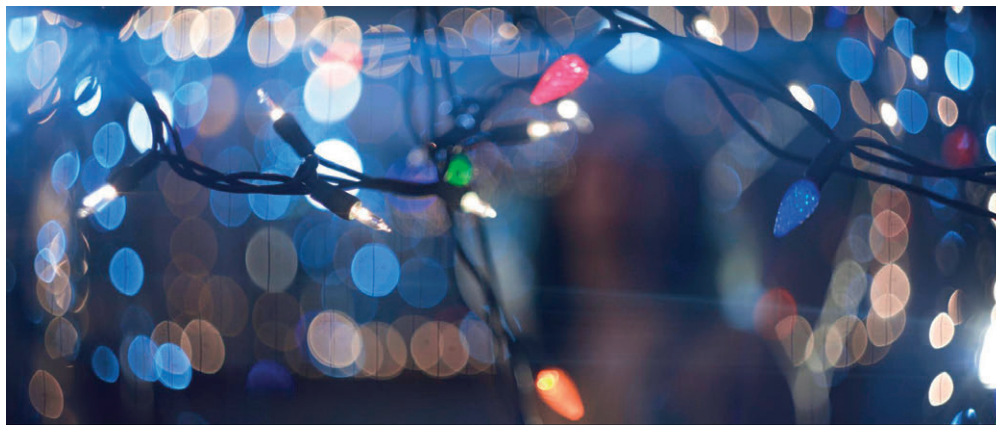
INSIDE THE LENS - $5

Another type you only have to align once, these are usually combined with an oval aperture insert and the line sits inside the lens, right by the aperture mechanism.

For this test I added just the line without the oval. The only cost is the price of the fishing line, but each lens has a different process of disassembling, so it could be a challenging build.

I used a 0.2mm string tinted blue inside the Helios 44-2 and I feel the tint was very bland while affecting the flares' colors. The splintery look was also noticeable here. Even if the string is perfectly aligned, you get lines shooting in different directions.

Bokeh gets a cut right down the middle and it is quite noticeable. Though, observe that the cut's positioning does not change across the frame like the rear adapter did.



QUICK REVIEW

This sums up all practical streak flare options I could come up with. Some are wildly original, others are tried and true DIY methods and some of them are also used in high budget productions.

Regarding the lines through bokeh, not all shots feature such noticeable out-of-focus highlights, so it is a case of picking the proper moments to mount the filter onto the lens in order to accommodate clean bokeh when that matters most, and streak flares when they are more important.

None of these are "the ultimate solution" so it is up to you to decide which one works best for your budget and your shots, and if none of them are up to your standards, check **Chapter 10**!

09. LET'S DO THIS

The time has come to open up the lenses and create your own set of anamorfakes. In this chapter I am going to cover how to modify six Rokinon/Samyang Cine lenses in EF mount, a set of budget Contax Zeiss and some of my favorite glass in the M42 line-up, a USSR set of six lenses. On top of all that I am adding steps for two oddballs, the Canon EF 50/1.4 USM and the Pentacon 29/2.8 in M42 mount.

Some of these tutorials are available at my channel and I am using frames from the videos here. Most of the tutorials are exclusive to this guide, though!

I chose EF as the "standard" mount for all of these tutorials because it has such a long flange distance that you can adapt them to almost any other camera system. Plus lots of REDs and Alexa's have EF mounts, so I would call it a "budget PL".

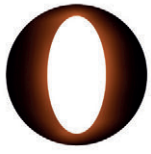
Before we start opening lenses, have everything ready at your work station. Get all of your tools within reach and work one lens at a time. Take notes of where each piece comes from and its orientation so the process of rebuilding is orderly.

STEP 0 - OVAL INSERTS

See what I did there, "0" and oval? Visual puns aside, you should have your aperture discs ready to go before you start opening any lenses. Get them laser cut with the templates provided in .\Modding Parts\Oval Apertures\

Depending on the thickness of the material used you might need to sand them down super thin. If you buy them from me on eBay, these already use a super thin acrylic and sanding is no longer needed.

Paint them using your sharpies. Go with black for a standard, clean oval, or use colors if you want special flares and color tinting. Paint both sides of the disc and let it dry for a minute or two. If you feel a second coat of ink would improve your paint job, go for it.



When using colors you will achieve the best results and not-so-strong tinting across the frame if you paint a gradient from your chosen color to black. It does not have to (and it will not) be as smooth as the example on the left, but the results are much nicer and controlled than a solid orange disc.

ROKINON / SAMYANG CINE DS LENSES - $2500/SET

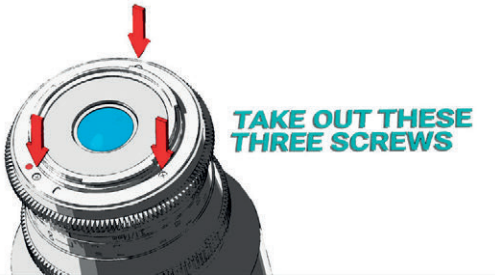
The Samyang or Rokinon Cine DS lenses are excellent candidates for anamorfaking. They are fairly cheap, always available, have a good construction, feature focus and iris gears, and deliver great image quality. I was pleasantly surprised when I opened them up because they have super straightforward insides that will get you in and out in no time.

Their only downside is that we will be mainly attaching the ovals to elements that have to be screwed back into place. This means that getting perfect alignment might require a few attempts.

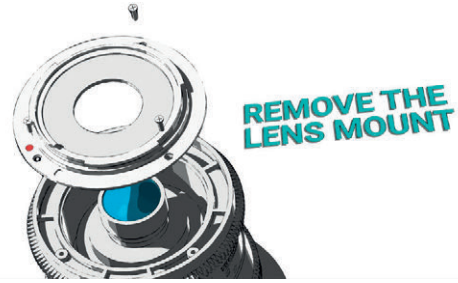
Rokinon Cine 24mm T1.5 exploded view showing where the oval goes

I only modded the six lenses I had at hand but I would not be surprised if all other lenses in the EF line-up follow similar steps. The extra ones would be the 16/2.2, 20/1.9 and 100/3.1 Macro.

#01. All these mods start with the same steps: Focus the lens to infinity. Take out the screws that hold the mount in place.



#02. Note how the mount is positioned to the rest of the lens, then take it out to access the inside of the body and rear groups.



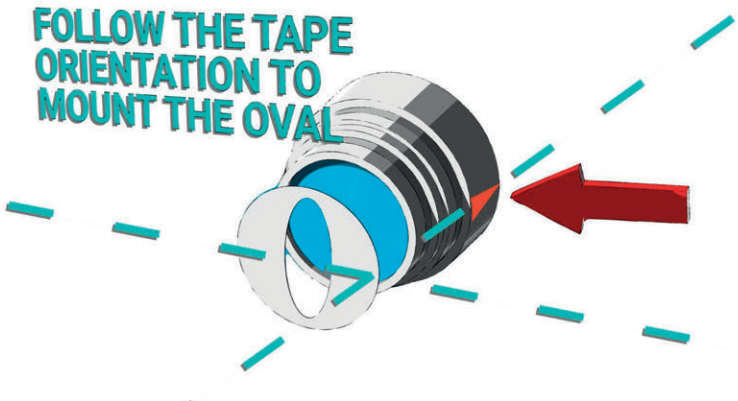
They end with the same steps in reverse. Put the mount back into position and lock it down with the three screws! Shall we begin?

ROKINON 14MM T3.1 CINE DS EF
BUY IT NOW! - $499

The 14mm in this set is an oddball: it is fantastic for its price, but the mod does not do much for it. You need a hardcore composition in order to showcase the ovals. In any case, just having ovals installed is enough to create the feeling that something is different with the out-of-focus areas and that is why I included it in this list.

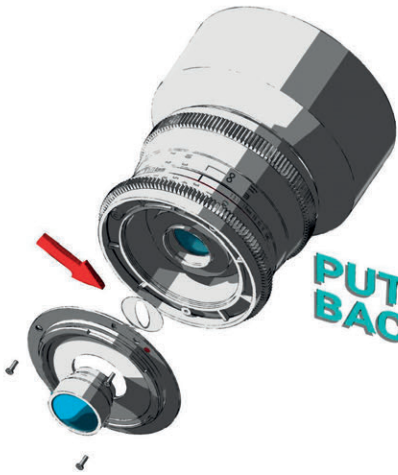
#01. Once you remove the lens mount, note the main marking for focus and aperture on the lens body. Follow a straight line from it and place a small tape triangle on the rear group. This will be key to aligning the oval insert later.

#02. Wearing gloves, get a good grip of the rear group and screw it out. Set it aside.



#03. The mark we placed on the rear group is roughly perpendicular to the lens' orientation when mounted to a camera. This means we will align the oval using its narrowest opening.

#04. Use two very small and thin pieces of double sided tape on the edges of the oval insert and overlay it on the inside of the rear group. The tape should only touch the housing edges, never the glass.



PUT THE LENS BACK TOGETHER

#05. Now rebuild the lens but beware: we are probably opening it back up again.

The chances that the oval is perfectly aligned are very slim. To speed up this process, lock the mount with just one screw and put the lens on your camera.

Check the oval bokeh alignment. If needed, remove the mount again and twist the oval insert in the required direction to make it straight. Once you are happy with the results, close the lens and lock the mount with all three screws.

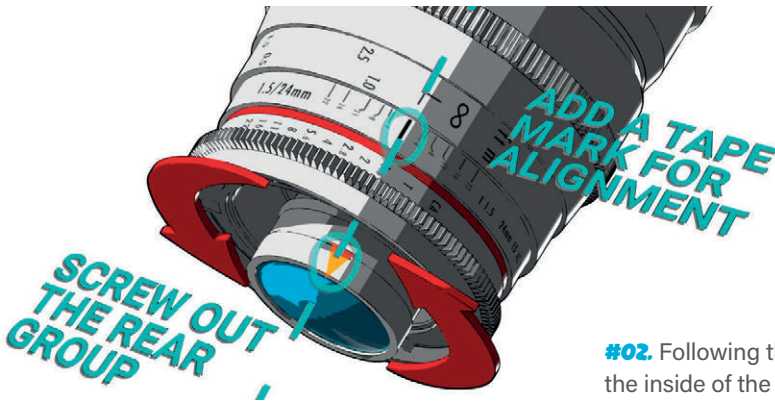
Use bright scenes to check the ovals by setting the lens at minimum focus and watching bokeh, like the examples below (not shot with the 14mm).

To give you some reference, I had to open and realign the oval at least three times for each lens in this set that uses this process.

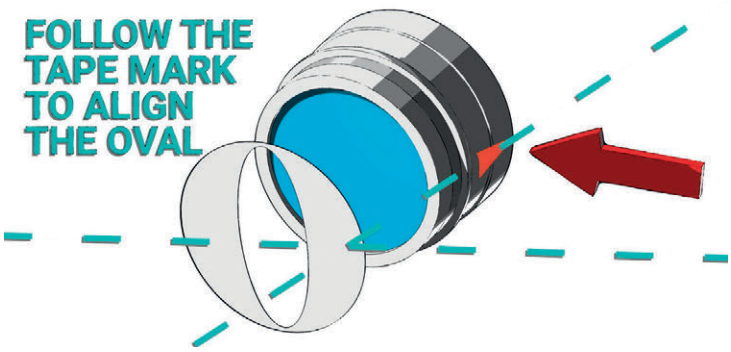
ROKINON 24MM T1.5 ED AS IF UMC CINE DS EF
BUY IT NOW! - $649

The 24mm was the first lens I anamorfaked in this set, months before the other ones, for the This is SCOPE video. It is a solid wide angle with super fast aperture, meaning that even after the oval cuts down a stop, you still get very pronounced bokeh for the out-of-focus areas.

#01. Just like we did with the 14mm, once the mount is removed, use the lens' markings to place a piece of tape onto the rear group. Then get a good grip and screw the group out.



#02. Following the tape mark, attach the oval to the inside of the rear group using two tiny pieces of double-sided tape. The oval should be perpendicular to the tape marking for alignment. Also make sure the double-sided tape only sticks to the metal edges of the housing and not to the glass itself.



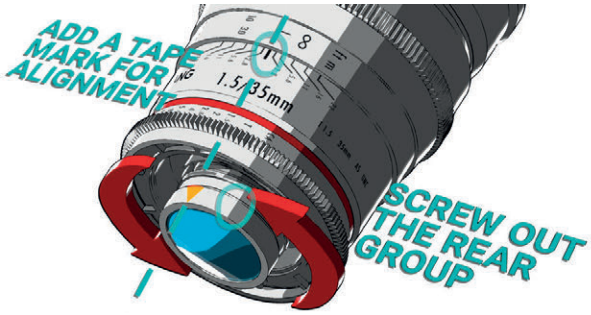
#03. Once that is done you are ready to close the lens back up. Screw the rear group into place but when attaching the mount, use a single screw. Connect the lens to a camera and check for oval alignment - twinkly lights are a blessing for these checks.

#04. Chances are you will need small adjustments. I only got good at this when modding my second set. After you get perfect alignment you are good to put back all the screws and move on to the next lens!

ROKINON 35MM T1.5 CINE DS EF
BUY IT NOW! - $599

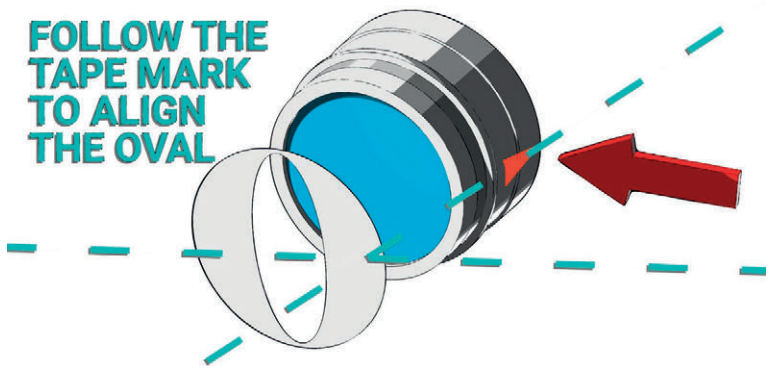
This 35mm is a very versatile lens. It has great minimum focus and features a fast aperture which boosts bokeh for the smart shooter. 35mm is a mandatory focal length in any set, acting as a normal lens on S35 cameras and a mild wide-angle on full frame.

Not surprisingly, the insides of the 35mm are almost identical to the 24mm. They have roughly the same size and build, so the steps here are the same.



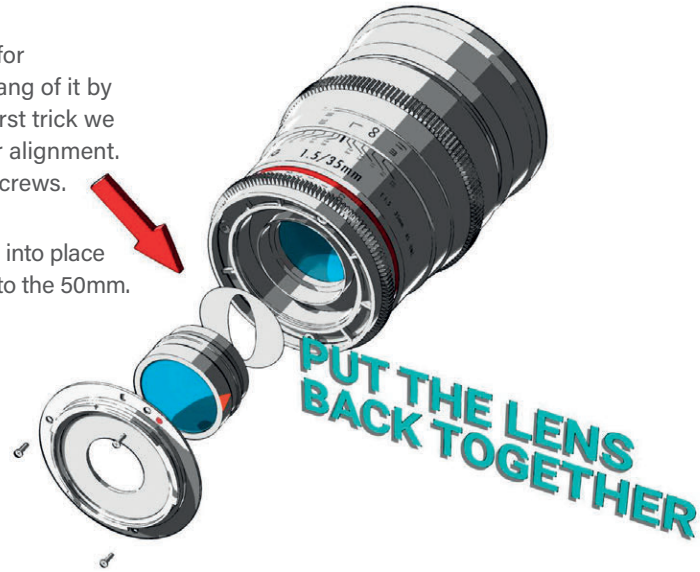
#01. Once the mount is out, follow the lens' markings to put a small tape triangle onto the rear group. Then unscrew the rear group out of the body, giving you access to the aperture mechanism.

#02. Following the triangle tape mark connect the oval to the side that faces the aperture. Use two tiny strips of double-sided tape to secure it, once more making sure they only stick to the housing and not the glass.



#03. Close the lens back up and check for alignment. You might have gotten the hang of it by now, but I recommend the one-screw-first trick we have been doing so far when I check for alignment. It saves a lot of time and stress on the screws.

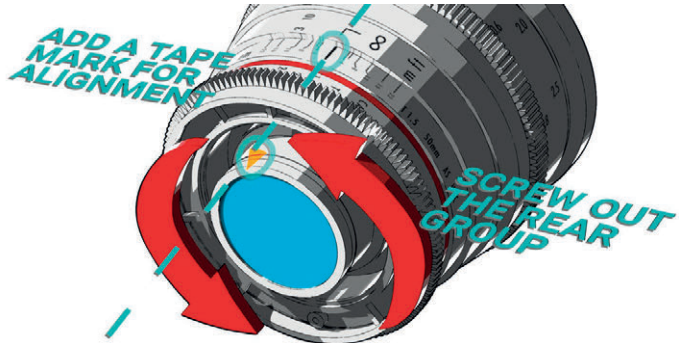
Once you have it perfect, lock everything into place with the remaining screws and move on to the 50mm.



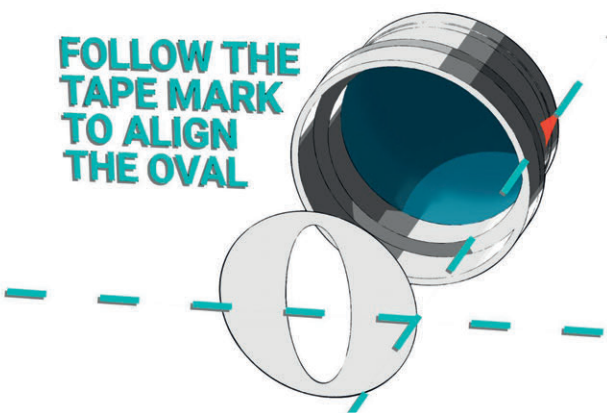
ROKINON 50MM T1.5 AS UMC CINE DS EF
BUY IT NOW! - $549

We are now stepping into big bokeh territory. This 50/1.5 turned 2.0 delivers great out of focus highlights. Great for medium and close-up shots, this lens will give you a lot of that anamorphic character you are after but not much distortion. This lens will have people wondering how you got that look.

#01. I feel like this is getting repetitive by now. Once the mount is out, mark the rear element, then unscrew it out.



#02. This one is shaped like a cup and you can reach all the way inside. Place the oval at the very back and secure it in place using little pieces of electrical tape on the sides.



#03. Screw it back into place, reinstall the mount and check alignment. If it looks good, move on. If not, take the lens apart again and adjust.

ALTERNATE OVAL PLACEMENT

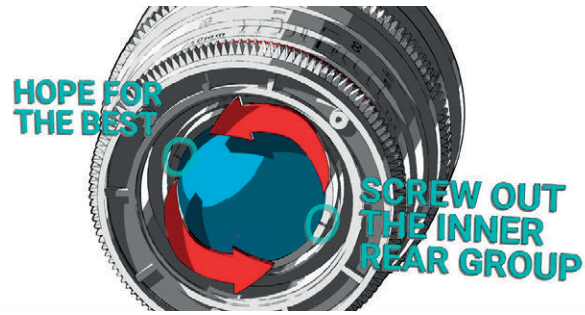
This time we are not going right up against the aperture, which is not the ideal scenario. 50mm is a longer focal length so the risks for unwanted artifacts are smaller - by artifacts I mean cut off bokeh shapes all around and vignetting on the sides of the shot.

There is a way to get the oval insert right against the aperture, but the pressure created rebuilding the lens turns the iris mechanism sticky - if you close down past 5.6 the iris will not open back up properly and it will require you to open the lens in order to release the pressure and fix the iris. In that process the aperture blades can get stuck onto the oval temporarily.

If you prefer to place the oval in the ideal position (which I did for my first mod) and you are aware of the risks and consequences, follow the next few steps:

The aperture disc used in this method is not the Rokinon 50/1.5, but the Jupiter 9 from the USSR set.

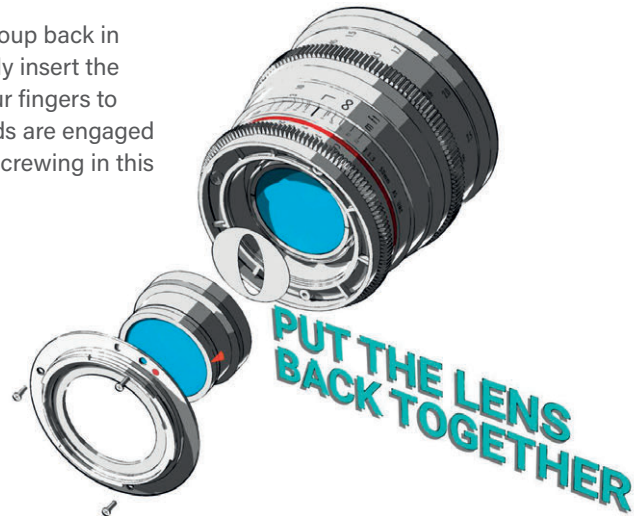
#02B. Once the cup-like group is removed, unscrew the inner group with a lens wrench. This group is much harder to mark for alignment so eyeball the whole thing and be prepared for lots of tweaking back and forth to align the oval.



#03B. Add very thin strips of double-sided tape - thinner than all the ones before - to the oval and stick it to the part of this group that faces the aperture mechanism.

#04B. Before screwing the group back in I recommend setting your iris wide open and locking the aperture ring with electrical tape to prevent rotation. If you want to see the side-effects, do not tape the aperture ring and play with it once the inner group is screwed back into place. Iris jams are not much fun.

#05B. Definitely wear gloves to screw the inner group back in place. While holding the lens body facing up, gently insert the inner group and secure it on the threads using your fingers to grip the protruding glass element. Once the threads are engaged you can put the lens back on the table and finish screwing in this group with the lens wrench. Do not overtighten!



#06B. If the ovals look alright at this stage, finish rebuilding the lens and check on camera. This is the painful part. If the ovals are slanted you are going to have to open the lens back up all the way and adjust the oval insert.

This alternate method for the 50mm easily took me more time than the other five lenses in the set combined. I recommend taking the easy path if you do not feel comfortable with all this lens handling!

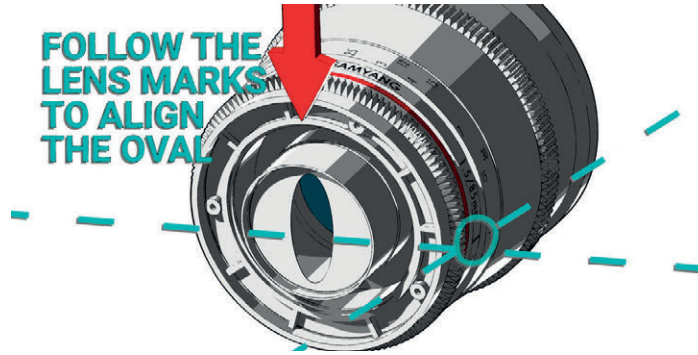
ROKINON 85MM T1.5 CINE DS EF
BUY IT NOW! - $299

Any 85mm is a great lens for portraits and close-ups; a fast aperture 85mm only does it in a dreamier way creating more separation between subject and background. Kick up a notch in production value by having oval bokeh on the background. This is also the territory where too fast of an aperture can make focus pulling a real challenge. Really consider which aperture value you want to install with the oval. The shot below was taken at T2, as fast as possible.

#01. The 85mm will give you a break of all the "place a tape marker to align the oval" process. After you remove the mount, go ahead and remove the little locking ring at the back.

#02. Carefully turn the lens upside down with your hand underneath it so all elements come out safely. There are two glass groups and a spacer between them. Keep track of their order and their layers too for when you are putting them back so nothing goes in backwards.

#03. With the stack of elements out of the way, you can get a clean view of the aperture and place the oval insert right there, aligning it by eye with the lens markings.



#04. Once the oval is in position, you can rotate it using the lens wrench.

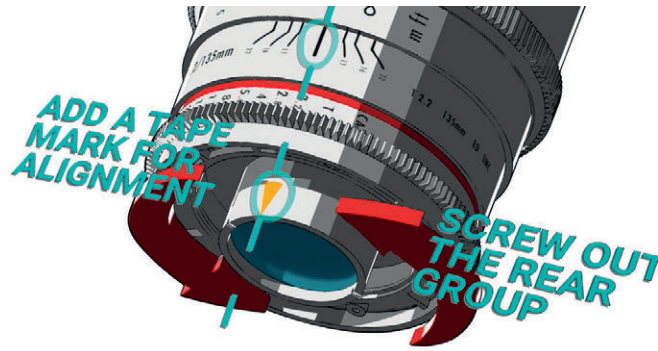
#05. Put the elements and locking ring back into position. The glasses and spacer can get slightly jammed depending on their angle of insertion, so I recommend putting them back one at a time, as straight down as you can. If they get stuck from tilting, free them with a little encouragement from the lens wrench to untilt them.

#06. The last step is remount the lens and connect to a camera to check for alignment - and repeat all previous steps if adjustment is needed.

ROKINON 135MM T2.2 CINE DS EF
BUY IT NOW! - $599

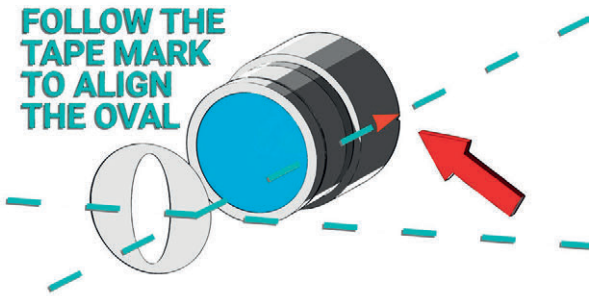
If the 85mm was already a great performer with the oval inserts, the 135mm absolutely shines. With the compression created by the longer focal length, backgrounds either melt away in beautiful bokeh for close-up shots or they show clear stretched out defocus on wider compositions like the one below. The little flying cotton fluffs were hit just right by the light and turned into floating ovals because of the focal length.

#01. After taking a break with tape marks for the 85mm, we are back at it. Now that the mount has been removed, mark where the rear group aligns with the lens body and unscrew it out.



#02. Add the oval to the innermost side, facing the aperture. No surprises here, keep doing the two little strips of double sided tape process, aiming for the

housing edges around the glass element. The glass element here bulges out just a bit, so the oval will touch it and bend around it. That is expected.



#03. Put the lens back together and lock the mount using one screw to check the oval's alignment. Out the whole set, the 135mm is the easiest one to spot misalignment.

#04. Once you are happy with your checks, put in the remaining screws and head out to shoot something using your freshly modified Rokinon/Samyang set!

CONTAX ZEISS LENSES - $1200/SET

Contax Zeiss is my favorite lens family. I fell in love with them years ago while looking for something cleaner than USSR optics but with an organic and vintage vibe. When following down that path, most people reach the Contax vs Leica dilemma. For budget reasons I picked Contax and I still have not changed my mind.

The set we are going to modify is the slower and cheaper set (f/2.8 lenses). Besides the oval inserts, there is a plethora of other mods you can do to optimize them for video, like swap the mounts from C/Y to EF, focus gears and standard fronts. I wrote an article about that a while ago, stuffed with useful information.

This budget Contax set is also a great match for gimbal work because there is very little difference in size and weight across the lenses (the 135 is the only exception). You can read some more about the Contax Zeiss line-up at the Contax Zeiss Survival Guide.

The five lenses we are going to modify can be found in both AE or MM versions. AEs are older and cheaper because they display a ninja-star shaped bokeh at their second stop (f/4), while MMs are newer and have cleaner shaped bokeh. I recommend using AEs for this process because they are cheaper - I got all five for less than $1200 - and we will bypass the ninja-star stop by using the oval inserts, making all lenses f/4 when wide open (except the 50mm, which will be f/2.4). Here is a small sample I shot with this set.

The only exception between AE's and MM's for this set is the 135mm. I could not figure out how to disassemble the AE version, so I had to switch for a MM. Lucky for us the price difference is not bad.

CONTAX ZEISS DISTAGON 28MM F/2.8
BUY IT NOW! - $200-280

The 28/2.8 is a classic lens in the Contax lineup, preferred by many over the 28/2 "Hollywood" due to its small size, cheaper price and stellar performance. Sharp even wide open, this is a better wide angle than my other wide anamorfakes, the Pentacon 29 or Mir 10A. Close focus is at 0.25m – just under a foot – and this is what gives this anamorfake an edge over actual anamorphics: it is really hard to do extreme close-up wide angles.

If the first steps below are incompatible with your lens due to the lack of a locking ring, you have a MM version and this tutorial is for the AE version. The steps for the 28mm f/2.8 MM are identical to the 35mm f/2.8 MM! Flip a few pages forward!

#01. With the lens mounted to a camera, note the aperture/focus line faces perfectly up. Place a piece of bright tape to mark that direction as the long axis of the oval insert.

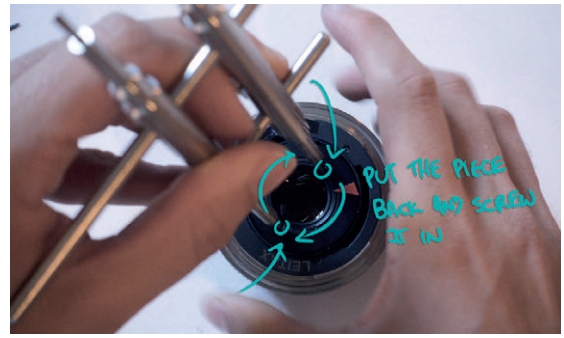
#02. Unscrew the outer locking ring and that will bring out the rear optical groups, giving you direct access to the aperture mechanism. Set the lens body aside.

#03. Get your aperture disc. When you put it over the inner glass element that goes right against the aperture, the disc is much bigger than the available space.

#04. Cut it to the best fit. Take small steps and do not worry too much about keeping the outside a perfect circle. Mine always look ugly.

#05. Now that you got it to the right size, orient the oval according to the mark we placed before disassembling this part. Use some electrical tape to secure the oval in place.

#06. The last bit is to put the lens back together. The aperture mechanism works fine and you can reverse the mod any time. If you do so, please take care to remove all the tape residue from the inner elements.



As a free upgrade to the lens, there is a file for 3d printing focus gears at . \Modding Parts\Focus Gears\CY_28_28.stl

There is nothing quite like a five-minute mod that works well.

CONTAX ZEISS DISTAGON 35MM F/2.8
BUY IT NOW! - $200-250

The slower 35mm f/2.8 budget Contax is a super strong performer among 35mm lenses - even when compared to Contax's own f/1.4 version. It has a much more compact and lightweight build, making it perfect for gimbal work. 35mm is close to 28mm but this focal length will allow you to do cleaner close-ups than its wider counterpart, not warping your subjects' faces much. Here is a video for this tutorial.

Some Contax lenses have two versions and the 35mm is one of them. This tutorial covers both of them. Find out which one yours is! The AE version takes way less disassembling (and is cheaper too), while the MM has more steps.

Before you start you can 3d print focus gears for this lens using the file at .\Modding Parts\Focus Gears\CY_35_28.stl and install them after the last step!

AE VERSION

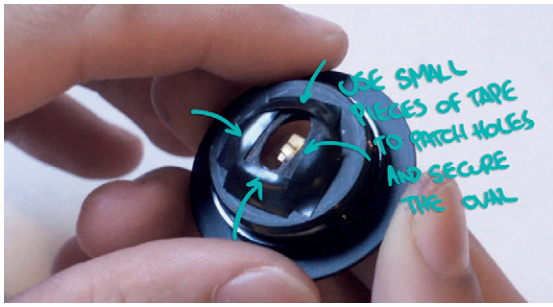
#01. With the lens mounted to a camera, note the aperture/focus line faces perfectly up. Place a piece of bright tape to mark that direction as the long axis of the oval insert.

#02. Unscrew the outer locking ring and that will bring out the rear optical groups, giving you direct access to the aperture mechanism. Set the lens body aside.

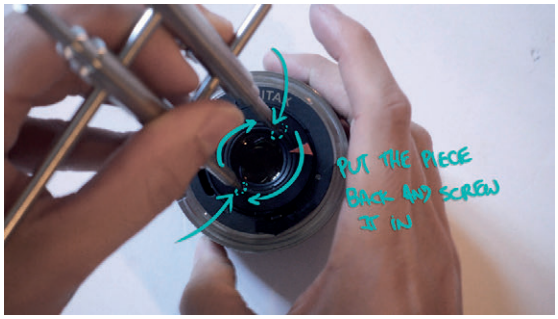
#03. Get your aperture disc. When you place it over the inner glass element, the disc is much bigger than the element's surface.

#04. Cut it to the best fit. Take small steps and do not worry too much about keeping the outside a perfect circle. Mine always look ugly.

#05. Now that you got it to the right size, orient the oval according to the mark we placed before disassembling this part. Use some electrical tape to secure the oval in place.



#06. The last bit is to put the lens back together. The aperture mechanism works fine and you can reverse the mod any time. If you do so, please take care to remove all the tape residue from the inner elements.



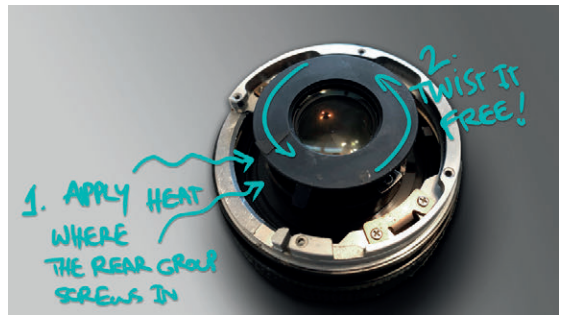
MM VERSION

I did not have the MM version of this lens so I got the photos and steps from Marshall Rogalski shortly after the original video tutorial was posted on my channel. The MM versions have a more closed-off build, requiring you to disasseble larger pieces of the lens to get to the aperture mechanism where the oval disc has to go.

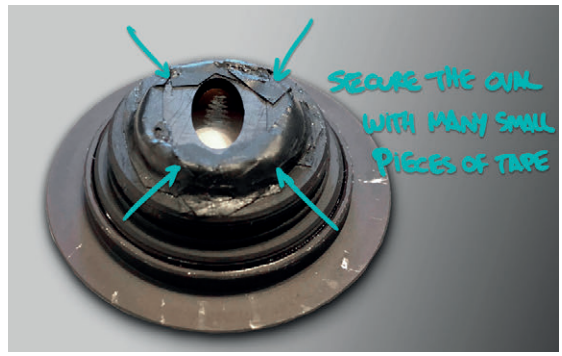
#01. Start by unscrewing the four screws on the back - the same ones used to install a Simmod or Leitax mount.

#02. Hold onto the outside aperture ring and gently lift the entire back part of the lens off. It will be wiggly but do not let it fall apart. Gently set it aside.

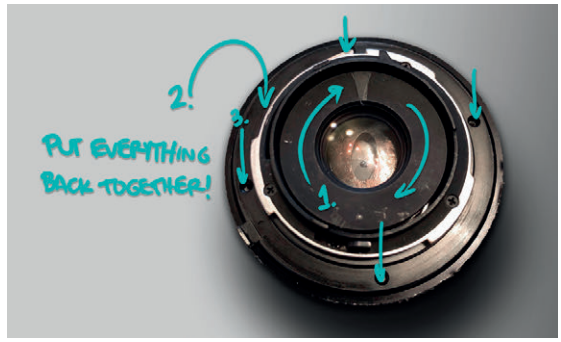
#03. This is by far the most difficult step: the manufacturer had glued where the rear element screws into the main element, so use a soldering iron or a heat gun to soften the adhesive. Apply heat to the underside of the matte black disc, where it makes contact with the lens body. Then grip the outer edges of the black disc and unscrew HARD counter-clockwise. It takes a fair amount of force to get it spinning, but once it loosened up the rest was easy.



#04. Get your aperture disc and trim it to the size of the inner element. Since the threads are so close to the aperture disc, it is best to use many tiny pieces of tape to secure the oval in place. Marshall even used the soldering iron to melt them all together.



#05. Lastly, screw the rear element back into place and reattach the back of the lens, making sure the aperture mechanism works as it should. If you have a mount adapter such as Simmod or Leitax, reinstall it using the four screws we removed in the first step. Otherwise, just put the screws back in and enjoy your C/Y mount.

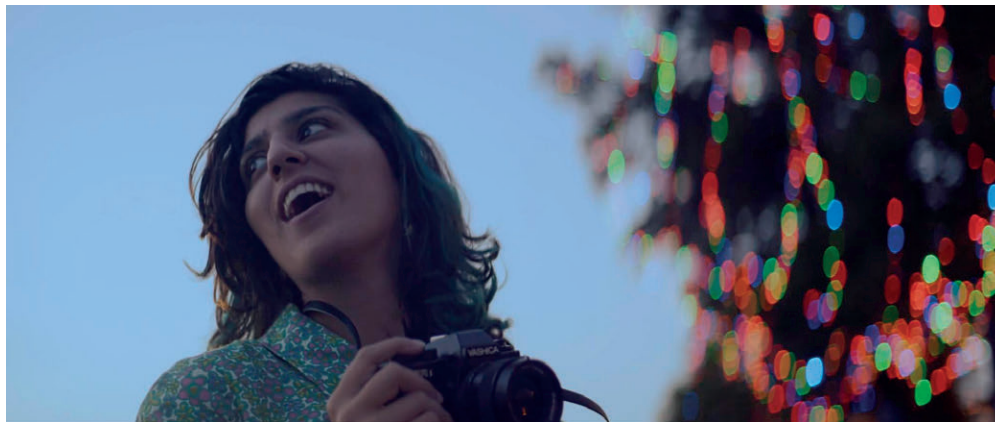


That covers everything you need to do for both versions of the lens and you should now be ready to go out and shoot with this amazing 35mm. Because of the aperture insert, the lens is now f/4 instead of the original f/2.8.

The mod for the 35mm is identical to the 28mm!

CONTAX ZEISS PLANAR 50MM F/1.7
BUY IT NOW! - $150-280

This 50 is the cheapest lens in the set and many people prefer its performance over twice-the-price 50/1.4 - plus the speed difference between them is less than a stop. It features a long focus throw in a tiny body. The 50mm focal length also shows much more pronounced oval bokeh.



Opposed to the two lenses before this one in the Contax lineup, we will not have to mark anything with tape to screw it back in place.

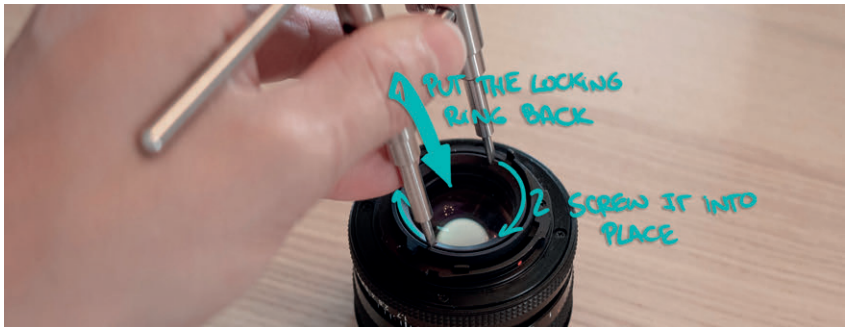
#01. Begin by focusing the lens to infinity and then use the lens wrench to remove this locking jacket from the back of the lens.

#02. This will unlock the entire rear group. They can also fall off the housing, so carefully reach in and lift this entire piece.

#03. Place the oval aperture over the bottom of the rear group and tape it to the sides of the element using very thin strips of electrical tape.

#04. Put the elements back into the body, aligning the oval with the markings on the top of the lens.

#05. Screw back in the locking jacket and you are done with this lens.



If you want to upgrade it, you can 3d print focus gears with the file at .\Modding Parts\Focus Gears\CY_50_17.stl

These Contax mods are simpler than I originally expected!

CONTAX ZEISS SONNAR 85MM F/2.8
BUY IT NOW! - $350-450

I had never tried this lens until I decided to write this guide. Using a Sonnar design means less elements than its bigger 85/1.4 Planar sibling. This lens delivers better contrast and flare control in spite of being two stops slower. It is much lighter and smaller than the 85/1.4, matching the rest of this set in size and weight. Resolution is consistent across all aperture values, razor sharp even when wide open.

We are back to marking the rear group with a tiny piece of tape. Take note of the lens' orientation when mounted to a camera and mark what is up. Hint: it is the little line mark between focus and iris values. Place a little triangle there.
MM and AE versions look slightly different but the only change is which side of the lens wrench you use.

MM version on the left, with two little holes on the rear group, AE version on the right, with the slots

#01. Use the lens wrench to take out the rear group and set the lens body aside.

#02. Place the oval insert inside the housing for the rear group and align it with the tape mark. This time we actually have some room in here, so use tiny bits of tape to lock it in place!

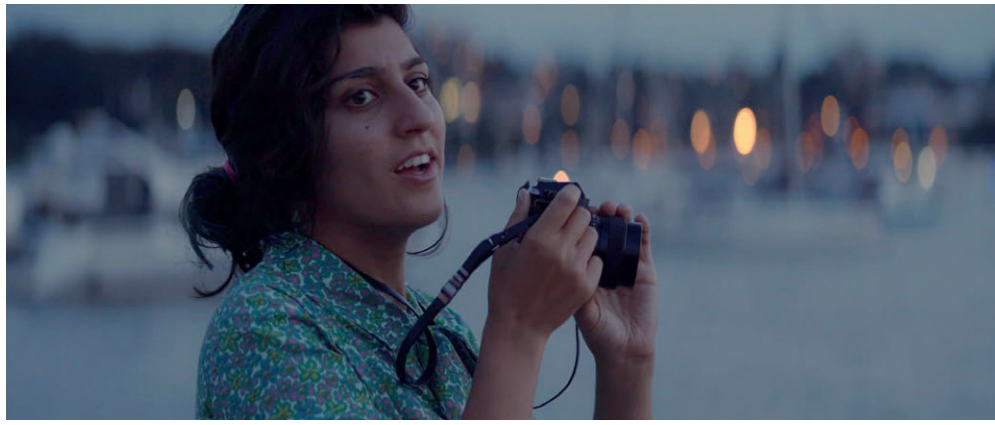
#03. Screw the rear group back in and you are done!

For the sake of upgrades, you can find a file for 3d printing focus gears for this 85mm at .\Modding Parts\Focus Gears\CY_85_28.stl

The longer focal length will give you more noticeable oval bokeh.

CONTAX ZEISS SONNAR 135MM F/2.8
BUY IT NOW! - $150-200

The 135/2.8 is almost a staple in any Contax Zeiss set. Cheaper than the f/2 version by $1000, this one performs just about the same as its expensive sister at f/2.8. The challenge is working with it even slower because of the oval insert. I would not recommend shooting dusk or night scenes on it, but the test video shows it is not impossible.



Before we start disassembling, it is worth noting that this lens was the most challenging among the other mods in this guide. I was not able to crack the AE version open so I had to go for the slightly more expensive MM. Fortunately that was much simpler, still made me more nervous than most modifications. Shall we?

#01. The first step is to remove the four screws that hold the mount in position. These can be stuck so make sure to put enough pressure on the screwdriver to take them out and not strip the heads.

#02. Next, lift the whole assembly with the aperture ring. There is a spring inside which will fight against you a little bit. During this step you can declick your lens if you want to do so. The spring that is pushing against the mount is holding a little ball bearing that makes the clicks. By separating the mount from the aperture ring, this little ball is likely to come flying out. I was able to lose mine, find it on the floor and put it back.

#03. I felt the de-clicked version without proper grease had too light of a touch to the aperture and it would become a problem soon.

That is the little ball bearing with the screw underneath. This will push up the mount part.

#04. Just like what we did for most lenses in this set, the piece where we are attaching the oval insert screws into place, so we have to mark its final orientation. Use the aperture line as a guide. Since this lens was so much trouble to disassemble (the little ball bearing made me nervous), I tried my best to get it right the first time, so I was very careful when placing this red triangle. I used tweezers to get it in place since my fingers were too big.

#05. After the mark is secured, make sure the lens is focused to infinity and screw out the tube with the glass element inside.

#06. Once the tube is out, carefully fold your oval and slip it inside of the tube through the opening that faced the aperture. The disc is just big enough to cover the glass. Twist the oval according to our triangle mark so it has the right orientation in the end.

#07. Use small pieces of electrical tape to secure it. I resorted to toothpicks to stick the tape properly.

#08. Before screwing the tube back into position, be sure to move back the iris mechanism to full open position. The mount and aperture ring will slot inside of that gap and that is very important to keep the lens functional.

#09. Once the aperture is wide open, screw back in the tube with the oval in it. Check orientation of our oval and triangle against the aperture line again. If it looks off, it is better to fix it now than later.

#10. If everything looks good, it is time to put the mount and aperture ring back. This is the most tense part for me because the little ball bearing kept trying to jump out. I held it firmly and slid the aperture lever into its shaft. Everything aligns neatly once it is in place.

You might need to rotate the mount a little bit to find the screw holes. Before finishing it up, check if the aperture mechanism works as it should and if the lens has the full aperture range. If the protrusion went out of place it is easy to have a lens that does not open or close all the way. If that is the case, just repeat the last step and make sure the aperture is wide open before you put the mount in place.

This wraps things up for the Contax 135mm as well as the Contax set.

Although trickier than the previous lenses in the set, this one is worth anamorfaking since it gives you the prettiest and largest bokeh while being razor sharp on focused subjects.

Feel free to upgrade it with the 3d-printed focus gears at \Modding Parts\Focus Gears\CY_135_28.stl

USSR LENSES - $800/SET

These lenses hold a special place in my heart because they were the first ones I dared to experiment with. They are simple and so cheap that you will not worry about messing up. I recommend playing with some of these before moving to the more expensive options! These mods are also more intense, involving tinting and paint. These are optional steps which can be skipped if you want a more neutral result.

Besides cheap, USSR glass has a ton of character: crazy flares, mad wide-angles, swirly bokeh and super organic vibes all while still built to last. They use only metal and glass, no plastic pieces. Keep in mind these are also the slowest lenses in the bunch when compared to the other sets.

The one thing that might give you trouble when modding older lenses are the small screws. Some of them are made of softer materials that will try and strip if you do not use a steady hand when taking them out. Take your time and be patient.

All the lenses in this set come in M42 mount and I highly recommend getting rotating M42 to EF adapters for them. This will save you tons of time when aligning the oval inserts and flares.

Helios 44-2, Mir 1B and Jupiter 9, the cheapest and most commonly modified USSR lenses!

MIR 20M - 20MM F/3.5
BUY IT NOW! - $200-250

Some might argue about the need for an anamorfake 20mm since bokeh is so small and everything is usually in focus. I had the lens sitting around , so I figured it would not hurt to experiment with it. You can find a video version of this tutorial here!

I like the Mir 20M by itself and since it is not a super fast lens to begin with, losing a stop because of the oval insert does not hurt it much. The ovals will not pop in bokeh. They aim to create a subconscious feeling that there is something different on the image, a subtle effect. Plus, if you do close up shots with this lens the ovals show up and the distortion makes up for memorable compositions.

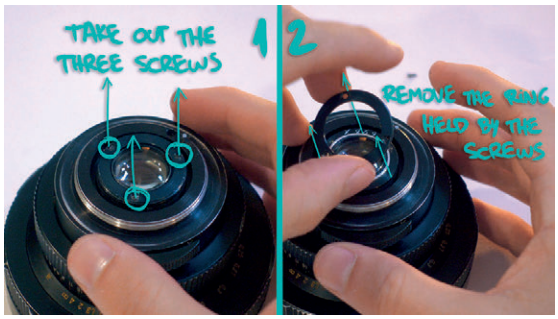
New version on the left, first generation on the right.

We are using a first generation Mir 20M, not the new version - see image above. The additional files of this guide have a model for 3d printing focus gears for the Mir 20M.

You can find it at .\Modding Parts\Focus Gears\USSR_Mir20M.stl

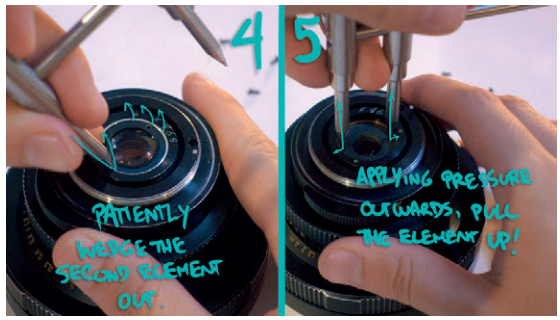
The flare line does not work well with this lens, creating horrible flares, so I am not putting it in. Prepare an oval aperture sized for the Mir 1B (not a typo) before you start opening up the lens.

#01. The first step is to focus it to infinity for easier access to the rear elements. Using the screwdriver take out these three screws and the ring they hold.



#02. Then remove the first glass element. This one is challenging because there is no easy edge to grab, but there is some technique to removing it: pluck it up using a flathead screwdriver and keep track of which side faces up.

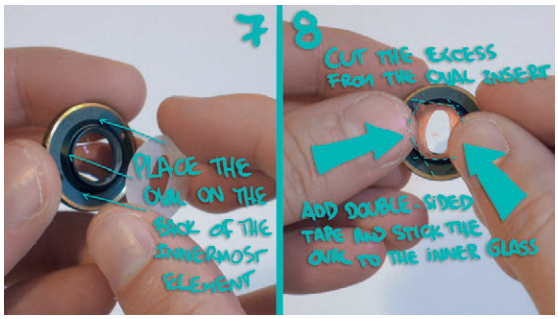
#03. Use the lens wrench to pluck out the next two elements.



#04. We reached the aperture level. The issue is we are on the wrong side of it: when adjusting the iris you can see there is a small rotation to this mechanism, making it impossible to align the oval bokeh for good. Also, this hole is tiny!

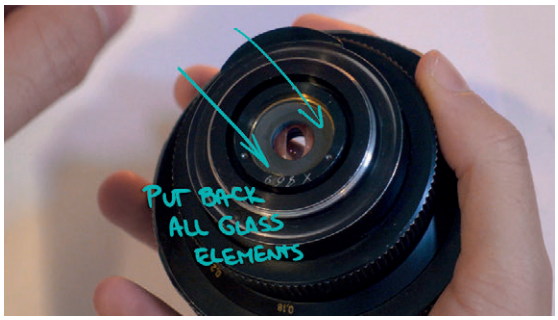
I am using an aperture disc for the Mir 1B. They are easier to handle in this case because they are slightly bigger than they need to be. We are going to stick the oval to the glass side instead of overlaying it on top of the aperture mechanism.

#05. Use scissors to cut the outside edges of the oval disc to match the size of the glass. This will turn the f/3.5 lens into f/5.0.



#06. Cover one side with a lot of double-sided tape and stick it to the glass. We shall embrace the mess of it. If we want to remove the oval in the future, we can wipe out the stickiness of the tape with no damage to the glass.

#07. Put all the elements back in place with the oval aperture facing down and tight with the aperture mechanism.



#08. The last step is to add the screws back. Careful: we added some thickness to the rear group by adding the oval insert so the glass elements might be a little pushed back. Do not overtighten when the screws start to resist, we do not want to crack the oval or snap any other pieces.

After the lens is closed again you can add the rotating M42 adapter to align the oval aperture with the lens mount and slip on the focus gears!

Here is the finished Mir 20M!

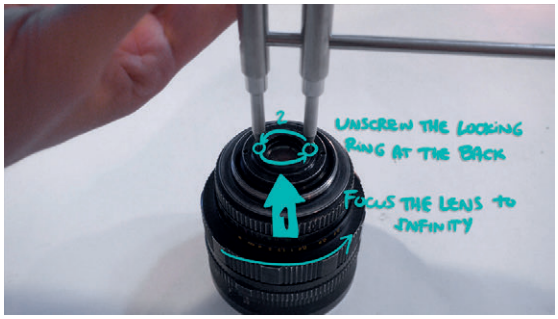
MIR 10A - 28MM F/3.5
BUY IT NOW! - $120-160

One of my favorite aspects of this mod is how the flare line turns out, super balanced with the footage and merging with the Mir 10A's lovely natural flares. The tint on the aperture gives a glow to strong light sources, which I think is very natural and organic. The ovals are very subtle in general, just adding a hint of anamorphic - but they really pop when you use the lens for a wide close up.

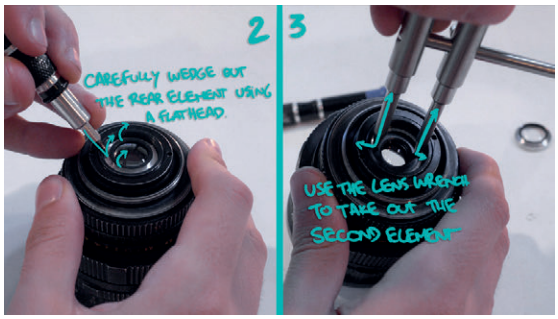
I am adding a tinted oval aperture and a flare line to the Mir 10A, but not sanding or painting anything inside of the lens. There is a video version of this tutorial

here, and you can get the file for 3d printing the focus gear in the additional files folder, at *.\Modding Parts\Focus Gears\USSR_Mir10A.stl*

#01. Start by focusing the lens to infinity in order to push the rear group as far out of the housing as possible. Then use the lens wrench to remove the inner locking ring.



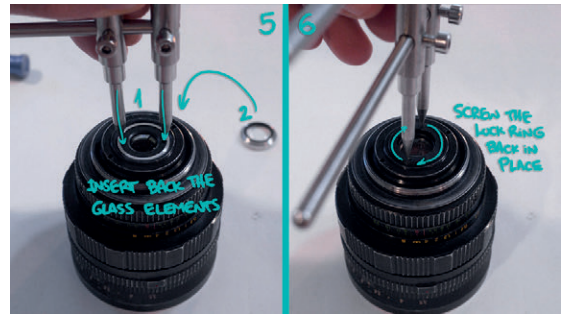
#02. This will give you access to a glass element with a metal frame around it. Welcome to the annoying part of this mod: using a flathead, patiently wedge it out of the housing noting which side faces outwards.



#03. Next is another one of these elements. The key to get it out is to keep both sides even. If they start to angle to each other, the element will get stuck. After that is taken care of also remember which side faces outwards. I like to keep the outwards-facing side facing up on the table.

#04. We have finally reached the aperture mechanism. I am using the same aperture disc I made for the Mir 1B, since the physical maximum aperture size is about the same. You should already have it prepared, painted and with a flare line if you want flares.

#05. Slot the disc in the lens and make sure the oval is right in the middle. This turns the original f/3.5 into f/5. Carefully drop the lens elements back in one by one and lock them in place with the ring.



SLIDE THE FOCUS GEAR ON, ALIGN THE OVALS AND FLARES BY USING THE ROTATING M42 ADAPTER, AND YOU ARE DONE!

Mir 10A with awesome flares and ready to rock.

MIR 1B - 37MM F/2.8
BUY IT NOW! - $40-80

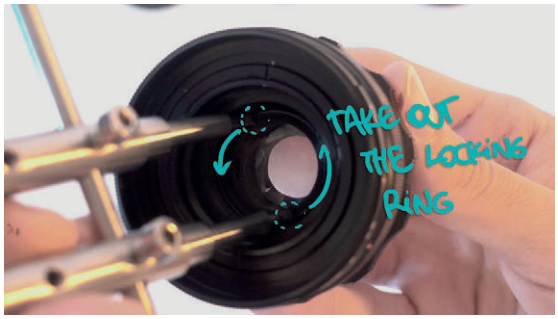
The Mir 1B always goes along with the Helios 44-2 and Jupiter 9 as the wide-angle of the base trio. It is a 37mm f/2.8 lens based off on the Zeiss Flektogon 35mm design. It has fairly low distortion and great capacity for flaring - and that is a good thing. The Mir 1B is the most fidgety lens in this set, so be patient in all steps. Things can go wrong.

If you prefer to see a video version of these steps, you can watch them here!

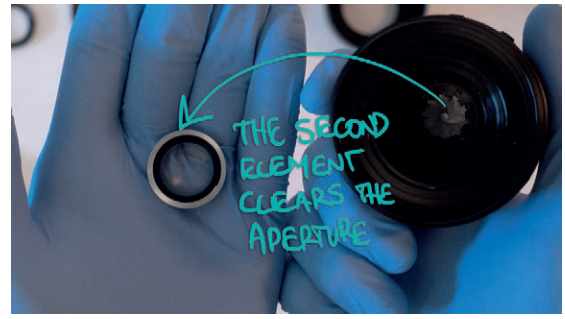
#01. Begin by holding the aperture and focus rings together. With your other hand get a firm grip of the front ring. Twist until it comes loose.

#02. Then, using the lens wrench remove the ID ring to take out the first glass element. Set it aside for now.

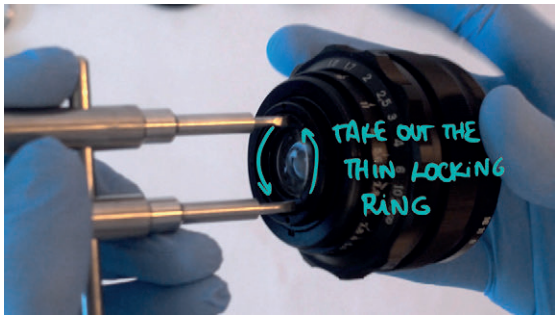
#03. Inside of the Mir, remove the small ring to release the two glass elements below.



#04. These two elements can be tricky. Wearing gloves, not-so-gently tap the lens on the palm of your hand and hope they pop out. This will give you a clear view of the aperture blades. If the elements refuse to come out, there is another way and we will come back to them in a moment.



#05. Go to the back of the lens and unscrew out the thin locking ring. This should allow for the two rear glass elements to come out.



#06. Repeat the process of tapping it on your hand and hope both of them come off. If they are not helping, use a flathead screwdriver to wedge them out like we did on the other Mir lenses.

#07. You need to take all these inner elements out, but if one side is stuck you can open the iris and push the glass out by sticking something through the blades. The only problem is you need to open at least ONE of the sides to do that.

The next few steps are all optional and are marked with italics You can pick and choose which ones you want to do and which ones to skip. They are meant for folks who want tinted flares or super low contrast with their Mir 1B's.

MINIMUM FOCUS IMPROVEMENT

Since you are already taking the lens it apart you might as well improve the Mir 1B's minimum focus. It is a technique explained by Max Westendorp in this video and it only takes a couple more minutes.

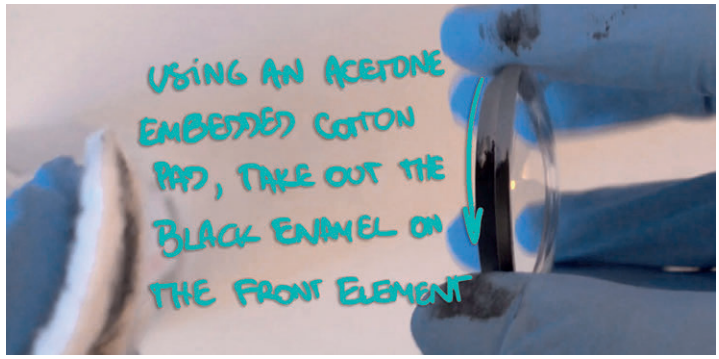
POLISHING THE GLASS

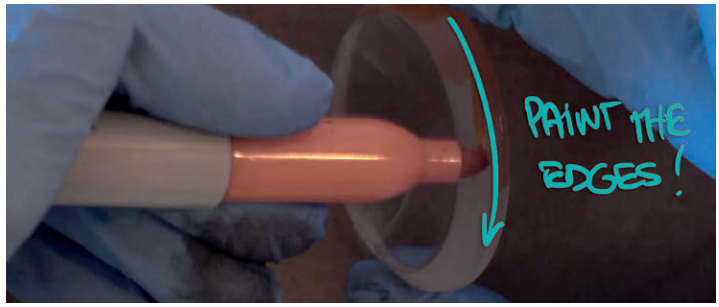
If you want to polish the glass, now is the perfect time to do it since you have them all laid out in front of you. Smear all elements with metal polish and rub them with a piece of cloth. This will tear away the coatings and introduce tons of micro scratches, increasing diffusion and blooming.

I prefer non-polished lenses, but I also shot lots of beautiful footage used polished samples. It is a personal choice.

ADDING PAINT TO INTERNAL GLASS ELEMENTS

Wearing gloves, imbue a cotton pad with acetone and scrub the black enamel on the front element of the Mir. Use more than one cotton pad if needed and make sure to wipe the glass clean too, since the black paint tends to smudge the front and back of the element while you are removing it.

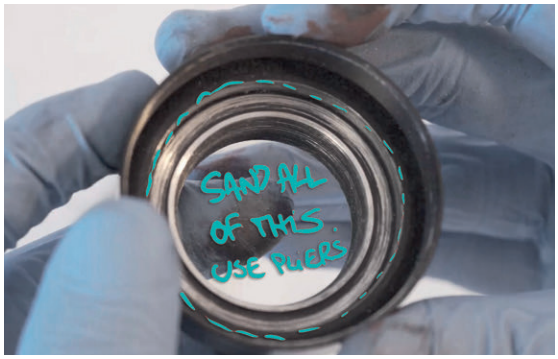




With your favorite color sharpie, paint the clean sides of the front element with many layers until the color is strong and saturated. Let it dry.

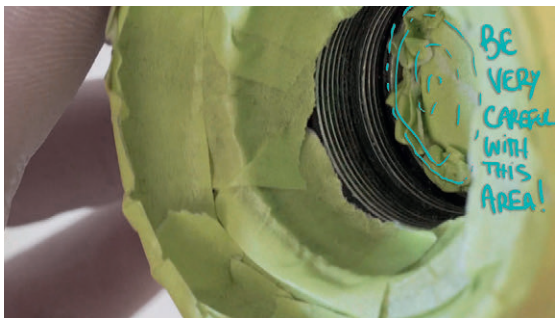
ADDING PAINT TO INTERNAL SURFACES

#01. *Using sandpaper, attack the inside of the front ring. You might need pliers to reach the tight corners.*



#02. *For the next part, use tape to protect the aperture blades. Since I do not want to take the Mir completely apart, we are gonna sand some of its insides and we do not want dust collecting on the aperture. This works better if you do it upside down. Sand the space between the front ring and the second optic group.*

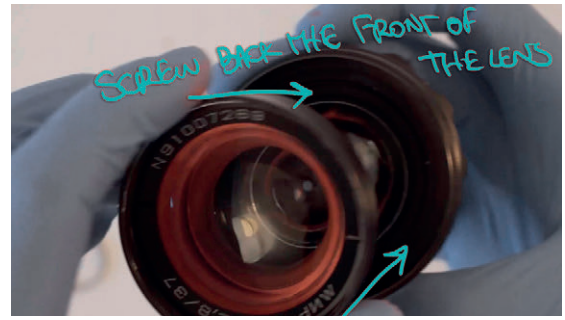
#03. *Mask everything you do not want to paint in the front ring.*
The process is a bit trickier for the inside of the lens. I added several layers of tape everywhere and made some fillers with tape and cardboard to secure the area where the optics are sliding back in.



#04. *Then I used more bits of tape to make sure no paint could drip in there. Take both parts outside and spray away - keep a distance so the paint does not pool. Let it dry, check if you have a good color. If not, spray some more.*

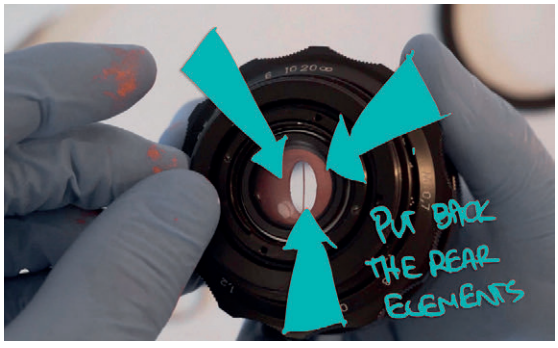
This ends the optional steps. Rebuilding the lens is a mandatory process if you intend to shoot anything in the near future! Gather all parts together. It is time to put this Mir 1B back together!

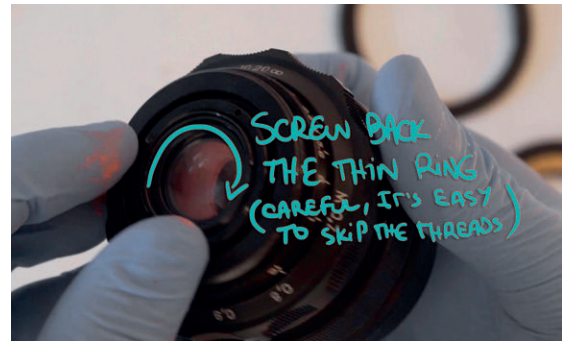
#08. Starting with the front, put in both glass pieces and then the locking ring. You should now have a stretch of orange in there. Fit the front element to the painted front piece and lock it with the ID ring. Screw them back to the body until everthing is tight.



#09. On the back of the aperture mechanism, add the oval insert. We will do it on the back because the front part rotates when you stop the lens down and that is no good.

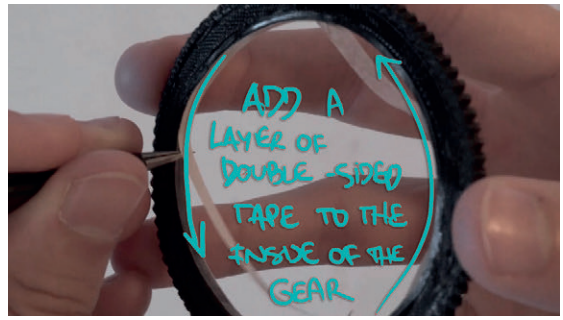
#10. With the aperture disc in place, fit back the glass elements and lock them with the small ring. Be careful during this step as it is easy to skip the threads on this ring.





If you are adding a 3d-printed focus ring you can find the file for it at .\Modding Parts\Focus Gears\USSR_Mir1B.stl

#12. The Mir has a super-thin focus ring and that makes it harder to work with. To better your odds, add a layer of double sided tape inside the focus gear. The double sided tape is meant to strengthen the grip on the Mir's focus ring. If you want a more permanent solution, you can use super glue, but I try to avoid using super glue near optics.



LAST STEP: ADD THE ROTATING M42 TO EF ADAPTER SO YOU CAN REALIGN OVAL AND FLARES QUICKLY!

Amber is my favorite tint to use with these lenses. Everything looks like golden hour!

HELIOS 44-2 - 58MM F/2
BUY IT NOW! - $40-60

My interest for the extensive modifications applied to these lenses - in particular the Helios 44-2 - picked up after I saw Victor Danell's photos using a modified Helios 44.

His process was to polish the glass elements with a strong abrasive and sand the inside of the barrel in order to increase how light bounces around. The resulting effect is a light and wonderful glow to the brighter areas of the frame.

I took it a step further adding the oval insert and flare line plus intense tint. I already explained the relevance of the Helios 44 before in this guide, so I will keep it at: this is the best entry lens for vintage glass. The same goes for modding. It is an extremely simple design and build that is hard to mess up. I have a few videos on this mod.

Golden hour provided by tinting the iris and internal elements.

#01. We will start with the front of th lens. The inside of the barrel has a slot for the lens wrench, just twist the lock ring out.

#02. Once the ring is removed, the front element pops right out. Set it aside. Give the lens a gentle shake and the second element will also drop on your hand with a spacer. Also put them aside and turn the lens around.

#03. Onto the back group, unscrew the glass out. There are two main versions of the Helios 44-2 and they only differ on the rear group. One is super easy and has all the rear group together inside its own little housing that pops off when unscrewed. From there, you can remove the two locking rings and separate all the elements, keeping track of their order and orientation.

#02B. The second version has only a locking ring on the back and when that is removed, all the elements come loose. This one requires more attention regarding the elements' order and which side faces up.

#04. You should have all the pieces spread out across your work area at this point, in which we will delve into optional steps - again in italics. The regular tutorial continues below.

POLISHING THE GLASS

Get a piece of cloth, add some metal polish to it and uncoat away at all four of the Helios 44's glass elements. This process creates tons of micro scratches on the glass which are the main cause for the glow - since light bounces through them.

To completely remove the polish I washed all the elements in water and let them dry on the wind for a bit.

ADDING PAINT TO INTERNAL GLASS ELEMENTS

#01. *Wearing gloves, use cotton pads embedded with acetone to rub the sides of all the glass and take off the black coating. We are replacing that with something much more interesting. You might get some black paint over the top and bottom of the lens, do not worry about it yet, just be sure to remove everything from the sides. Some of the ones I did were pretty hard to take off, so be persistent.*

#02. *Now that they are clear, get your sharpie collection out and keep the gloves on. Carefully paint the sides of each group - paint the ones that did not have black coatings too. The ink will take a bit of time to dry. Add several layers until you have a nice, saturated color. The new tint will make your lens look amazing from the outside and it adds an extra layer to tinting the flares.*

ADDING PAINT TO INTERNAL SURFACES

#03. *Get the front spacer and the rear spacer or the rear group housing, depending on the version of the Helios you are working, and scrub their insides with a tough grit sandpaper. The goal is to take away the black paint and make the surface rough so another paint can adhere to it.*

#04. *With the masking tape, cut small pieces enough to cover ALL the threads and just leave the sanded metal exposed. Be very thorough and add many layers in the process because the paint will mess up the threads if it gets there.*

#04. *Spray away until you have a good coat of color. When they dry, take out the tape and get these spacers ready to be placed back inside the lens.*

Gather all your parts together because this is the end of the optional steps and it is time to rebuild the Helios 44-2.

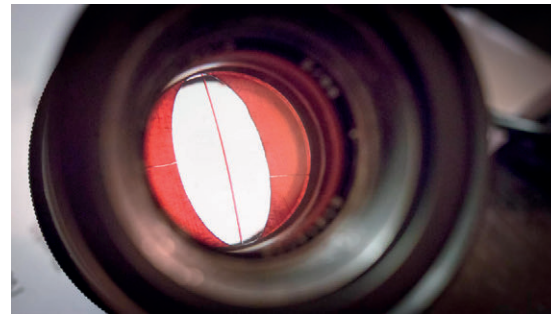
#05. Starting with the front, put back the inner glass element, then the painted spacer, front element and screw it tightly back.

#06. If your lens was the nice one with the rear group housing, put that back together too, but do not screw it back into the lens yet.

#07. Insert the oval aperture on the back of the iris mechanism and close the lens back up.

#08. This is the moment when the annoying non-housed rear group has an advantage. Just slot in the elements in the right order and lock them with the ring on the end.

#09. If you have the rear housing version, this step might take a few tries. When screwing the housing back into the lens, the aperture disc is prone to rotating or moving around. Do it very slowly and steady. You want this to be perfect. Also, do not overtighten. Depending on the oval aperture material, the disc can crack inside the lens.



Once the lens is rebuilt, all that is missing is the focus gear - *you can find the file for 3d printing at .\Modding Parts\Focus Gears\USSR_Helios44-2.stl* - and the rotating M42 adapter so you can align the oval bokeh and streak flares.

This Helios 44-2 will attract curious looks when you go out to shoot with it.

JUPITER 9 - 85MM F/2
BUY IT NOW! - $110-150

The Jupiter 9 is a quite famous fast little 85mm from the Soviet Union. It goes really well with the Helios 44-2, delivering astonishing images. From all my Russian primes, I believe the Jupiter 9 is the one I like the most. It features a

preset ring for aperture and focuses down to 0.8m. If you like moving images better, there is a video version of this tutorial on my channel!

#01. Start by opening the lens. The Jupiter is pretty much a beefed up version of the Helios. To give you a better grip, set the lens to minimum focus. Get a firm grip of the focus ring and rotate the front block – the one with the aperture rings. That will set the optical block free. Put the focusing helicoid aside.

#02. Like the Helios 44, unscrew the rear optical block, it is big enough that it does not require a tool.

#03. The next steps are optional and involve tinting the insides of the lens and glass, as well as polishing. They involve even more disassembling and if you just want an oval insert and flares, skip all that is written in italics below.

FURTHER DISASSEMBLING FOR MODIFICATIONS

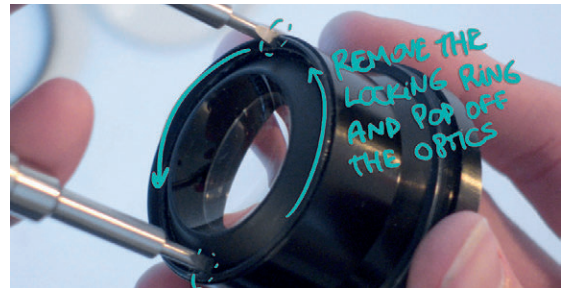
#01. *Grab the frontmost part of the optical block and twist it off to release the front optics. It is a tight lip and grip gloves might be useful.*

#02. *Repeat the process for the second set of tiny screws revealed when you removed the first ring. Once they are loose enough, slide up and remove the ring.*

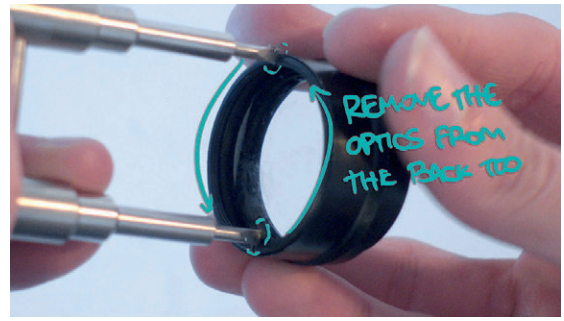
#03. *If nothing budges, there is a locking screw at work. Half the Jupiters I disassembled had this, the other half did not. To reach and remove this pestering screw do the following: find the three tiny screws at the base of the aperture ring. Loosen them enough to slide and remove the ring.*

#04. *This will reveal the tiny lock screw that is giving you trouble separating the front optical group. Take it out. I never bothered putting it back in, as it just makes the disassembly process more convoluted.*

#05. *Put back the rings you removed and then move on! Use the lens wrench to remove the two glass elements in the front optical block. There is a locking ring on the front and one at the back.*



#06. *The rear group only has one glass piece. Take it out using the lens wrench too.*

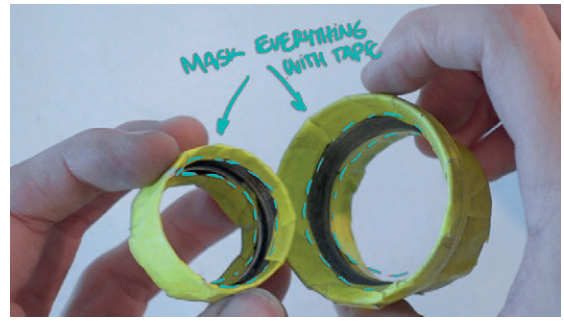


ADDING PAINT TO INTERNAL SURFACES

#07. *We will start with spray painting so it dries off while we work on the glass. With the metal parts at hand for both the front and rear blocks, sand the paint on the space between the front-most and middle elements.*

#08. *Repeat the process for the rear tube.*

#09. *Mask each piece carefully, making sure paint will not stick to anything else. I rip small pieces of tape into even smaller pieces so they cover all the tiny edges. Then I overlay thicker pieces over this patchwork.*



#10. *Wearing gloves, go outside and spray orange over these two pieces. Give it a few minutes and add another layer of paint just to make sure you have a bright and saturated color.*

#11. *Once the paint is dry and you have a good color on these two pieces, remove the masking tape and set them aside until it is time to rebuild the lens.*

POLISHING THE GLASS

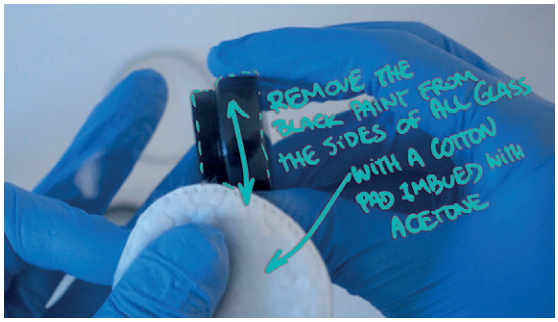
You should have all three glass elements (two from the front, one from the back) spread out in front of you now.

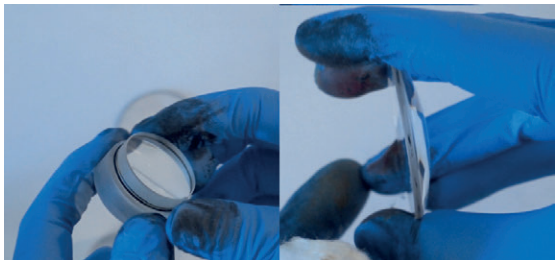
Using a piece of cloth and metal polish you can remove the coatings from the glass and add a ton of microscratches making the lens a tad softer, diffusing the light, lowering contrast and creating bloom around strong highlights in the frame. Once you are done, wash away the metal polish from the glass elements using water.

ADDING PAINT TO INTERNAL GLASS ELEMENTS

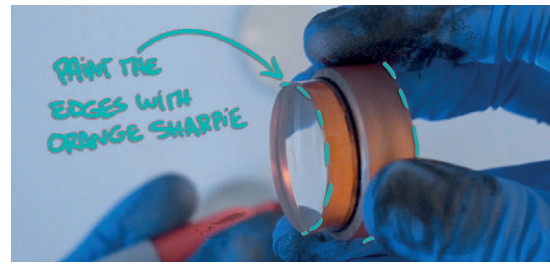
#12. *Still wearing gloves, pour a little bit of acetone on a cotton pad and melt away the black paint that covers the sides of the optics. This will help introducing more color to the flares and lower overall contrast.*

#13. *Keep your window open while you do that since acetone's smell is very strong and the fumes are not the best thing to be breathing in.*

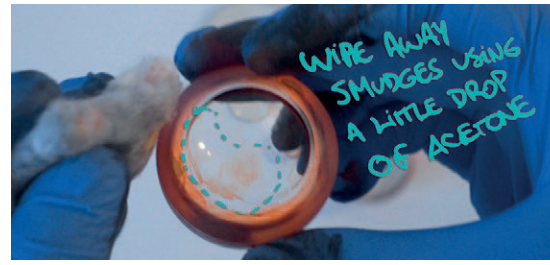




#14. *With your color-picked sharpie, paint the sides of all glass elements you just wiped clean. This adds a level of radioactive shine to the lens and influences the tint some more.*



#15. *If you have any mishaps and add some orange to the front or back of the glass, just wipe it off with a tiny bit of acetone. After this step I always do a lens-cleaning round, wiping away all the smudges, oil, grease and fingerprints away.*



#16. *Put all the glass elements back into their housings and reinstall the front group into its original place. All the work is now aimed at the rear group. This is the end of the optional steps.*

#05. Get your aperture disc and add two thin strips of double-sided tape to the top and the bottom of the disc. I like to use tweezers to remove the white layer of the double-sided tape. Stick the aperture disc to the inner side of the rear group - it should fit perfectly.

#06. Now screw the rear block behind the iris mechanism, where it was in the first place!

#07. Get the focusing helicoid back and screw the optical block back in. The lens should be as it was when you first started (plus a lot of paint or at least an oval disc inside).

There are two versions of the Jupiter 9. They are shown below and each one has a different file for focus gears.

You can find them at .\Modding Parts\Focus Gears\USSR_Jupiter9_typeXXX.stl - the type is determined by which model you have.

On the left, with the indented focus ring, type 1, on the right, with the smooth focus ring, type 2

#08. Just like the Mir 1B, I recommend adding a layer of double-sided tape to the inside of the focus gear, just to make the grip firmer with the lens body, especially with the finger indents on the type 1 Jupiter 9 since the contact area is smaller.

THE LAST STEP IS TO ADD THE ROTATING M42 ADAPTER AND ALIGN THE DIRECTION OF THE OVAL AND FLARES. NOW YOU ARE READY TO SHOOT.

This is the third lens in the basic USSR set (Mir 1B, Helios 44-2 and Jupiter 9). I have shot lots of material using these three and if your budget is really tight, these should suffice for most projects.

The Jupiter 9 is probably my favorite lens in the USSR line-up. Compact and fast, it delivers beautiful bokeh.

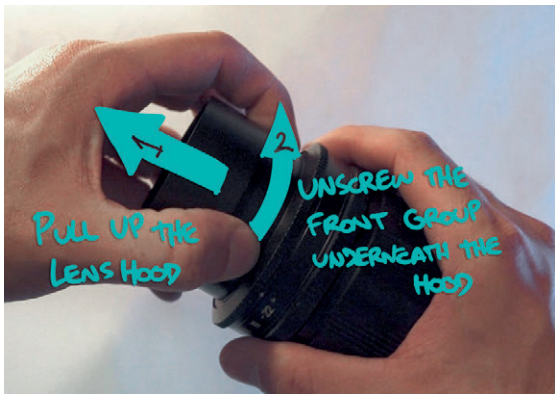
TAIR 11A - 135MM F/2.8
BUY IT NOW! $120-170

The Tair 11A is our telephoto choice, a gorgeous 135mm f/2.8 with 20 aperture blades and smooth aperture ring. I went much lighter with the modding on this one because not much change was needed to drastically affect the footage. I used a solid orange aperture which is quite intense. Smooth out yours if you want less tint. Check out the video version of this tutorial if you prefer moving images!

I will start by saying I love long lenses. My favorite focal lengths are all north of 85mm and the Tair 11A has a special place. It is sharp wide open and is an awesome Soviet lens – which brings a ton of character into the picture.

The Tair 11A uses the largest oval I made so far, so get yours ready according to your preference of paint and flares. I also have a file for 3d printing focus gears for this lens.

You can find it at .\Modding Parts\Focus Gears\USSR_Tair11A.stl



#01. The first step is to pull up the lens hood and get a good grip on the body underneath it. Now unscrew it out. This will give you access to the aperture mechanism. Yes, it was that easy. Drop in the aperture disc and close the lens back up.

THE LAST STEP IS TO ADD THE M42 TO EF ADAPTER AND ADJUST IT SO THE OVAL IS FACING THE RIGHT DIRECTION!

Tair 11A with amber oval insert and backup neutral insert

ODDBALLS

I already had the tutorials for the two lenses in the following section and they can be useful in various situations, if not just to widen the scope of ready-to-go options for you to explore.

The Canon 50mm is the only AF lens in this entire list; the Pentacon 29 is a super cheap wide-angle that offers amazing close focus capabilities.

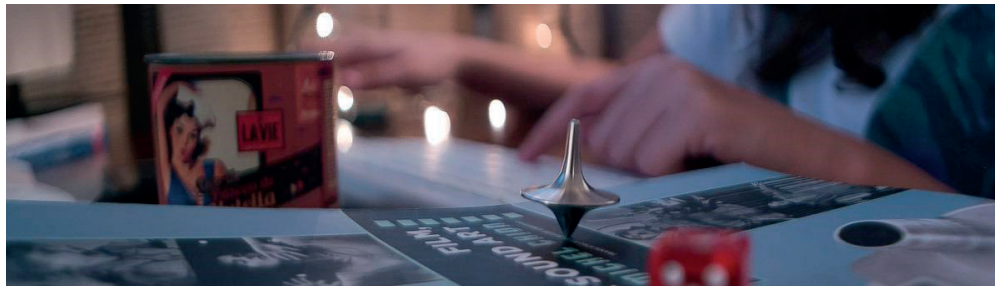
These two also have some sentimental value because they were mods that I learned from other folks. The Pentacon was the second-ever lens I modified, following close instructions from Amir Safari and he even sent me the oval inserts through the mail when I lived in Brazil.

I learned about the Canon 50mm mod through a subscriber on my channel and he also sent me detailed info on all the steps as well as some results he got with his lens.

Pentacon Electric 29mm f/2.8, this one was modified back in 2013

PENTACON AUTO OR ELECTRIC 29MM F/2.8
BUY IT NOW! - $60-90

The Pentacon 29mm f/2.8 is an inexpensive and fairly common wide angle lens. It was my first take on finding a good wide angle to anamorfake - video here. They are cheap around Europe. I have not played with the "Made in G.D.R". version and although getting it open uses a different method, the iris should be the same size.



The Pentacon 29 is cheaper than the Mir 10A in the USSR set and it still gives you a decent wide angle feeling with anamorphic bokeh and super close focus capabilities (the Pentacon focuses down to 0.25m, just under 1ft). This one blooms a bit more than the Mir when it comes to highlights, but nothing terrible.

#01. The first step is to remove the ID ring (the one with the lens' name and info on it).

#02. It is useful to add markings to the front group regarding the oval's orientation - like I did with the Canon EF 50mm and the Rokinons -, otherwise you will be opening up the lens, adjusting the oval and closing it back a number of times until it is perfect.

#03. Once that is done, unscrew out the front block. That gives us clear access to the aperture, where we are going to attach the oval insert.

#04. Tape the oval disk to the back of the front block, following the orientation guides.

#05. When you are through putting the oval aperture, close the lens back up and you are good to go.

#06. If you decided to skip getting the oval aligned inside the lens, just add a trusty rotating M42 adapter and if you want full upgrades, there is a STL file for 3d printing a focus gear *at .\Modding Parts\Focus Gears\Others_Pentacon29.stl*

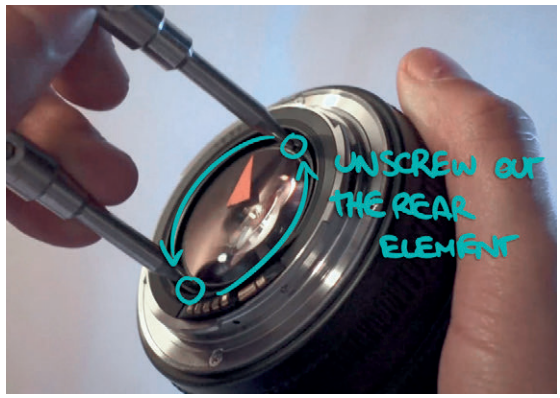
The Pentacon 29 is a cheap and easy wide angle to mod, giving great bokeh at super close focus

CANON EF 50MM F/1.4 USM
BUY IT NOW! - $330

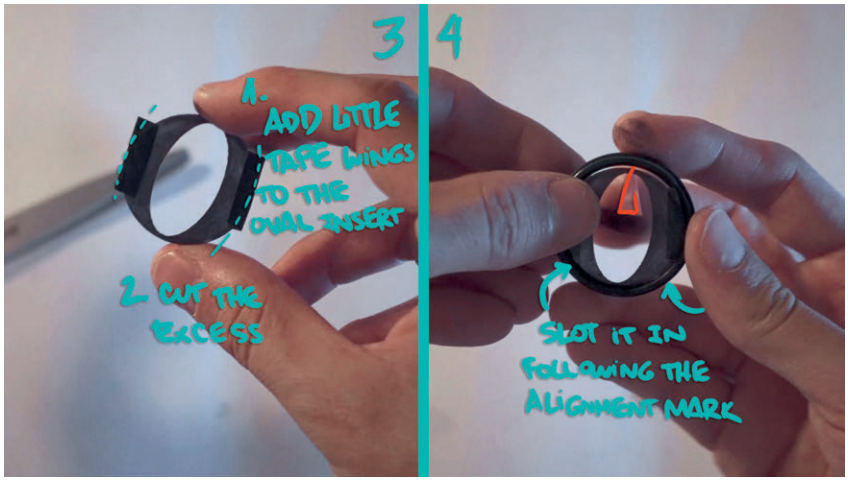
The Canon 50mm f/1.4 is one of the most common and relatively inexpensive lenses for many starting photographers/video shooters. "You need the kit lens and a fast 50". This one was my first fast lens. It is much cleaner and contrasty than most options in this list of mods and this is the only autofocus anamorfake mod I ever made. I got all the info for it from Grant Gilmore, so thanks Grant! Video version here!

#01. In order to get alignment right in our first try I will put marks on the lens of how the element screws back in. First I will put a bright tape triangle on the 50mm mark that faces up. Now add another triangle on the rear element pointing directly to the first triangle. The oval has to be in line with this triangle on the rear element to land perfect alignment.

#02. On the rear side of the Canon, with the pointy end of the lens wrench, twist out the rear element. It comes off easily. The element right below it might try to come off too if you turn the lens upside down, so be warned that the concave side is the one pointing up. Put this loose element back in carefully if you happen to take it off.



#03. Since the disc is a tiny bit smaller than the diameter of the rear element, add a small piece of electrical tape on each side and cut the excess. This will make your life easier for spinning the disc into alignment. All you need to do is slot the aperture disc on the empty side of the element we took off. It fits perfectly in there.



#04. Then, put the lens back together. Remember to clean your glass before shooting to wipe out fingerprints and tape residue.

Unlike the other anamorfake mods, I am not putting the oval directly over the aperture mechanism, which is not ideal. If this was a wide angle the results would not come without a consequence like dark edges or massive vignetting.

That being said, this mod works pretty well and takes only a few minutes to execute. The coolest part is that you can use autofocus as usual with the added benefit of anamorphic bokeh!

Canon 50mm f/1.4 turned f/2.0 and full of oval bokeh goodness!

10. POST PROCESSING

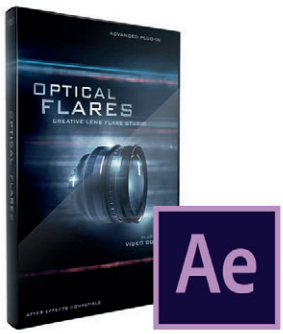
Even after you are past the production stage and your project is in the can, there are still many ways to create the anamorphic look solely in post. Some of them - bokeh manipulation in particular - require a lot of time and fiddling, but changing aspect ratio is super easy and adding distortion can make it fairly convincing. Flares can be done in a few different ways and I will briefly cover VideoCopilot's Optical Flares ($124.95), Red Giant's Magic Bullet Looks ($399) and using flare footage (although I am not a big fan of that method).

Resolve, Premiere, After Effects and FCPx will be covered. Some have more uses than others.

We will go over two different methods for adjusting aspect ratio - cropping and masking - in Adobe Premiere, After Effects, Davinci Resolve and Final Cut Pro X.

Distortion can be easily done in After Effects since it offers separate controls for vertical and horizontal distortion. Davinci Resolve is a bit more convoluted and requires a little eyeballing. Premiere gives you a single "Lens Distortion" filter, which is nowhere near the vast controls from After Effects. Still, it is better than nothing - which is what Final Cut Pro offers: no distortion effects there.

Bokeh is more intense than the two processes above and will only be covered in After Effects and Resolve since it involves rotoscoping. This method for oval bokeh works super well for still frames, but getting it to work in motion will require either meticulous planning of shots to minimize the movement of your subjects, or advanced skills in rotoscoping/matte manipulation and 2d tracking.



Flares are mainly covered in After Effects, as **VideoCopilot's Optical Flares** is not compatible with Resolve. Resolve has its own Lens Flare effect, but I am not familiar enough with it to explain how to use it. **Red Giant's Magic Bullet Looks** is compatible with FPCx, Premiere, Resolve and After Effects. There is also a quick overview using blend modes and flare footage.

VASHIMORPHIC40!

It is important to acknowledge the VashiMorphic40 by Vashi Nedomansky. This is the only free template available online aiming to simulate anamorphic distortion and artifacts in After Effects - specifically from Panavision's Primo 40. You can visit the link above for a full written and video tutorial plus downloads. The VashiMorphic40 works through a combination of Bezier Warp, vignette and blur and it has been downloaded thousands of times.

VashiMorphic40: The original template was published in 2013 and updated in 2018

The downside is that it is only meant to work in After Effects, but if that is what you are using, this will save you a ton of time!

Helios 44-2 with oval bokeh mod on Canon EOS 5D Mk III + VashiMorphic40

2.35:1 OR 2.40:1? WHICH ASPECT RATIO IS PROPER CINEMASCOPE?

The next topic to cover is - surprise! - a mathematical one. A film's aspect ratio is determined by its horizontal measurement divided by its vertical measurement. Since we are working with digital images, these measurements will be given in pixels. 16:9 aspect ratio means that for every 16 horizontal units, there are 9 vertical units. You can also simplify it by dividing 16 by 9, which we will round up to 1.78, and that means that the horizontal dimension of the frame is 1.78x longer than the vertical dimension. This can be demonstrated with a division of familiar numbers:

1920 ÷ 1080 = 1.7777... ≈ 1.78

Through history, film has gone through multiple different aspect ratios for various reasons. Here are a few famous ones and their uses: 1:1 (Instagram), 4:3 (old time movies and TVs), 1.85:1 (Academy), 2.35:1 and 2.40:1 (CinemaScope).

When the anamorphic format started its Hollywood life, it was a time of hardship. Movie theaters were losing audiences to television content, so they had to crank up the experience of going to a theater. They did so by making the screen impossibly wide when compared to tiny 4:3 TVs.

"Which aspect ratio is proper CinemaScope?". There are a lot of numbers thrown around the web: 2.66:1, 2.55:1, 2.40:1, 2.39:1, 2.35:1. Due to film sizes, sound strips and other technical issues you can learn about here, CinemaScope meant 2.35:1. In 1970 technical specifications were updated and the format was expanded to 2.40:1.

This leaves you open to decide which one you prefer. I tend to stick to 2.40:1 because I like wider frames better. The difference is tiny though, only 16 pixels at FullHD:

1920 x 800 = 2.40:1
1920 x 816 = 2.35:1

Now that we are settled with history, time to get into software!

MASKING VS CROPPING

When it comes to conforming 16:9 footage to a new aspect ratio you have two options: masking and cropping. I find cropping vastly superior, but masking has its uses. Both of them involve discarding parts of the original frame, roughly 25%.

1080 x 0.75 = 810
(close enough to 800 and 816)

Masking is also known as Letterboxing and it consists of inserting black bars on the top and bottom of the frame. This does not effectively change the dimensions of your video. As simple as that. You can find both 2.35:1 and 2.40:1 templates in the **Appendix** folder for 1080p, 4K UHD and 8K UHD 16:9 base footage. There is also an After Effects script that will work for any footage aspect ratio!

Check the file ./Appendix/10_Masking/autoMask_readme.txt for more information on how to use the script! You can find a video tutorial for autoMask in ./Videos/10_PostProcessing/AE_autoMask.mp4

All you need to do is drop one of these masks on a video track above your footage and make the mask duration the same as your video. No need for a tutorial on that!

Masking will not change the original size of your frame. "The Way Station", dir: Janin Palahicky.

On the advantages of masking: until recently YouTube did not properly display videos in different aspect ratios. It always added black bars to deliver the default 16:9 player window. In early 2018 that was adjusted and YouTube plays video using their upload aspect ratio, but not all bases were covered. YouTube still does not allow you to use the “End Screen” feature if your video has an aspect ratio different from 16:9.

Some festivals and output formats/codecs also require specific image resolutions. In those cases, masking is your only solution.

Cropping requires changing settings on your timeline. You will adjust the timeline’s height based on the frame’s width. Cropping also works flawlessly even if your base footage is not 16:9 - while masks will require adjustments (except that handy autoMask script).

Cropping makes your frame smaller. “The Way Station”, dir: Janin Palahicky.

To find out your new settings, you need to choose an aspect ratio and solve the equation below.

frameH = frameW ÷ aspectRatio

Here is an example using a base resolution that is not 16:9. Let's say I shot my project in C4K (4096 x 2160px) and I want to output it with a 2.40:1 aspect ratio.

frameH = **4096 ÷ 2.4**
frameH = **1706px**

Now I know my timeline width will be the base 4096px, but the height has to be adjusted from 2160 to 1706px.

> **COMPUTERS LOVE EVEN NUMBERS SO WE ARE GOOD ON THAT, BUT MORE THAN THAT, COMPUTERS LOVE MULTIPLES OF 8. I HIGHLY RECOMMEND ADJUSTING THE RESULTS BY A FEW PIXELS TO GET A HEIGHT THAT IS DIVISIBLE BY 8 (IN THIS CASE I COULD GO TO 1704 OR 1712).**

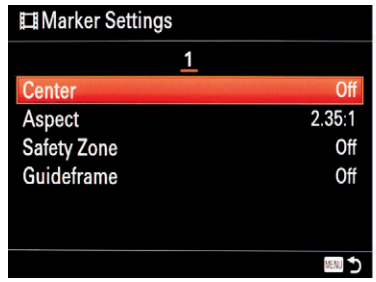
FRAME GUIDES AND CROP MARKS

Deciding the aspect ratio of your project before you shoot is extremely useful. Many cameras and most monitors offer some type of overlay on the image that displays your desired final aspect ratio.

The name and location in the menu can change for each manufacturer, so dig deep! Sony calls them Marker Settings, Canon shows them through MagicLantern, Panasonic allows you to draw your own, so on and so forth.

If the number offered by your camera is not exactly what you want, it is still better than no guide! For example, if I want my project to be in 2.40:1, but my Sony camera only gives me a 2.35:1 overlay, that is close enough to what I want.

Sony's "Marker Settings" menu and the overlay on the screen.

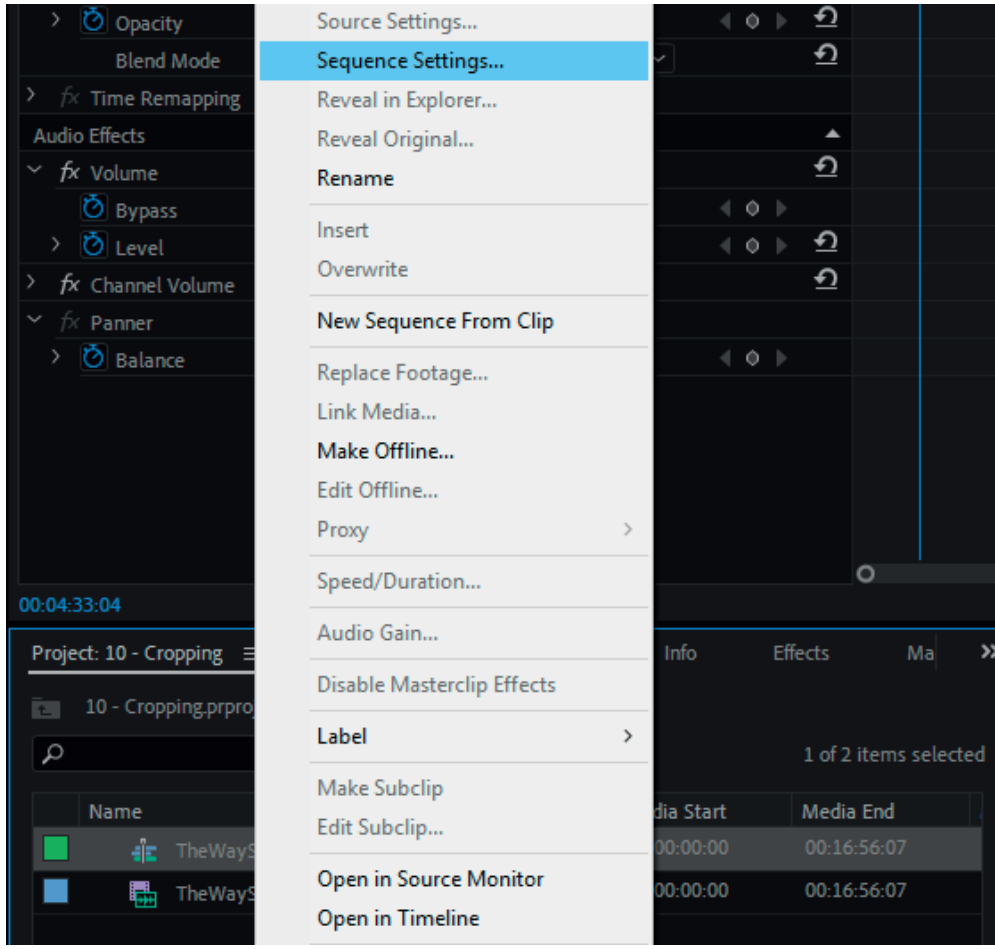


This helps immensely and guarantees you are not leaving anything important out of your final output.

In the Appendix folder you can find accurate cropmarks to load on your MagicLanter Canon camera.

ADOBE PREMIERE

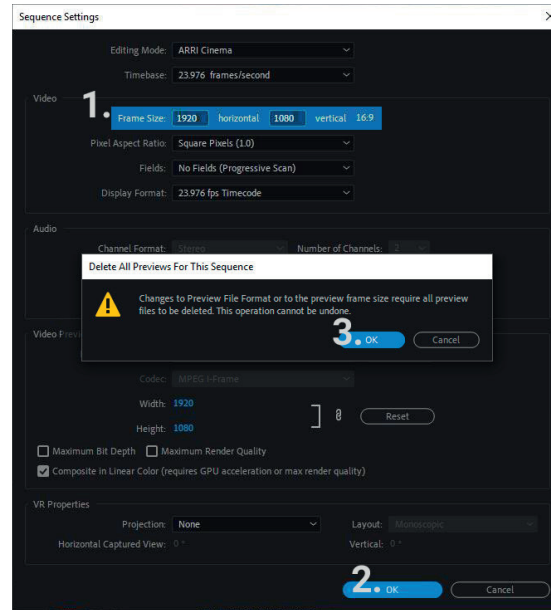
You can find a video version of these steps in ./Videos/10_PostProcessing/Premiere_Crop.mp4



Once the project is created and loaded, you can move all your footage onto a sequence using the footage's base settings. Now you right-click the sequence in the "Project" tab and select "Sequence Settings".

That will bring up a window with the timeline's dimensions. From there you follow the equations demonstrated earlier. Get the width and divide by the desired aspect ratio, do any adjustments to get a number that is divisible by 8, and input that onto the height.

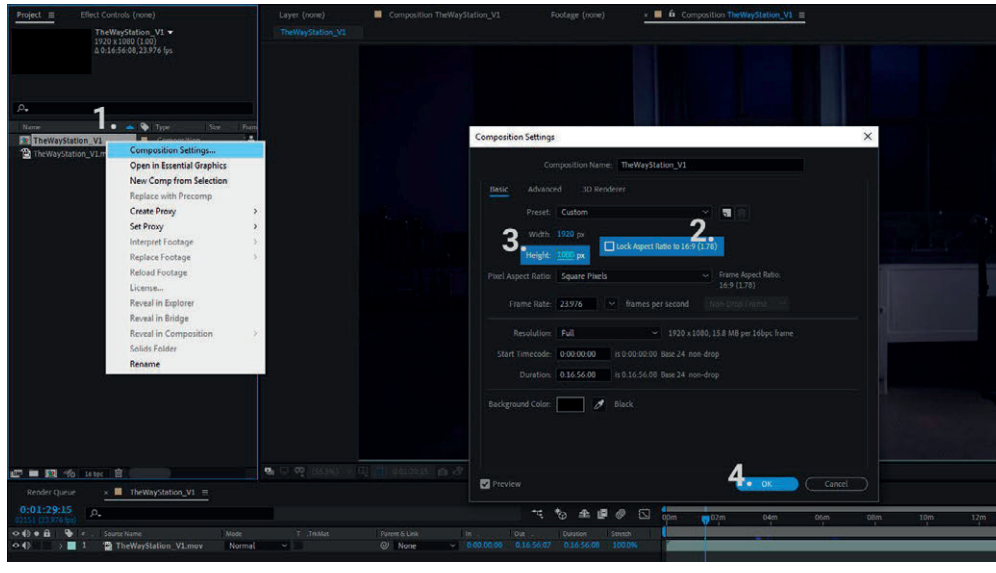
Hit "Ok". A pop-up will jump in your face saying your previews will have to be re-done. Well, nothing we can do about that. Hit "Ok" again and your "Program" window should be rescaled to match the new settings.



ADOBE AFTER EFFECTS

You can find a video version of these steps in ./Videos/10_PostProcessing/AE_Crop.mp4

Create your project, import the footage and drag/drop it into a new Composition. From there, right-click the composition on the "Project" tab and select "Composition Settings" (hotkey: Ctrl + K).



In the Comp Settings double check that the box “Lock Aspect Ratio to 16:9 (1.78)” is unchecked. Get the “Width”, divide it by the aspect ratio you want, make sure the result is divisible by 8 and insert the number onto the “Height” field. Press “Ok”.

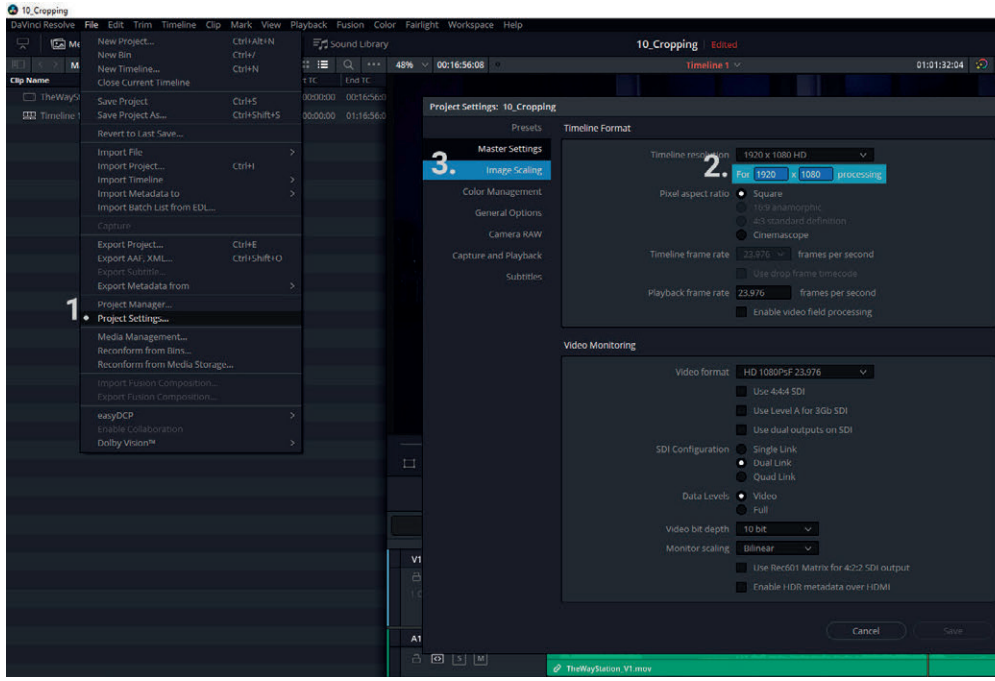
You are done!

DAVINCI RESOLVE

You can find a video version of these steps in ./Videos/10_PostProcessing/Resolve_Crop.mp4

Create your project, bring in your footage and create a timeline conformed to that footage. With that loaded, go into “File > Project Settings”. Resolve works in a slightly different fashion than Premiere and After Effects: while in the Adobe suite you can have multiple timelines with different resolutions and aspect ratios, in Resolve all timelines within a project are conformed to the settings defined in “Project Settings”

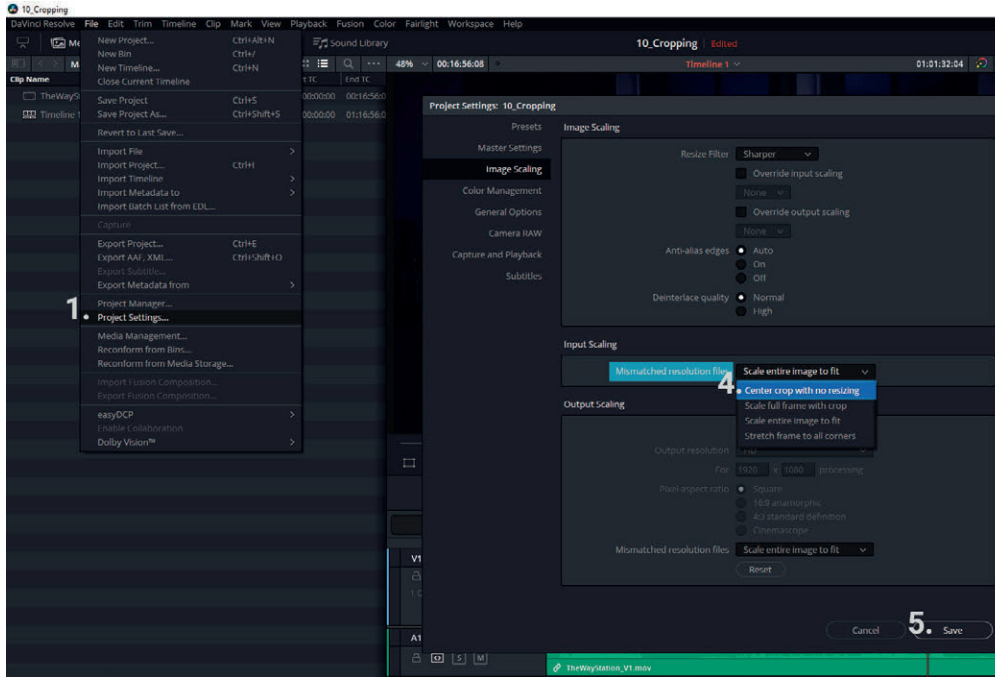
In “Master Settings” you can find the “Timeline Resolution” we are going to change. Pick your width, divide by the desired aspect ratio, make sure the result is divisible by 8 and input it on the height.



Before you are done here, quickly jump to “Image Scaling”. This section defines what happens to footage that does match exactly the numbers set in the “Master Settings”, like what we are about to experience.

“Input Scaling” is set by default to “Scale entire image to fit”. Since your footage no longer matches the timeline resolution (we changed it in the previous step), with “Scale entire image to fit” your shots will be redimensioned to fit inside the new timeline size - if you are curious to see what that means, hit “Save”. Spoiler alert: It will add black borders on the right and left sides of the frame in order to fit the height. Not quite what we had in mind with our aspect ratio change.

Set “Input Scaling” to “Center crop with no resizing” and then hit “Save” for proper display, cropping out the top and bottom of the frame.

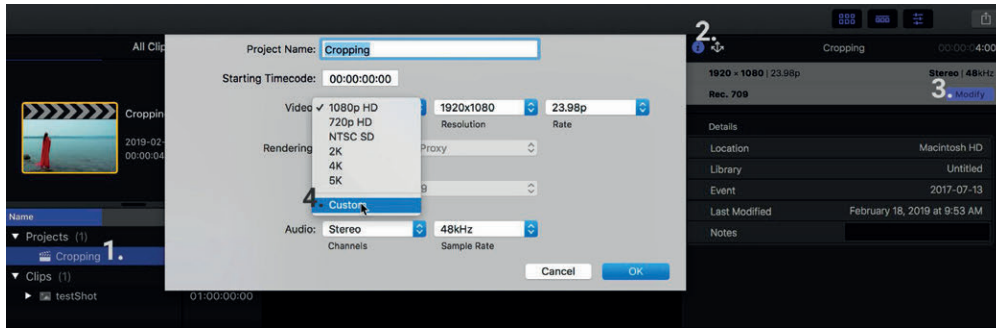


This change affects all the timelines within this project.

FINAL CUT PRO X

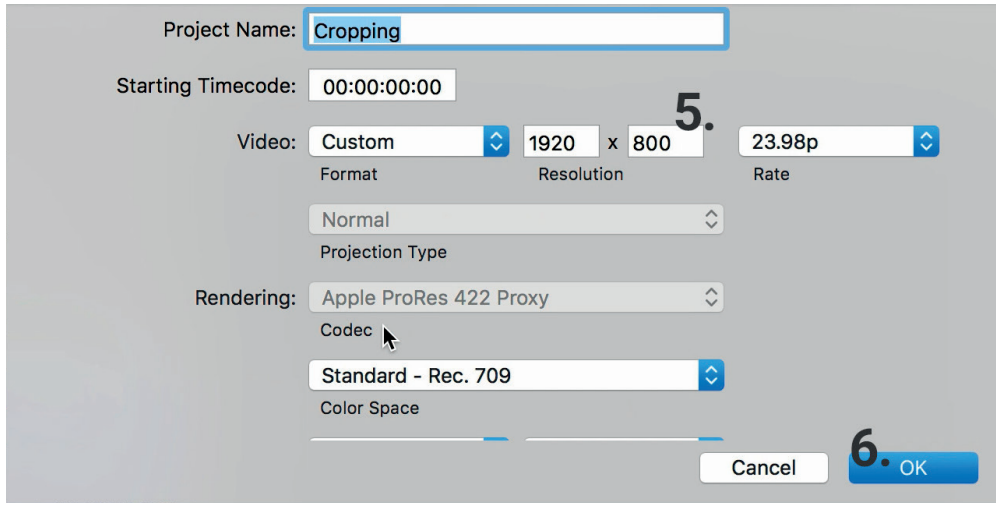
You can find a video version of these steps in ./Videos/10_PostProcessing/FCPx_Crop.mp4

Final Cut Pro has different names for everything. Once you set up your Library and create a Project (a "timeline" for everyone else), you can select that Project. The "Inspector" tab will display its properties, including resolution. Click on "modify" to call the Project's settings.

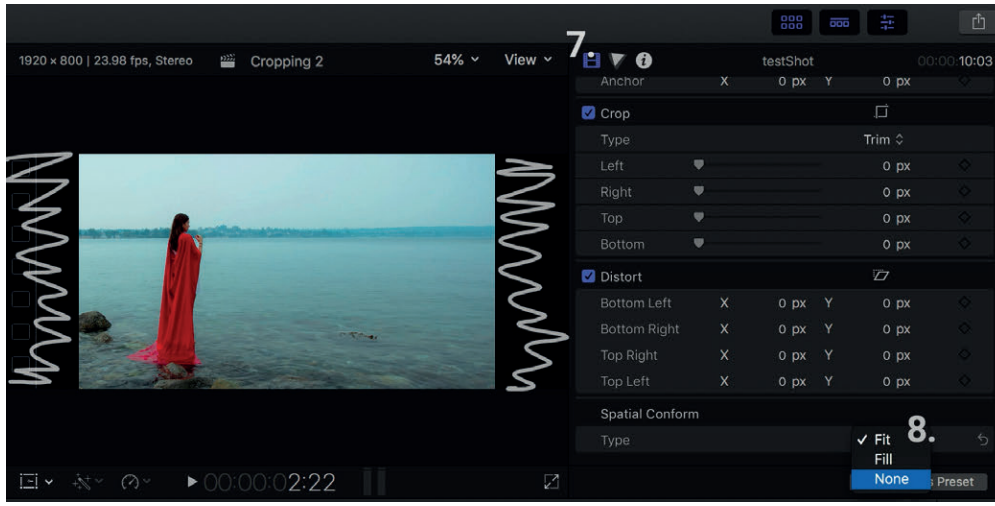


On the new pop-up you can change the video resolution picking the "Custom" option on the menu and typing in the desired value. Hit "Ok".

This is the same pop up that shows when you are creating a "File > New > Project", so you can have your timeline set up to the correct aspect ratio from the very start.



All media on a Project follows a "Spatial Conform" rule. The default is "Fit", which rescales everything to fit inside the frame. So if your footage is 16:9, as soon as you change the Project's aspect ratio, your media will show black bars on the sides.

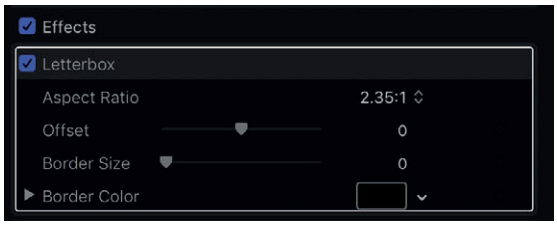


Select the media and, on the "Inspector" tab, scroll all the way to the bottom of the list, finding "Spatial Conform". Change the "Fit" setting to "None" and the media will rescale to the proper size, eliminating the black edges and cropping out the top and bottom of the frame!

MASKING IN FCPX

You can find a video version of these steps in ./Videos/10_PostProcessing/FCPx_Letterbox.mp4

FCPx has a handy effect much like my After Effects autoMask tool. In the "Effects" tab select "Letterbox" inside "Stylize" and apply it to your footage.



The effect gives you a few different aspect ratios to choose from, including 2.35:1. "Letterbox" gives you an "Offset" slider that helps reframing your shot and a "Border Size" slider which does small tweaks to the aspect ratio.

DISTORTION

Most lenses have some sort of distortion. It is easy to see it when you think of fisheye lenses: the bulging effect that makes the shot look like it is popping out of the screen is an extreme example of barrel distortion.

Anamorphics have a very unique type of distortion that is a trademark of the look. It looks similar to barrel distortion but stronger in horizontal lines than vertical lines.

Canon 8-15mm f/4L @ 15mm f/4 on the Canon EOS 5D Mk III, before and after distortion correction - pOrtal

In this chapter we are doing the the opposite of what I did in the sample images above. We are adding distortion instead of removing it, and only on the horizontal axis.

This distortion is much more noticeable on wide-angles because you see more of the environment around the character. Look at those bendy horizontal lines at top and bottom of Wes Anderson's shots!

The Life Aquatic with Steve Zissou

The Royal Tennenbaums

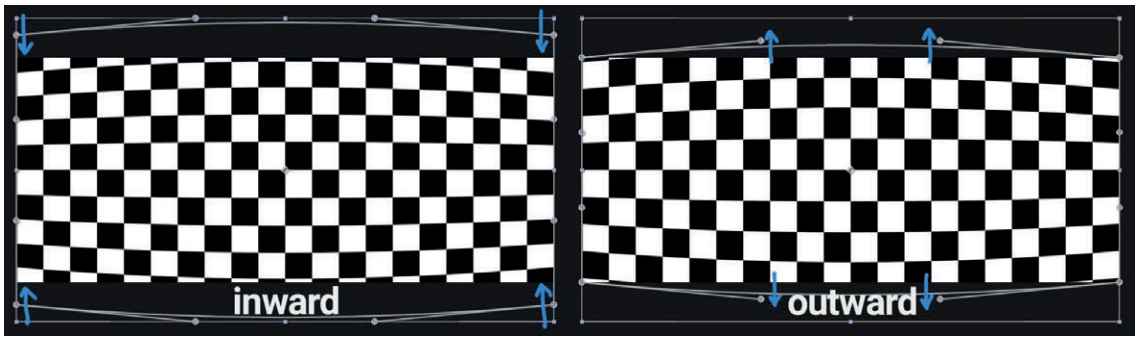
Rushmore

The Darjeeling Limited

Wes Anderson is a big fan of the Panavision Primo 40mm Anamorphic - thanks for the hint, Vashi!

There are two approaches to distortion. The first and most traditional one is to distort your final frame, pushing pixels outward and keeping the corners locked to avoid more cropping.

Our strategy however will benefit from having a crop/mask applied: we can pull some of the original corner information back into the shot to achieve the bendy horizontal lines - yet keeping the center of the frame untouched.



The blue arrows indicate where distortion is being originated.

While I am using the same value for both distortions, the results are quite different. The inward approach maximizes the use of your footage and and holds a more natural look. If you look at the squares around the center area, the outward ones are stretched, while corners stay neutral. On the inward distortion, the center is neutral and the corners are compressed.

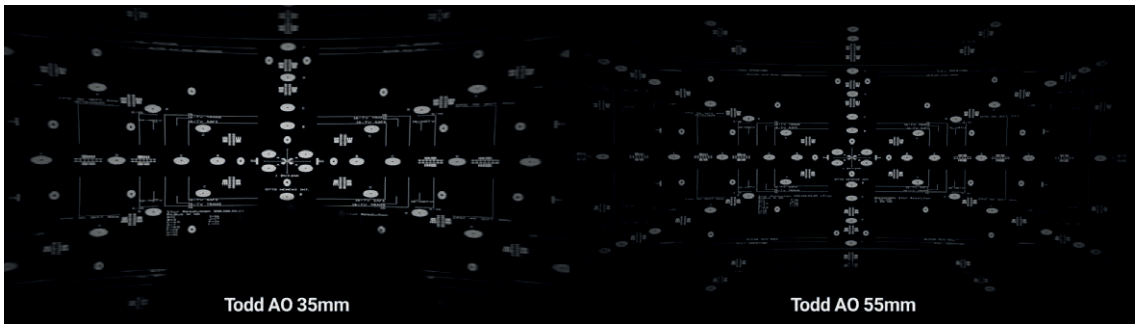
Unfortunately not all of the software covered in this guide works the same way. The best and most reliable results can be achieved in After Effects through the effect Bezier Warp - thanks to the preset I painstakingly created and included in the **Appendix folder**.

Davinci Resolve has us jumping inside the “Fusion” tab to play with the “Grid Warper”, avoiding what seemed to be the best option, the “Warper” effect, which is actually very limited.

Premiere does not offer any way to control vertical and horizontal distortion separately so the results are not ideal. Yet, any distortion is better than no distortion so I will cover this option briefly.

Final Cut Pro offers no native “Lens Distortion” effect, displacement maps or user-controlled warps, blocking us from achieving distortion efficiently.

Lastly, while you are in control of the amount of distortion being added, keep in mind distortion should be stronger in wide angles and get more and more controlled as your focal length grows longer. You can see this on ShareGrid’s test charts below - taken from The Ultimate Anamorphic Lens Test.



Contrast has been increased for easier observation of distortion lines. Original images here!

This last example also shows that not all anamorphics curve outwards (barrel distortion). These Todd AO lenses show a type of pincushion distortion, the opposite of what we saw on the Panavision Primo’s used in the VashiMorphic40 - again, only in the horizontal axis.

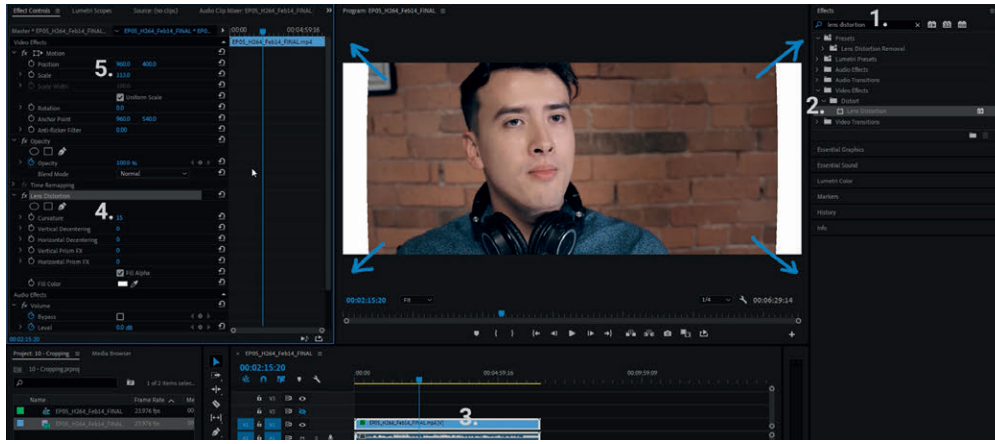
ADOBE PREMIERE

You can find a video version of these steps in ./Videos/10_PostProcessing/Premiere_Distortion.mp4

We are starting from the point where we already have our project and timeline set up with the final aspect ratio - 2.4:1 in this case.

Go to the “Effects” tab and pick “Lens Distortion” from the “Distort” folder. Avoid all the default presets that show up for GoPros and drones. The “Lens Distortion” effect was created to counter bad distortion (such as GoPros and drones!), yet, here we are trying to force it upon our shots!

With the effect applied, go to the footage’s “Effect Controls” tab and play with the Curvature slider. Positive values induce barrel distortion, negative create pincushion.



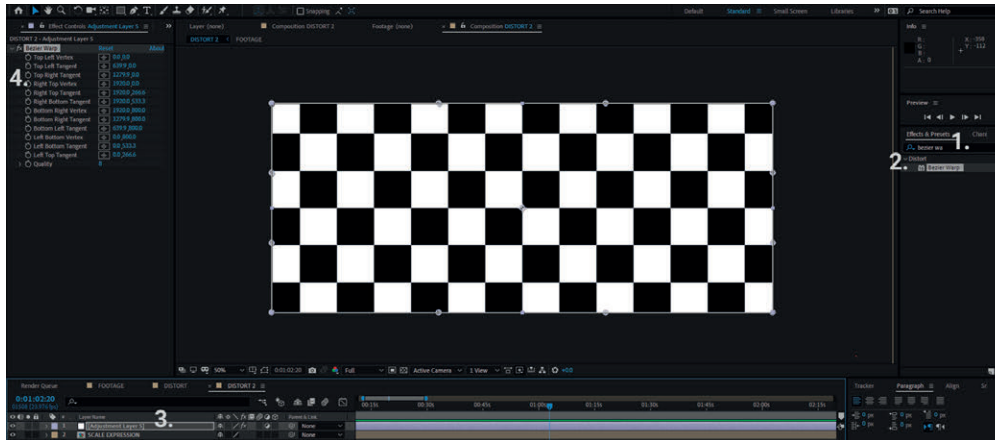
When applying barrel distortion, the sides of the frame start to break. The solution is to scale up the shot to the smallest value that covers the gaps. For example, if I do Curvature 15, the smallest Scale value I can use is 113%. The scaling causes almost no losses. Check below for two versions of the same frame, one without distortion and scale 100%, the other with distortion and scaled up.

Original frame on the left, distorted and scaled frame on the right. Watch the brick lines in the background.

ADOBE AFTER EFFECTS

You can find a video version of these steps in ./Videos/10_PostProcessing/AE_Distortion.mp4

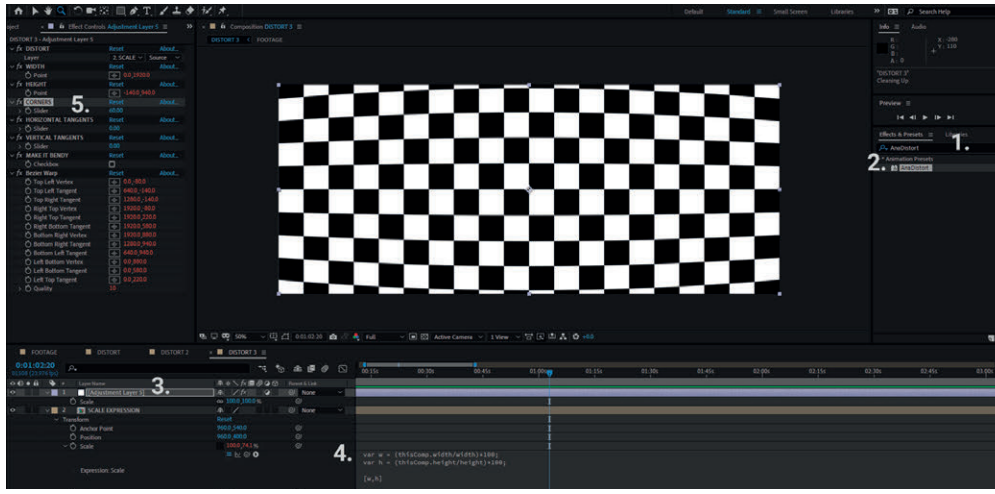
After Effects is the most versatile option for distortion. I already have my Comp set up, aspect ratio 2.4:1. On the "Effects & Presets" tab pick "Bezier Warp" and apply it to an Adjustment Layer over the footage. Ultimately all you have to do now is play with the control points on the screen. If you are accuracy-obsessed like me, you will want to make sure the numbers are also matching on all vertexes and tangents.



If you want something better than the standard "Bezier Warp", install the **AnaDistort** preset included in the **Appendix** folder. It will give you accurate controls over distortion and simplified sliders to play with. It also utilizes the cropped top and bottom of the frame for the distortion, effectively compressing the sides, keeping the center intact and giving you those nice and smooth curved horizontals.

Once you have it installed, just search for "AnaDistort" in your "Effects & Presets" tab and apply it to an Adjustment Later over the footage. It will require you to apply an expression to the "Scale" parameter of your footage. In order to do that select your shot on the timeline, open the "Transform" parameter and Alt + click the stopwatch by "Scale". Paste the following:

```
var w = (thisComp.width/width)*100;
var h = (thisComp.height/height)*100;
[w,h]
```



Now go back to the distortion Adjustment Layer and fiddle with the "CORNERS" slider until you find a look you like. Positive values give you barrel-type distortion, negative values give you pincushion.

Check the file *./Appendix/10_Distortion/AnaDistort_readme.txt* for more information on each slider of the preset and the necessary steps to install it.

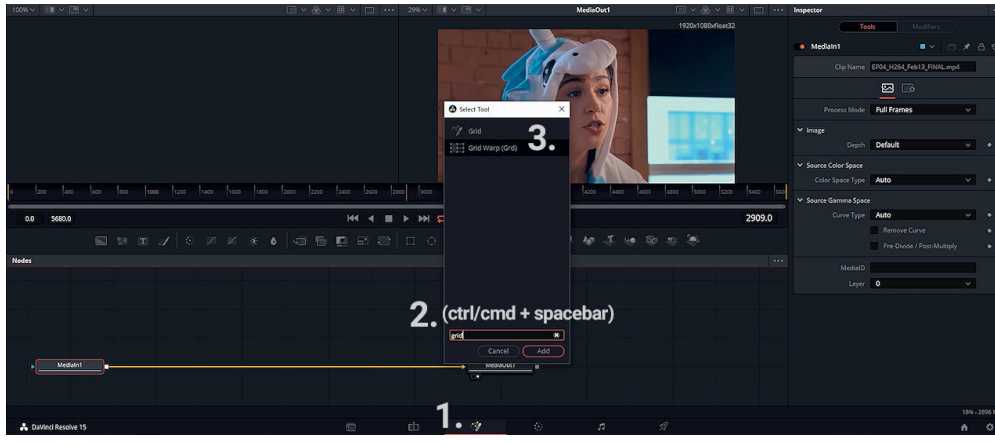
You can find a video tutorial for AnaDistort in ./Videos/10_PostProcessing/AE_AnaDistort.mov

DAVINCI RESOLVE

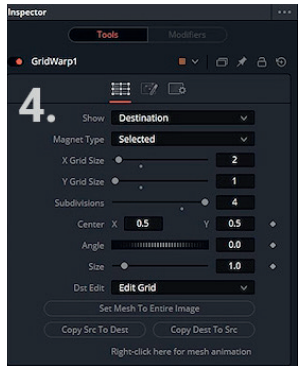
You can find a video version of these steps in ./Videos/10_PostProcessing/Resolve_Distortion.mp4

Resolve offers an intermediate solution between Premiere and After Effects. It is not as rudimentary as Premiere but not as accurate and controlled as After Effects. The process takes place in the "Fusion" tab, so let's get started.

Have your project set up with the proper aspect ratio and bring the footage onto the timeline, cropping the top and bottom. Now head over to "Fusion".

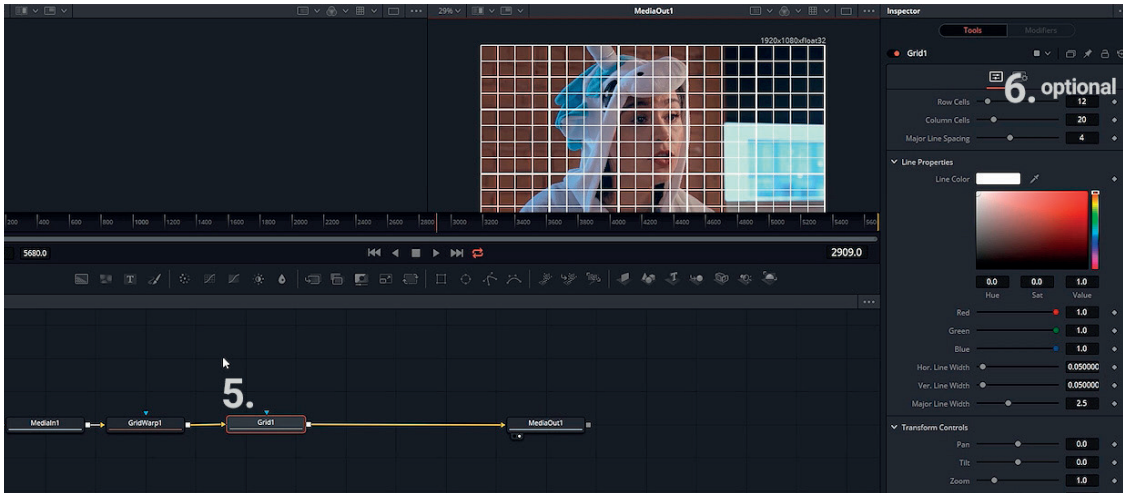


The first thing to do here is to set up our distortion tool. Hit Ctrl + Spacebar (Cmd + Spacebar on a Mac) to bring up the "Select Tool" list and pick "Grid Warp" - you can just type it in. This will lay down the distortion mesh we are applying to the shot.

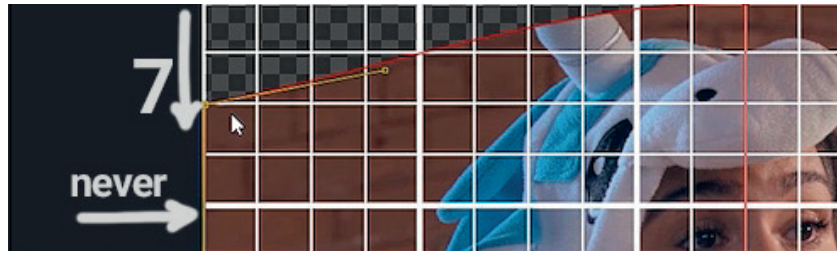


There are way too many control points in the default grid though. Time to change some settings. "Magnet Type" goes to "Selected", "X Grid Size" comes from 8 to 2, "Y Grid Size" comes from 8 to 1, "Subdivisions" maxes at 4. The distorting will come a little bit later.

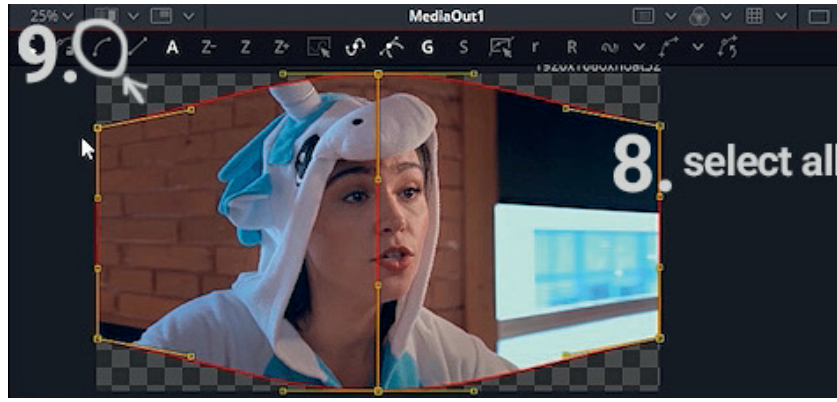
Next, add a "Grid" node. This second grid will be our measuring tool. We are using these lines to dial in the actual distortion. Feel free to tweak the "Row Cells" value for more precise targets.



Go back to the "Grid Warp" and pick one of the corners. I like to do it in the following order: top left, top right, bottom right, bottom left. Always skip the ones in the middle. Anyway, pick a corner and move it down or up while counting the white grid units. Then repeat the process for the three remaining corners and use the same number of units. This guarantees the distortion is even over all four corners. Try as hard as you can not to move the points horizontally! Horizontal movement introduces vertical distortion and we want none of it!



Once you are happy with the positioning of your corners, zoom out and select all the Grid Warp's points (including the middle ones). Go to the icon with the little curve and click it to smooth out the tangents. This makes a big difference by making your distortion more rounded and organic.



Disable the "Grid" node and jump to the "Color" tab. If the result looks good with the crop, move on! If you have black corners, go back to "Fusion", re-enable the grid and adjust your points until no more black corners show up.

Original frame on the top, distorted and scaled frame on the bottom. A lot of crop has been brought back in.

OVAL BOKEH

At this point in this book it is needless to say that the oval bokeh is one of the most coveted anamorphic traits. The so-called "waterfall bokeh" is a big boost in production value, giving each shot a paint-like quality.

"Moonlight", 2016. Dir: Barry Jenkins, DP: James Laxton. Hawk V-Lite Anamorphics.

In previous chapters I went over how to achieve anamorphic bokeh through the use of aperture discs, but if you really want to do it in post-production, there is still hope. "The Lego Movie", 2014, is a great example that the anamorphic look can be achieved completely through the use of digital tools. This article by Expanded Cinematography on "The Lego Movie" is absolutely worth reading!

And, I mean, if animation can do it, there has got to be a way we can do it too! The downside: It is an incredibly time-consuming process and I would only recommend it for short shots, or pick-ups that did not use the same gear used during principal photography.

"The Lego Movie", 2014. Dir: Phil Lord and Christopher Miller, DP: Pablo Plaisted. Nuke + LegoLens.

Why is the process so time-consuming? Because it involves rotoscoping the elements you want to keep in focus. And if you want to be really accurate you will want to break down your out-of-focus areas in different layers with different levels of defocus - a simplified simulation of reality.

Rotoscoping is one of the oldest techniques in the visual effects world and it means to manually separate elements in the footage through the use of animated masks. There are plenty of ways to go about making those masks. This is a short and straightforward tutorial. I am not teaching rotoscoping in this guide. One could write an entire book on the subject.

The best shots for cheating oval bokeh are the ones with considerable depth of field since it is easier for the software to pick up the highlights and blow them up accordingly. Be careful on your approach though: it is really easy to overdo this effect and end up with a shot that looks unrealistic because depth of field feels too shallow for the framing.

In order to keep this tutorial short, I am going to cover the process on a still frame for After Effects and do a moving shot in Resolve.

ADOBE AFTER EFFECTS

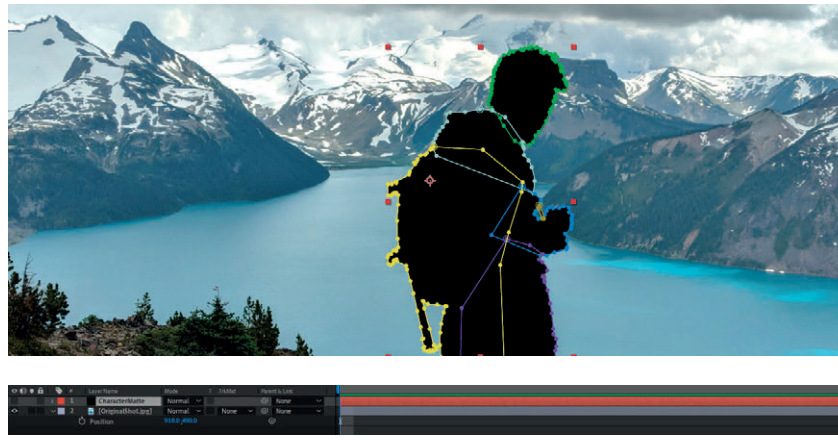
You can find a video version of these steps in ./Videos/10_PostProcessing/AE_Bokeh.mp4
The footage for this tutorial is in ./Videos/10_PostProcessing/AE_Bokeh_Tutorial/

The first step is to choose your shot. I am picking a medium-wide with good foreground/background separation and lots of depth of field. This was shot on my phone, so it is quite noisy. The aspect ratio is also very wrong.

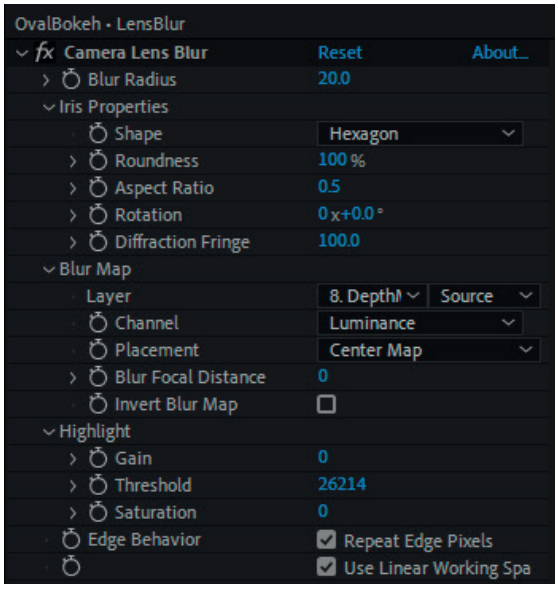
The first thing I will do is bring it into a 2.40:1 Composition and find a framing I like by moving the image around.

The “Camera Lens Blur” filter has a feature that allows us to use a separate layer as a depth map of sorts. To make good use of this feature we will separate each element of the shot. Create a black “Solid” for each element and rotoscope them on their own layers. For this one I am rotoscoping my character.

The way I like to do this is by disabling the visibility of the “Solid” but keeping the layer selected so my masks are applied to it and I see them. When I am done, I just have to flick on “Visibility” and I can check my masks. Names are very important. So the character mask will be “CharacterMatte”.



Quick roto shapes breakdown + layer naming.



Create a new “Adjustment Layer” and from the “Effects and Presets” tab pick “Camera Lens Blur”. Tweak settings! “Roundness” goes to 100%, this makes the “Shape” dropdown useless since now we have a perfect circle for defocus. “Aspect Ratio” should be 0.5 for 2x bokeh, but you can make it smaller for even thinner ovals.

Turn on “Repeat Edge Pixels” in “Edge Behavior” at the bottom.

In “Blur Map” for “Layer” select the “CharacterMatte” layer and pick “Masks” from the second dropdown.

“Channel” should be “Alpha” and tick the “Invert Blur Map” box. Go back to the top and fiddle with “Blur Radius” until you are happy.

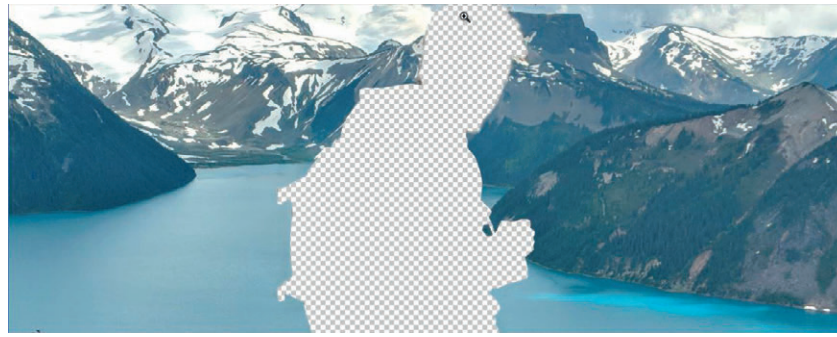
"Highlight" lets you control how bright your bokeh should be. "Threshold" sets the limit of which areas are actual highlights, "Gain" boosts them up and "Saturation"... saturates them. In this shot I have a lot of bright peaks on the mountains so I will refrain from using this feature, but it can make a big difference for darker shots.

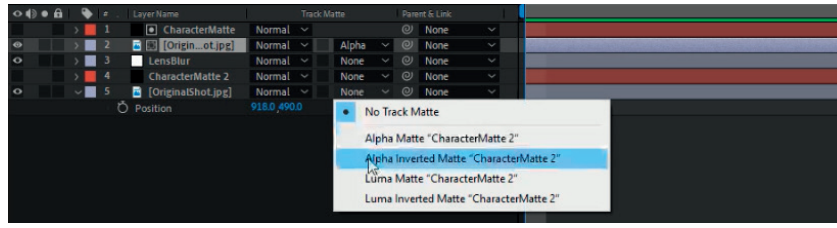
This is the end of the short version of this tutorial. If you pay close attention to the edges around the character, you will see that some of his colors are bleeding through the blur. His backpack creates a dark shadow on the background and the area around his hand has a soft skin-toned tinge to it. These artifacts are quite noticeable and I want them out. Let's add some more layers and complexity.

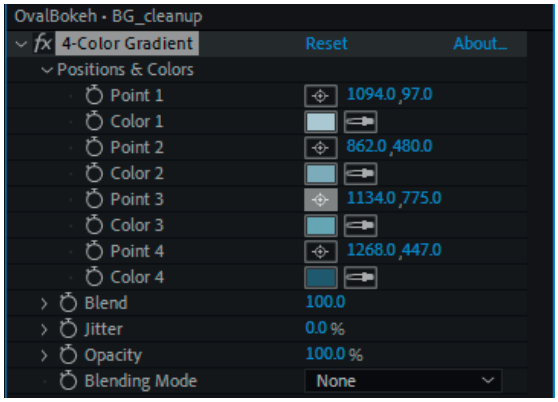
Let's begin by separating the character layer (the one that is going to be in focus) from the background. Duplicate the original shot layer, place it under the "CharacterMatte" and change its ".TrkMat" to "Alpha Matte 'CharacterMatte'".

Then we are doing a sloppy version of what is called a “clean plate” in visual effects. We are removing our character from the background so his colors do not bleed through the lens blur. For now, disable the visibility of the character layers.

There are many ways to remove our character from the background. The fastest one I could come up with was duplicating the original shot and the “CharacterMatte” layers. Put them at the bottom of the stack and choose “Alpha Inverted Matte” for the original shot’s “.TrkMat”. This creates a hole in the image and now we are going to fill it up.







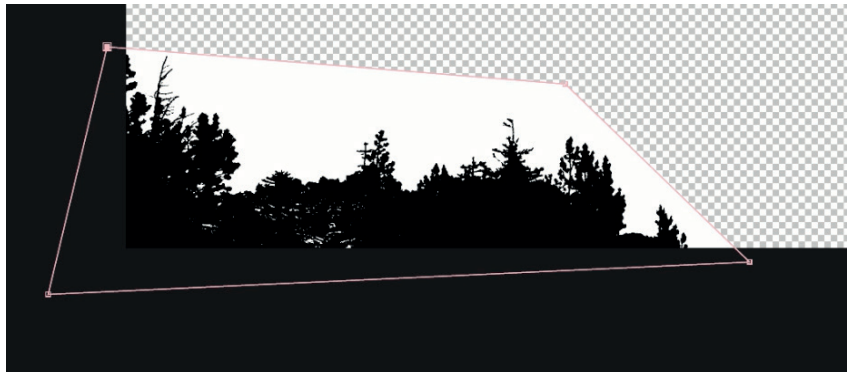
Place a new “Solid” under the original shot layer and add a “4-Color Gradient” effect to it. We are now looking for colors near the areas that are giving us trouble. Namely his backpack, hands, face and hair. Each of the “Point” controllers in the gradient is going to be placed near those areas and I am using the color picker to select colors from the background that look like those areas. This is not an exact science.

Background cleanup using "4-Color Gradient" effect.

Now slide the "Adjustment Layer" with the "Camera Lens Blur" in between the clean background and the character cutout and turn on visibility for the character. He will fill up the strange sky-looking spot, and the lens blur will give much cleaner results.

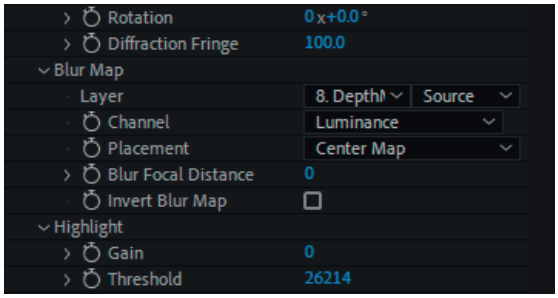
I feel we can kick this up a notch by making better use of the "Blur Map" feature in the "Camera Lens Blur". I want a separate layer with the trees in the bottom left corner, but I also do not want to rotoscope them. Lucky for us, the luminance contrast between the trees and the lake is pretty intense and this calls for a Luma Matte (look at all these VFX terms flying around).

Duplicate the original shot layer, draw a quick mask around the trees - this mask is called a garbage matte - and pull in a "CC Threshold" effect. Now lower the threshold until the lake is white and the trees are black.



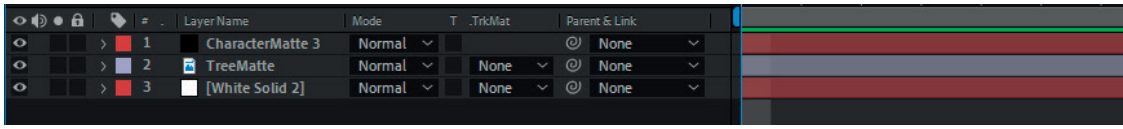
Garbage matte and Luma Key for the trees in the bottom left corner.

Call this layer "TreeMatte". Then duplicate the "CharacterMatte" and precomp (Ctrl/Cmd + Shift + C) both mattes together - select "Move all attributes into the new composition". Name this precomp "DepthMap", disable its visibility and move it to the bottom of the stack. Leave it be for a second.



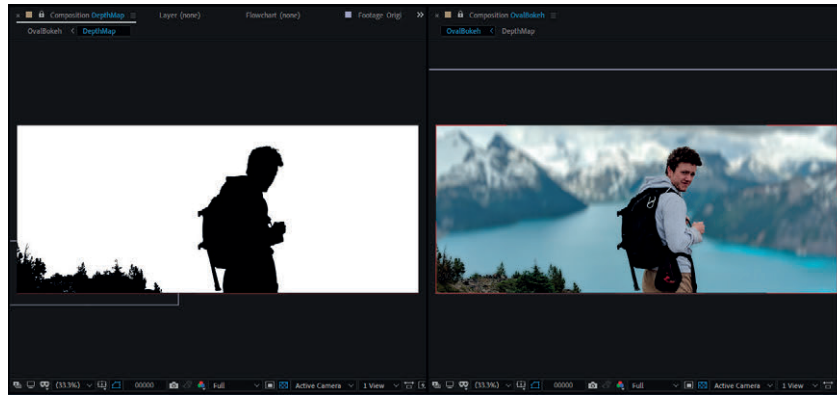
Go back to the "Camera Lens Blur" and change some of the "Blur Map" selections. "Layer" should become "DepthMap", just the "Source", and "Channel" switches to "Luminance", also disable the "Invert Blur Map" checkbox. Keep the rest as it is. Focus will be weird for a second.

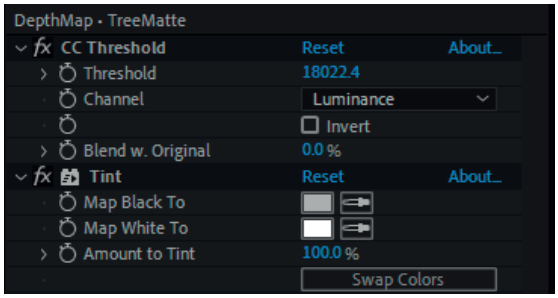
On your "Comp Viewer", click the composition's name and select "New Comp Viewer". This will double our current composition on the screen so we can keep an eye on changes. Now double click the "DepthMap" precomp. One of the viewers should show the main comp and the other one is the trees on a white corner and everything else black.



First of all, create a new white "Solid" and put it at the bottom of the stack. This should make things look better in terms of focus, but now both the character and the trees are sharp, plus the trees have a dark halo around them.

Split comp viewer: DepthMap on the left, final comp on the right. Strange focus artifacts around the trees.





Next, add a "Tint" effect to "TreeMatte" and in the "Map Black To" color, pick a light shade of grey. This grey controls how much blur is applied to the trees. In the "Camera Lens Blur" "Blur Map", white means far from camera, black means close to camera and the greys in between set the distance. So a light grey will be closer to the background - more blurred -, while a dark grey is closer to the foreground - less blurred.

Final Depth Map, the trees are a light grey for more defocus.

For final touches I am going to brighten up my character a little bit, and add some vignette! Now imagine doing all of this taking into account movement. Yes, it is a lot of trouble.

DAVINCI RESOLVE

You can find the files for this tutorial in ./Videos/10_PostProcessing/Resolve_Bokeh/BaseShot.mov
You can find a video version of these steps in ./Videos/10_PostProcessing/Resolve_Bokeh.mp4

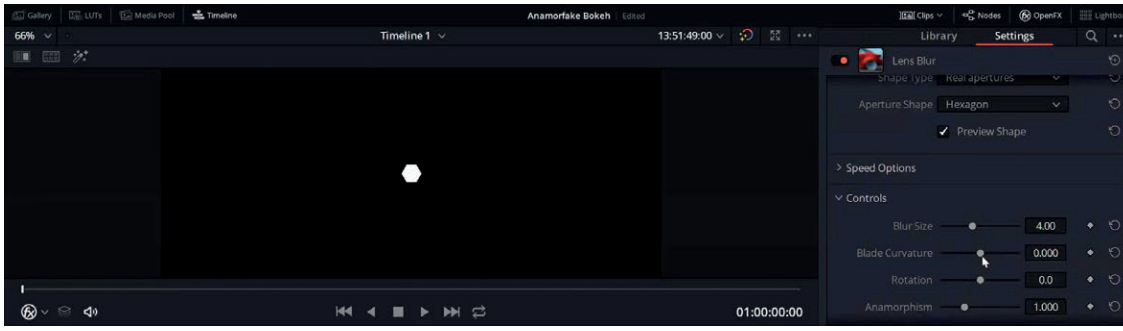
Time to create oval bokeh in Resolve. In this tutorial we will be doing a lot of tracking and masking, so get comfortable and familiar with those techniques!

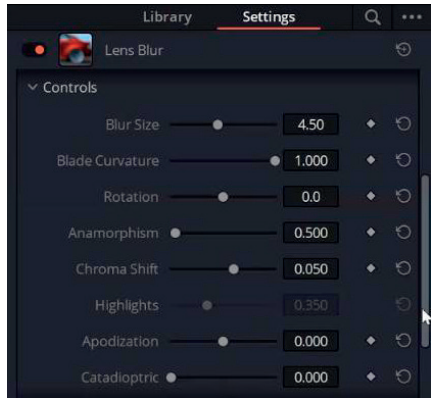
Start by importing your footage and setting the aspect ratio for your project as seen in the previous chapters. The proper aspect ratio is key element to make the effect more convincing.

This is our original shot. Notice the circles for out of focus highlights. It is time to change that!

Pick your shot and move into the “Color” tab. Before we get into rotoscoping I will go over my settings for the bokeh.

Add a new node and, from the “OpenFX Library”, drag in the “Lens Blur”. In its settings, mark the checkbox for “Preview Shape” as this will be handy to create our bokeh accurately. The frame goes black and in the center you see the shape that is driving bokeh right now.

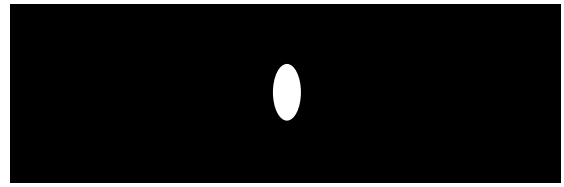


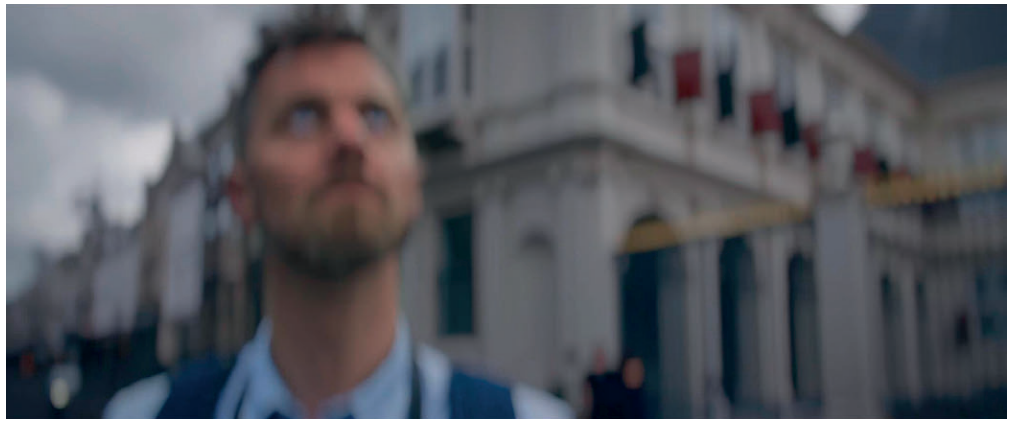


After fiddling with the settings I went with the ones shown here. “Blade Curvature” set to the maximum value makes the shape a perfect circle, then “Anamorphism” turns into an oval. “Chroma Shift” introduces a little bit of chromatic aberration, which I felt was appropriate.

“Highlights” is greyed out because we are in “Preview Shape” mode, but we will crank it all the way up too, so the ovals are brighter and more pronounced against muted colors.

Apodization and Catadioptric are characteristics of particular lens designs, so we will leave these untouched this time. You should have something like this in your preview panel. If you are happy with the results, uncheck “Preview Shape” and crank “Highlights” to 1.



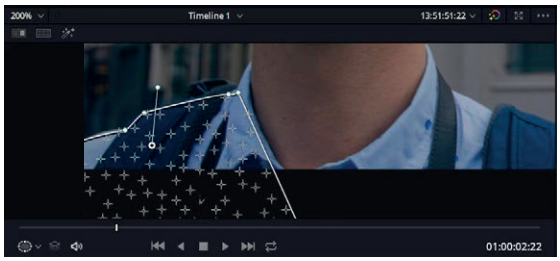


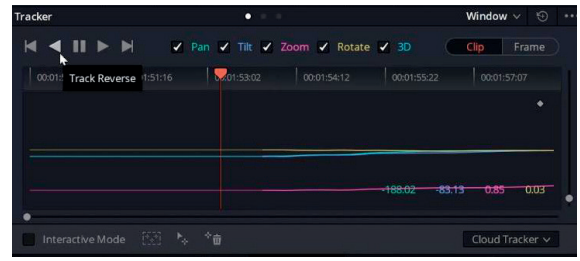
The oval highlights are noticeable enough now, especially when compared to the original shot. Time to start rotoscoping to separate the subject from the blur.

Watch the shot a few times and observe how each part of your subject moves. The goal when rotoscoping something in motion is to break it in smaller pieces that can be treated separately. It might seem counter-productive to have half a dozen masks instead of just one, but breaking it up makes the results more consistent and easier to manage.

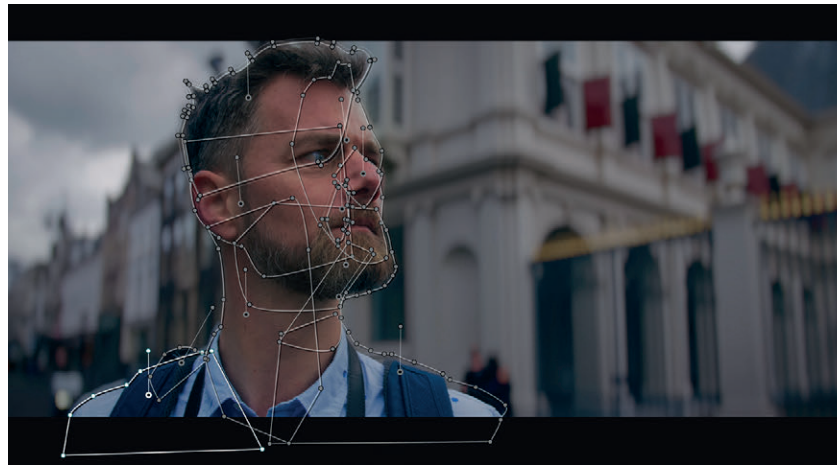
Here is a tutorial about the essentials of rotoscoping - it is After Effects, but the principle applies to any software.

Once you decide where to start, make your first mask. I am starting with his left-side shoulder on the first frame of the shot. Once the shape is done, go to the “Tracker” tab and track the mask all the way through (forward or backward, it does not matter). I kept all the checkboxes on, but you have to judge these on a shot-by-shot basis.





It does not stick perfectly, but that is part of the process. The tracker is only meant to make life easier, but not to solve all of the world's problems. Now rinse and repeat until you have a tracked shape for all of your subject. I ended up with eleven masks.



After all the masks are there and tracked, I like to disable all of them and go through each one individually, adjusting the points as needed to make them stick completely to the part of the frame I was tracking. For example, the shoulder had a perfect shape in my end frame, but when I tracked backwards, some of it fell out of place.

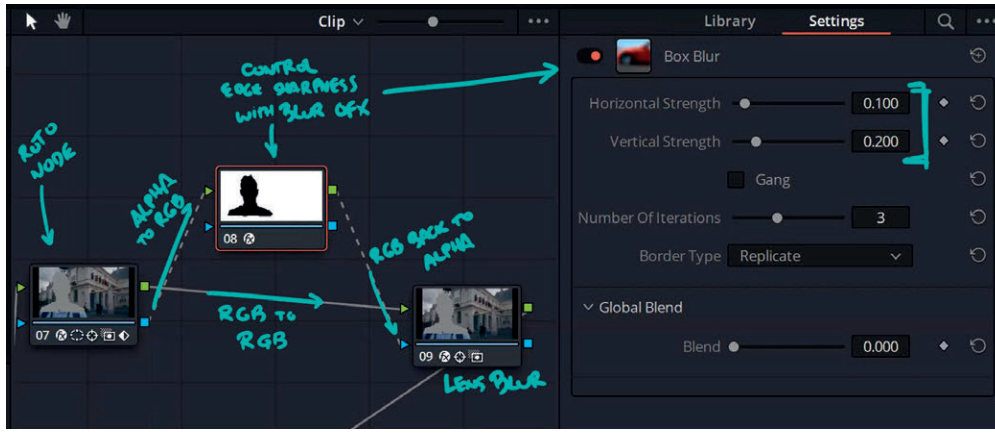
I went in and adjusted the individual points of the mask to bring them to the correct shape, creating a new keyframe for that mask. Resolve interpolates between both keyframes, but check the entire shot to make sure the middle is not drifting either.

For reference, this process took me about one hour and I have some previous experience doing this. So do not expect it to be fast. This is a time-consuming solution.

This gives you a solid alpha channel around your subject, but such hard edges might not work in your favor, making your roto job a little too noticeable. This is where I hit Resolve's limitations and had to create my own solution.

Adjusting the masks individually is a painfully convoluted process, so create a new node and send the alpha channel of your roto to its RGB channels (blue connector to green connector). Then add a regular old "Box Blur" to this new alpha node and adjust it to your liking. I favored adding more blur vertically than horizontally since scope affects more the vertical than horizontal axis of the image.

To bring this back into the main line, cross the RGB of the alpha node to the alpha of the node with the "Lens Blur" OFX, like the diagram below, making a small web with these three nodes.



When I checked the Lens Blur node though, my alpha was reversed and the lens blur was being applied to the character and not to the background. So the last thing I had to do on the Lens Blur node was to invert the alpha, on the "Key" tab, "Key Output", by clicking the little circle inside the square switch.

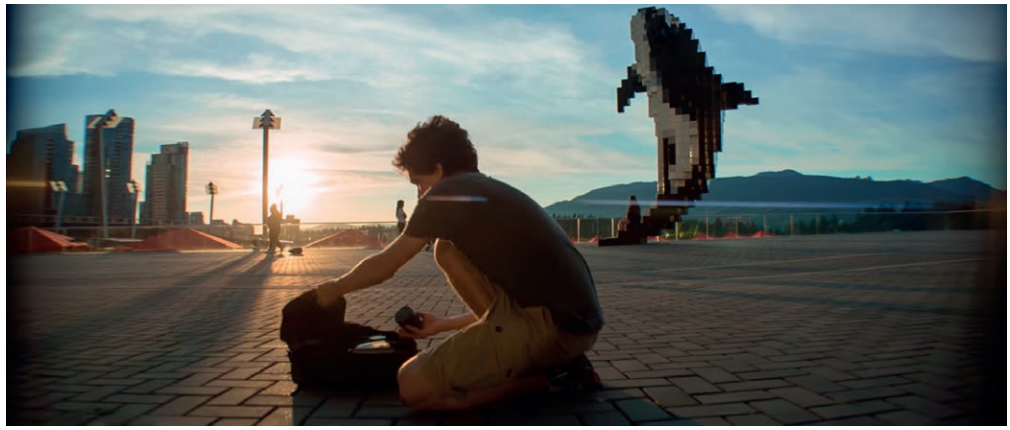
THIS GAVE ME WHAT WE WERE AIMING FROM THE START!

If you are watching for small imperfections there are some to be found. The brighter edges around the character over a darker background are consequence of the Lens Blur (see his shoulder on the left?). I am sure these would go unnoticed in a video, but this paragraph is for nitpicking.

The only way I can see of solving this problem is to create a clean plate like what we did in the After Effects tutorial, except Resolve does not have easy tools to do so. Or, at least, not yet. You can follow pretty much the same steps described here in After Effects too for moving shots!

LENS FLARES

Flares are the staple quality of anamorphics that makes spherical lens users drool. Unfortunately, cheating flares with wires introduces one (or more) lines through your bokeh. To some that is acceptable, to others post is the best option and this is where we are!



Contax Zeiss 28mm f/2.8 + Panasonic LA7200 1.33x Anamorphic Adapter.

The two more detailed approaches to creating lens flares in post-production in this guide are achieved with After Effects. **VideoCopilot's Optical Flares** is solely compatible with AE, but **Red Giant's Magic Bullet Looks** works with Premiere, After Effects, Resolve and FCPx. I am using After Effects, but you can use anything else!

Canon EF 16-35mm f/2.8L + VideoCopilot's Optical Flares.

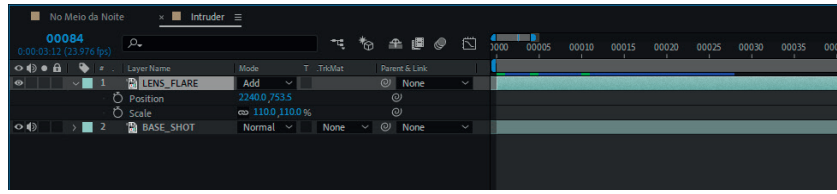
Fortunately, these are not the only options and there is a much simpler approach that can work for quick shots and boosting your production value. These flares can be achieved in any editing software that supports blending modes - FCPx, Premiere, After Effects and Resolve are covered!

FLARE FOOTAGE + BLEND MODES

You can find video versions of the steps for each software in ./Videos/10_PostProcessing/Flares/

In order to begin you need stock footage of lens flares over a black background. To give you a quick start here are 17 free ones from PremiumBeat and 120 paid ones from Rocketstock. There are many other libraries online, search up!

Once you have your timeline set up, import the flares into the project and pick the ones you like best. Drag one on a video layer above your shot and change its blend mode to “Add” or “Screen” in order to preserve the highlights of the flare footage and discard the shadows.



You can change many things about your stock flare such as scale and position through the use of regular “Transform” settings and adjust its contrast and saturation with a little color correction applied only to the flare layer - or even change its color completely by rotating the hue wheel!

The two separate base shots and their resulting combination through blending modes.

The downside to this technique is the flares are usually not in sync with the actual camera movement. They are shot by shining a flashlight at the lens right

outside of the frame. Because of this movement mismatch you can end up spending a lot of time trying to find the best combination between the shot and the available stock footage.

The blending mode method aims to add cool effects to quick shots but are very far from physically accurate. Think trailers, promos, music videos and other quick-paced edits.

Why not add some flares on top of the fake bokeh?

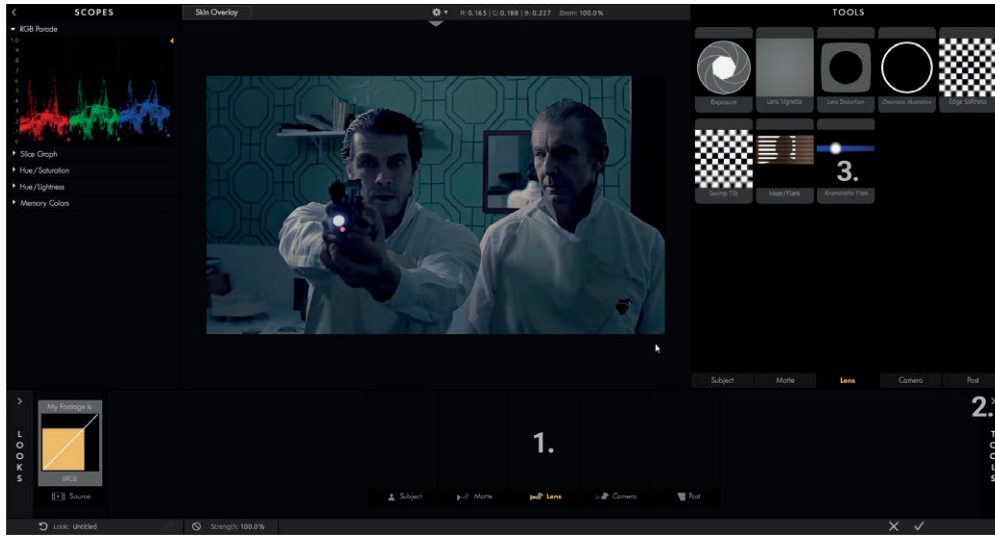
RED GIANT'S MAGIC BULLET LOOKS

You can find a video version of these steps in ./Videos/10_PostProcessing/RGMB_Looks_Flares.mp4

Looks is part of Red Giant's Magic Bullet Suite and it can be purchased separately for $399. The full suite costs $899. This price tag is hard to justify if you are looking just for lens flares, but Looks gives you many tools to simulate various lens, film and post-processing effects. I have been using it consistently on all of my projects since 2012. Also, a single license works for all the compatible software.

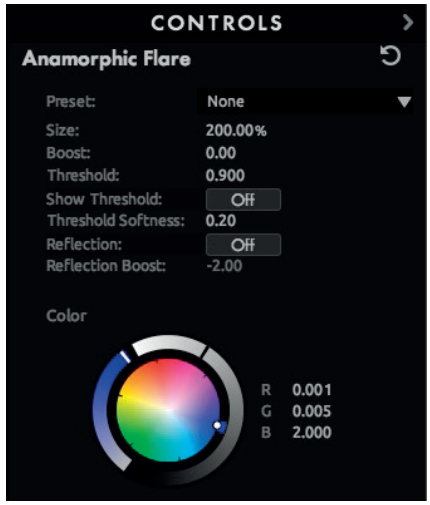
Achieving anamorphic flares in it is a simple process but the level of detail in the flares leaves a lot to be desired. It is more like a streak. The fun part is that it connects to the highlights/blown-out areas of your shot, making the flare move in conjunction with the camera.

Load your shot onto the timeline and then apply the "Looks" effect on it. Once you open the plugin's interface, navigate to "Lens", then "Tools" and drag "Anamorphic Flare" to the "Lens" section at the bottom.



Now the right-side panel will show you parameters to control the flare. There are a few presets in terms of style and color but if you made this far into this guide, I am guessing you like to know your tools.

"Size" controls the length of the streak, allowing you to have shorter streaks or longer ones that cross the entire frame. "Boost" controls intensity and "Threshold" sets the limit of how much highlights should project lens flares. A higher threshold will only flare the strongest values close to white while a

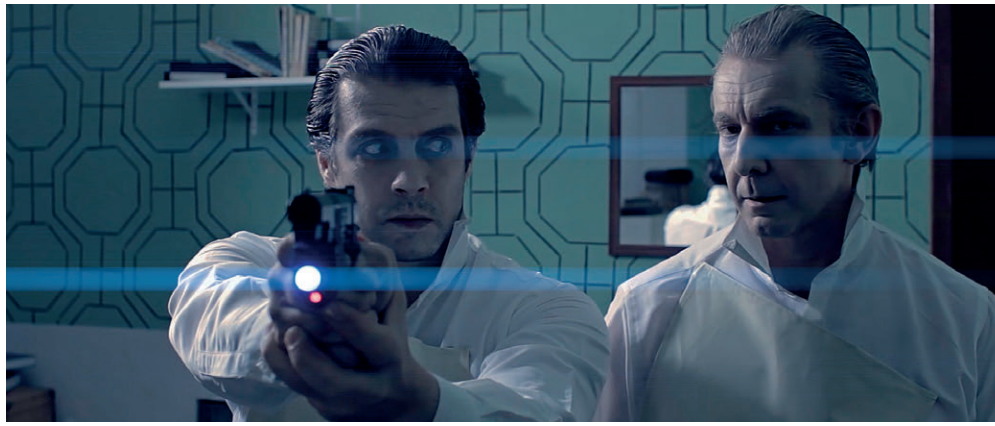


lower threshold will affect not-so-bright areas. If you want visual control of what is being kept as a highlight, turn on the "Show Threshold" box and adjust your "Threshold" values. "Threshold Softness" applies a blur to the resulting image, smoothing out the results.

Turning on "Reflection" adds a second streak which is created by internal reflections between the lenses and other filters. Reflections are always noticeable when using single focus solutions and anamorphic adapters. I believe "Reflections" kicks up a notch in the believability factor of this plugin. And you can control how strong the reflections are with "Reflection Boost".

The "Color" wheel lets you pick the hue, brightness and saturation of your flare. These values have to be pretty cranked up to come through, so do not be surprised if you are constantly at the very edges of the wheel for a little tinge

of color. You can even adjust the Red, Green and Blue values separately when settling on a color. Last, hit the checkmark at the bottom right corner and you are out of the "Looks" panel.



Red Giant's Magic Bullet Looks Anamorphic Flare.

VIDEO COPILOT'S OPTICAL FLARES

You can find a video version of these steps in ./Videos/10_PostProcessing/VC_Optical_Flares.mp4

Optical Flares is one of the best investments you can make if you ever consider adding flares in post-production. The level of customization and control is enormous, the price is great (\$165 including two bundles of flares) and there are lots of detailed tutorials about all of its features and what you can do with it.

Its biggest advantage over the previous methods is that the flares here are multi-dimensional, composed of several different layers that move at different pace, giving a much better impression of what a real flare does. I have used Optical Flares in countless projects, including some that were actually shot shot as a mix of anamorphic and spherical and I wanted to match the look of the lenses.

Zona SSP - Episode 0, Part I - LOMO Foton-A 37-140mm f/4.5 2x Anamorphic - 2013

Up & Away - Contax Zeiss taking lenses + Iscorama Pre 36 1.5x Anamorphic - 2017

One of the tutorials below tackles a simpler “put a flare here” approach for creating a custom flare and adding it to specific elements in the shot. The other tutorial uses a similar approach Red Giant’s Magic Bullet Looks through the use of a luminance map. The difference between Optical Flares and Magic Bullet Looks is that with Optical Flares we are controlling and adjusting the map by hand and tweaking the flares to a style that suits our shots, while Magic Bullet Looks is a bit more automated and less controllable.

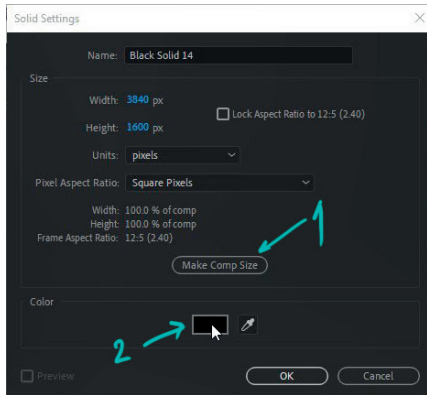
The base footage for this tutorial is from my April Fools video, This is SCOPE, shot with an anamorfake Rokinon 24mm, Letus 1.33x AnamorphX Pro and a lot of post production, especially for lens flares.

Up & Away - Contax Zeiss taking lenses + Iscorama Pre 36 1.5x Anamorphic - 2017

2D FLARES

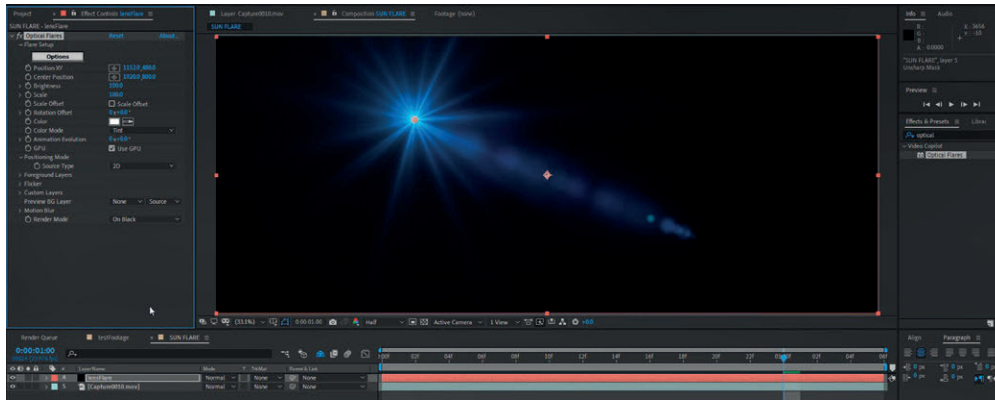
You can find the files for this tutorial in ./Videos/10_PostProcessing/AE_OpticalFlares_Tutorial/2D.mp4
You can find a video version of these steps in ./Videos/10_PostProcessing/OpticalFlares_2D.mp4

In this tutorial we are customizing a flare from the Optical Flares library and locking it into place on a shot. The whole process involves a bit of linking properties, blend modes, tracking and null objects.

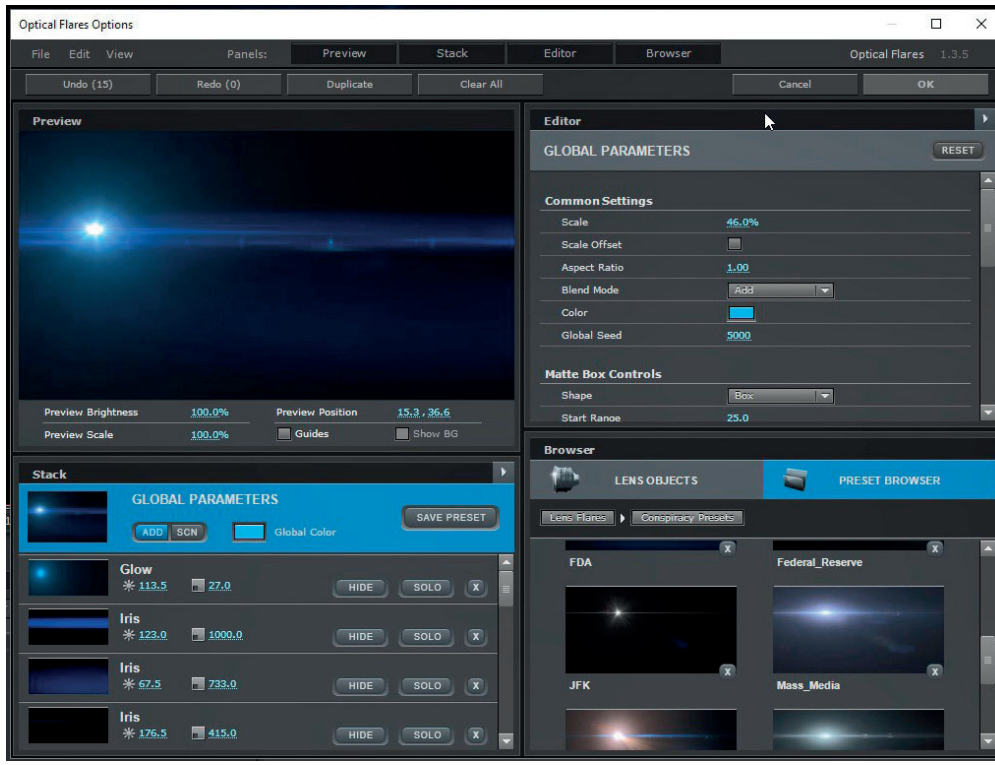


The first step is to bring the shot into After Effects and put it on a new Composition. From there, create a new Solid (Ctrl/Cmd + Y), hit the “Make Comp Size” button on the “Solid Settings” dialog that shows up and pick the color black for it. For the sake of organization I am naming this solid as “lensFlare”.

Then, from the “Effects and Presets” list pick “Optical Flares” and apply it to our “lensFlare” solid. This will give you a default boring blue flare filling the frame.



Click the big “Options” button inside the plugin’s parameters and that will bring up a separate window for adjusting and customizing the flare. On the bottom right side you will find the “Browser”, my favorite spot to start. Go through the folders until you find a flare you like. For this tutorial I am going with “FDA”.



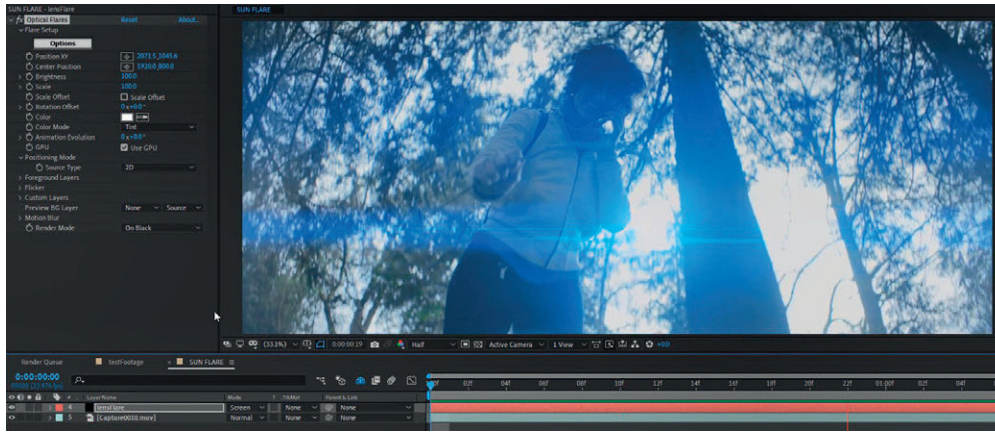
The “Preview” window on the top left will update as you go through the list. You can also click and drag on the viewer to move the flare around and see how it behaves as well as change its master brightness and scale (controlling these parameters will be very useful later).

After you have a preset you are happy with, you can fiddle with its settings. The “Stack” on the bottom left side shows all the elements that compose the flare, each one following its own color, scale, brightness, opacity, position and many other parameters. You can quickly hide and solo elements to see what is their role on the overall flare look.

Once you click one of the elements, you can change all of its internal parameters with the numbers in the “Editor” on the top right side. You have infinite possibilities.

For this tutorial, I find the preset glow a bit too intense, so I will dial down the brightness of that element separately (the number by the little sun symbol).

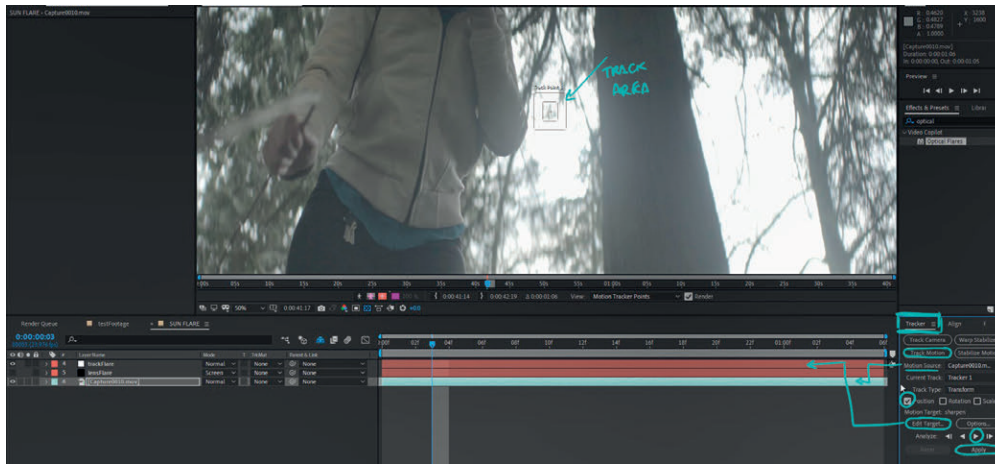
Once you hit “OK” the flare will update on the black solid and it is time to change its blend mode to “Screen” (preferred) or “Add”. Now you have a crude rendering of the flare over the shot. To make things better I will move the “Position XY” parameter to the spot I want the flare to land in the shot: the sun among the branches.



Looks terrible, right? It looks even worse when it is in movement. Time to start tracking. For processing sake, turn off the visibility of “lensFlare” layer.

Create a “Null Object” (Ctrl/Cmd + Alt + Shift + Y) - again, for organization I like to rename it as “trackFlare”. Go into the “Tracker” window, select your base shot on the Comp (or pick it in the “Motion Source” list) then click “Track Motion”. From the checkboxes mark only “Position”

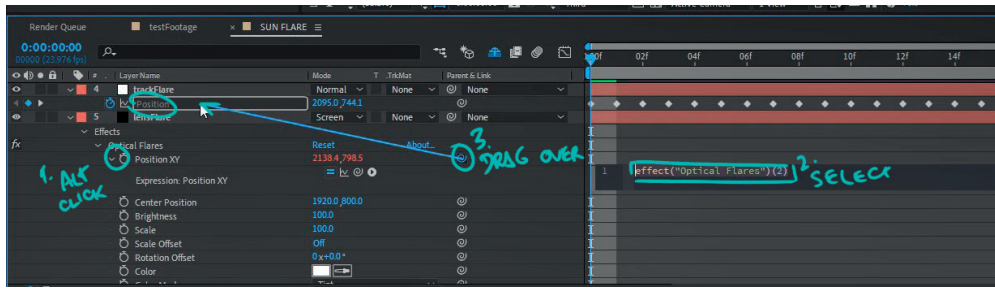
Place the tracker in the area you want the flare to hold onto. High contrast corners are ideal, so pick a leaves/sky area and hit the play button in the “Analyze” section of the “Tracker” window. This is going to scan the entire shot, frame by frame, holding the track area locked to the place you selected. Fast movements, obstructions and other things will make the track slip, so pay close attention while After Effects does its job.



I had bad luck and the little stick in Ari's hands crossed the area I was tracking. For that section, I went frame by frame adjusting the position of the tracker. Be prepared to do the same for your shots. It is not necessarily hard, but can be quite time-consuming.

After the shot is tracked, click the "Edit Target..." box and select the "trackFlare" layer from the list. Once you click "Apply" a new dialog will show up, asking if you want the information on the X and Y axis. Confirm it.

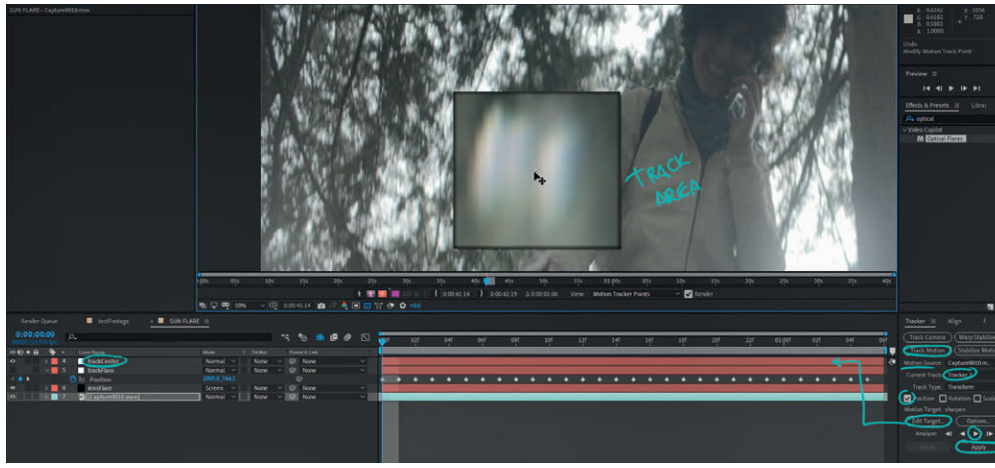
Now if you scrub through the shot you will see that the "trackFlare" stays locked to the leaves we wanted. Good. Reactivate the "lensFlare" layer and while holding down the alt key, click the stopwatch on the "Position XY" parameter of the Optical Flares effect. This will bring up the expression editor for that parameter. Leave that section open.



Click the "trackFlare" layer and hit the "P" key to bring its "Position" attribute. Now shuffle back to the expression editor in the "Position XY", select all the text there and then use the pick-whip on the left by clicking and dragging it to the "Position" attribute of the "trackFlare" layer. This parents the "Position XY" of "lensFlare" to the "Position" of "trackFlare", making it move according to the camera move.

This makes the whole thing look a lot better already. Now the flare looks pasted into place, like an image taken from Google, or a stock footage flare that we froze into place. Optical Flares has a great 3D feel to it like a real lens flare, so we should take advantage of that.

Create another "Null Object" and name it "trackCenter". Jump back into the "Tracker" window and change the "Current Track" to "Tracker 2". Repeat the process we used for the first track but pick "trackCenter" from the "Edit Target..." list and when picking the area to track, look for something across the frame. I am going for a break in the leaves towards the top left corner.



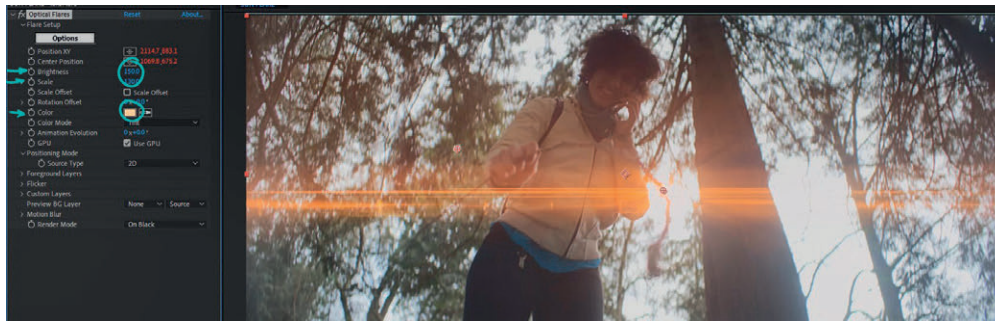
Once that is tracked and applied to "trackCenter", go back into the Optical Flares' parameters and alt click the "Center Position" stopwatch. Repeat the process we used for "Position XY", but connect the "Center Position" value to the "Position" parameter of the "trackCenter" layer. Properties that have their values controlled by expressions show up in red - this is how "Position XY" and "Center Position" should show.

Since those two areas of the shot move at slightly different speeds when the camera moves, this will create depth within the flare, with elements parallaxing with each other, like no stock footage flare will ever do.

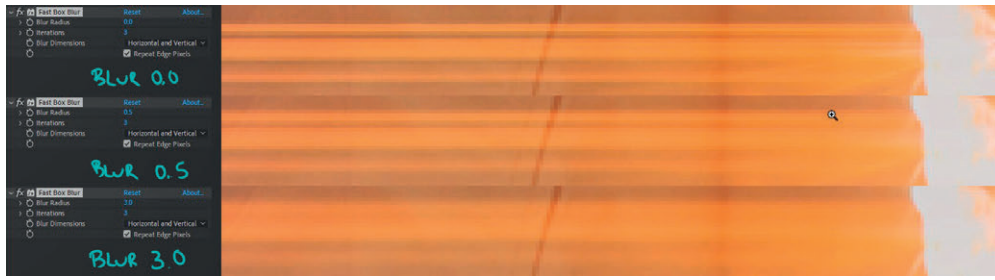
This makes me happy with the way the flare wobbles around and we can move to more fun steps like changing the flare color, size and intensity, plus a few last adjustments to blend it better with the original footage.

In the Optical Flares options, click the color field and pick a color of your preference. I am leaning towards the golden/amber.

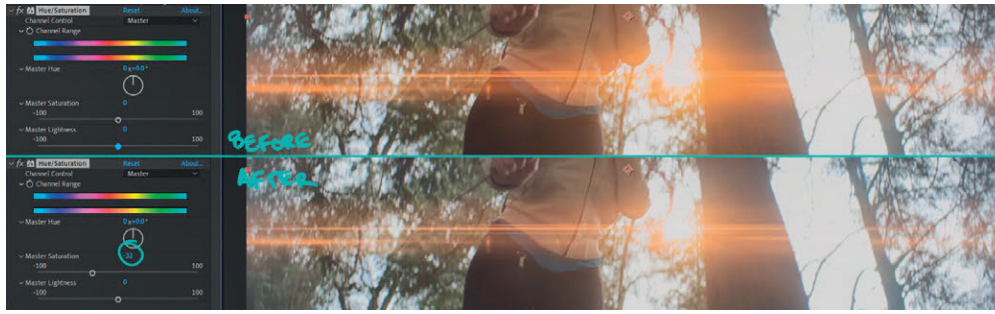
Good enough to start. I will adjust the flare's "Scale" and "Brightness" parameters until I find something I like better. These controllers can be used very effectively when you want to simulate the distance of the light source to the lens (closer/stronger lights produce thicker flares, farther/weaker sources make thin streaks).



Once you are happy with those, zoom in all the way into your flare. Notice how the edges of the flare are razor sharp compared to the footage? This is a dead giveaway of CGI. Computers do not make mistakes. So throw in a quick "Fast Box Blur", check "Repeat Edge Pixels" and adjust the "Blur Radius" value until the edges are smoother.



Lastly, I want to push my flares towards yellow instead of red. I failed when using the Optical Flares color picker so I will add in a "Hue/Saturation" effect to the "lensFlare" layer to make fine adjustments to color, saturation and lightness.



As soon as you are done with this, you can say your shot is finished. We made our own flare, locked it onto the original footage, added some lens depth to it, changed its color, smoothed out the perfect CGI edges and fine-tuned its colors. Congratulations, you did a pretty extensive process in just a few minutes!

You will get faster as you go along working on your own shots - and this is the simplest way to add a flare to flare-less footage. Which brings us to...

LUMINANCE-BASED FLARES

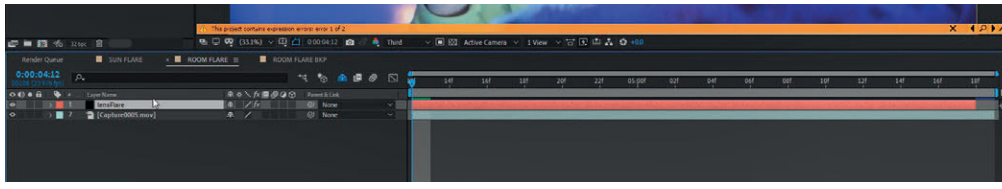
You can find the files for this tutorial in
./Videos/10_PostProcessing/AE_OpticalFlares_Tutorial/Luminance.mp4

You can find a video version of these steps in
./Videos/10_PostProcessing/OpticalFlares_Luminance.mp4

Think of Luminance-based flares as smarter flares. They take into account the brightness (luminance) in each part of the image to decide where to flare up. Brighter areas will make stronger flares and you are the one determining how bright an area has to be for it to flare.

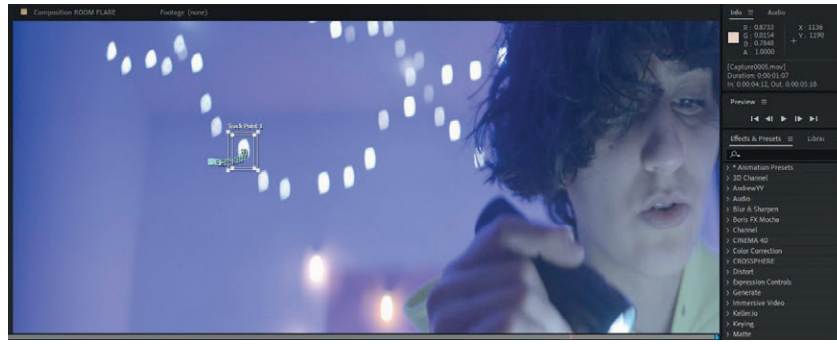
The immediate advantage of this method is that the flares stick to certain areas of the frame without need for tracking each light source like we did for the 2D flares.

To speed things up I am going to copy the flare layer from the previous tutorial. As soon as you paste it in the new composition, you will see an Expression Error (little orange line on the bottom of the Comp Viewer). This shows up because some of our flare properties were linked to tracking data and that data still does not exist in this composition.



So it is time to get started with tracking the environment so we can have parallaxing within the flare itself. Before you do so, create a Null Object (Ctrl/Cmd + Shift + Alt + Y) and name it "trackRoom". This null is going to hold the tracking data for us to connect the flare layer later.

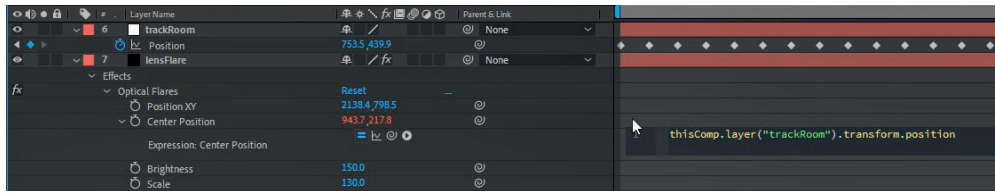
For tracking, favor high contrast areas and easily identifiable patters. For this shot I am using the blown-out bokeh in the background. These are not obstructed in any part of the shot, so tracking is smooth.



Once tracking is finished, click "Edit Target" on the Tracker panel and select the "trackRoom" null to receive X and Y data. This will put "trackRoom" right on the path from the light.

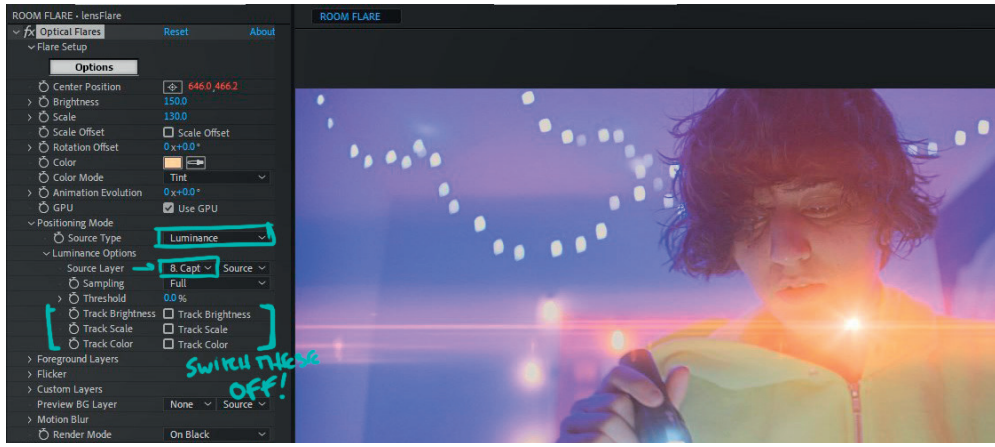
Now go to the Effects tab for the flare layer. If you copied this layer from the previous tutorial too, there should be two fields with red numbers in them ("Position XY" and "Center Position"). Alt + click the stopwatch for both of them to get rid of those links.

Moving forward, Alt + click the stopwatch for "Center Position", opening the expression field for that property. Use the pickwhip to connect the flare's "Center Position" to "trackRoom"'s position. The expression field should look like this:



Still on the Effects tab for the flares, go to "Positioning Mode" and change it to "Luminance", which is the point of this tutorial. No flares will show up at first because we have to connect the effect to a source layer for it to analyze the Luminance values. For the "Source Layer" property, pick your footage layer and as soon as you do that, the whole shot is likely to blow up with giant flares.

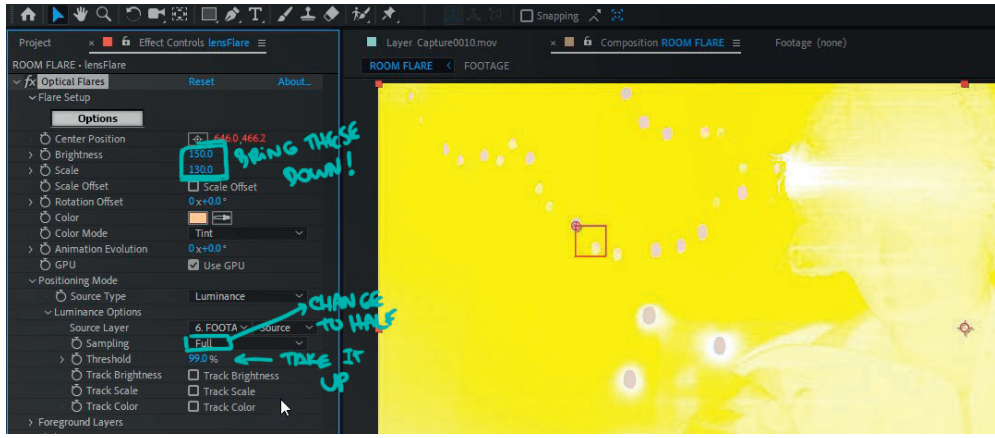
The “Threshold” slider is set to 0.0% which is the reason the explosion happens. This slider is our main controller for the amount of flares we want, and what intensity of lights should be making them. This is a rough slider, so start with a value like 99% to see some change.



By default, the checkboxes for “Track Brightness”, “Track Scale” and “Track Color” are on. They do exactly what they mean, and create flares that match the same level of brightness, size and color of the highlights on the base shot.

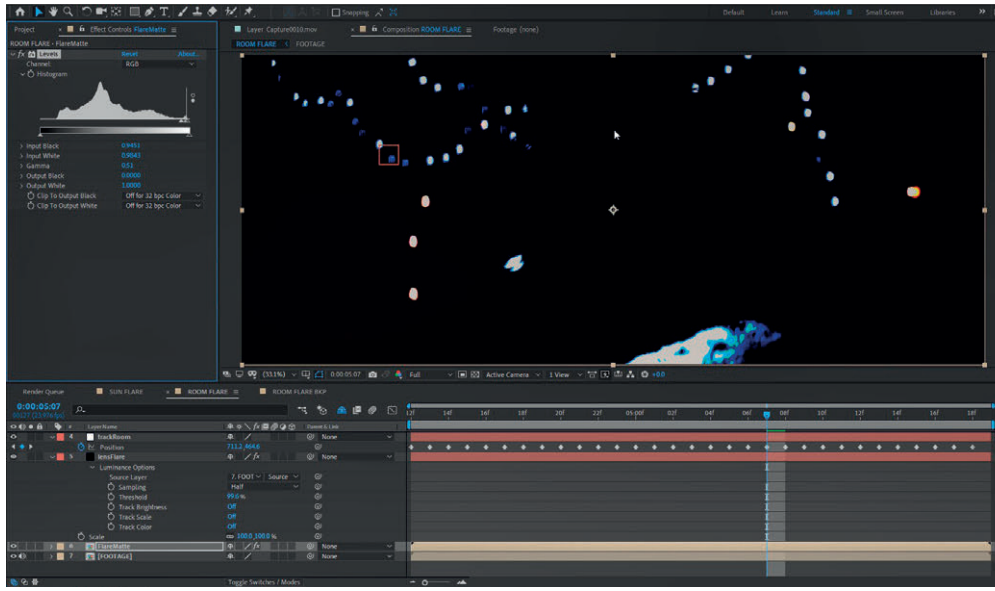
If we are being serious about this, we need to disable “Track Color” right away because most anamorphics flare only one color, and not according to the light source. The other two checkboxes can be useful, but I am keeping them off for now, which gives us a very blown up shot.

We counter it by bringing down the overall “Brightness” and “Scale” properties of the flare, change the “Sampling” in “Luminance Options” from “Full” to “Half” and cranking even higher the “Threshold” value. My shot looked decent at 99.6%.



As soon as you play the shot you will see that things are not going quite as planned. My flares were flickering in and out of existence, changing sources and making a big mess. Unnacceptable. That happens because the controls offered by the Optical Flares plugin to define what should flare and what should not are very simplistic.

Fortunately, there are ways around it. This is the plan: duplicate the footage layer, apply a “Levels” effect to it and crank it so the only visible things are the highlights you want to flare. Your shot should look somewhat like the one below. In an attempt at being organized, name this new layer “flareMatte”

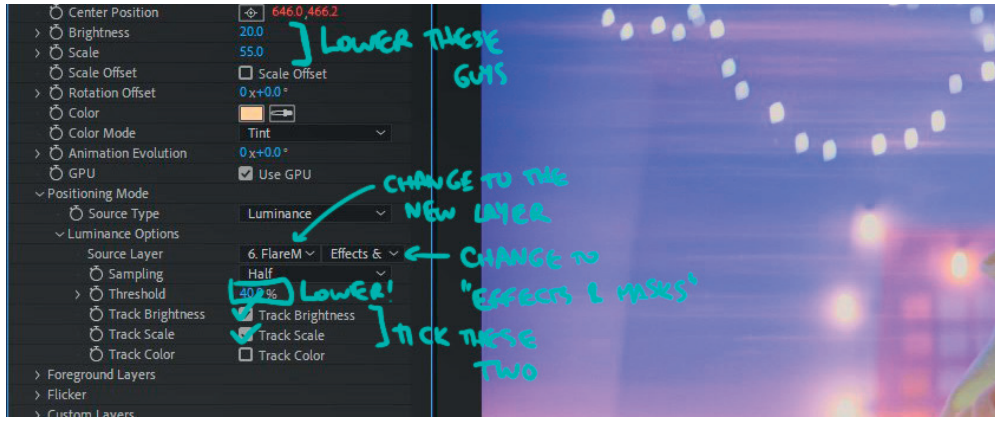


We can clearly see all the twinkly lights on the wall, the flashlight and a big patch where the stuffed animal is. That big patch is going to give us trouble, but let’s focus on the background first.

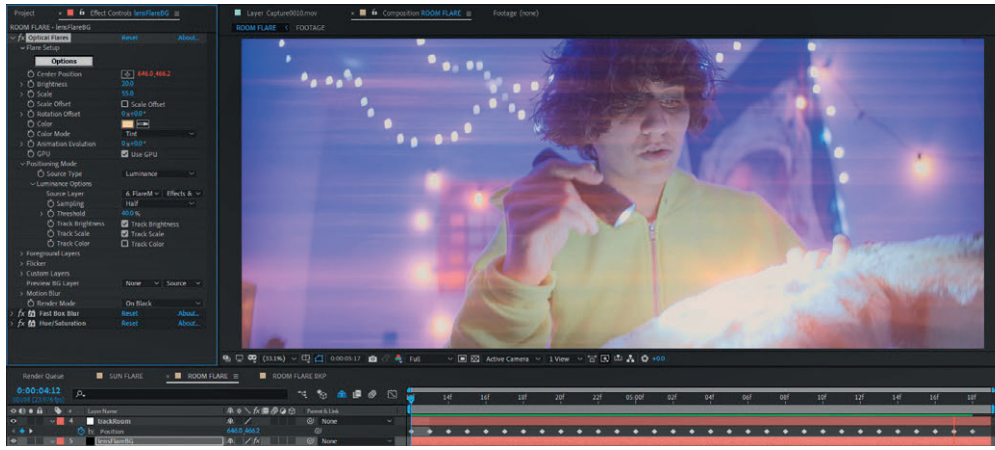
Disable visibility for the “flareMatte” layer and rename the “lensFlare” layer to “lensFlareBG”.

Change the “Source Layer” in the Optical Flares to our new “flareMatte” and change the dropdown right next to it from “Source” to “Effects & Masks” so our Levels correction is taken into consideration.

This will give us a much better rendering of where the flares should be, although a bit too intense. We can dial the “Threshold” down to 40% so we have more consistency over time, reducing the sparkling flares we had on our first pass.



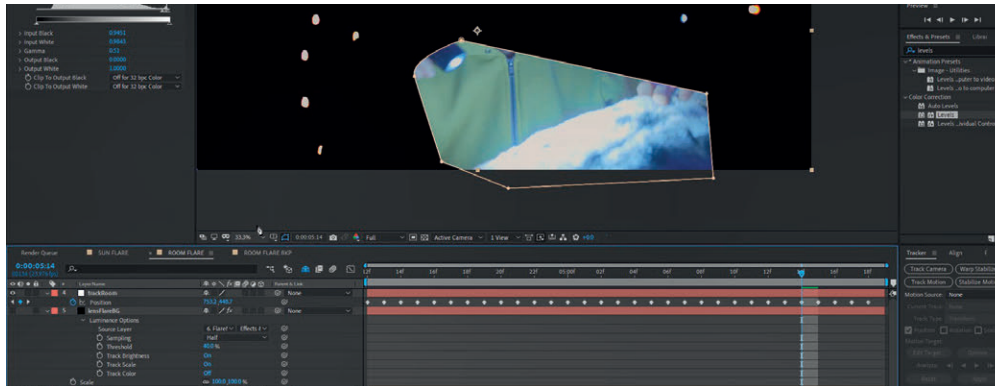
Enable back “Track Brightness” and “Track Scale” so the size and intensity of the “flareMatte” controls the resulting flares, and dial down the overall “Brightness” and “Scale”. I set mine to 20 and 55, respectively.



The background is looking pretty good, but that stuffed animal is giving us a massive flare. Time to get creative with our control levels. We are going to remove this highlight (along with the flashlight) from the flaring layer, then add the flashlight back to another layer so we can adjust background and flashlight separately.

Re-enable visibility on the “flareMatte” layer, select the Pen tool (G) and draw a mask around the flashlight and stuffed animal highlights. Then change its type to “Subtract”.

To make things easier to see, disable the visibility of the footage layer underneath “flareMatte”, and “lensFlareBG” above it, giving you a clean black background.



The next step is to animate this mask so the highlights never leak from it. With the “flareMatte” layer selected, expand its properties (or quickly hit M twice) to see the Mask’s controls. Activate the stopwatch for “Mask Path” to add a keyframe and then scrub to the last frame.

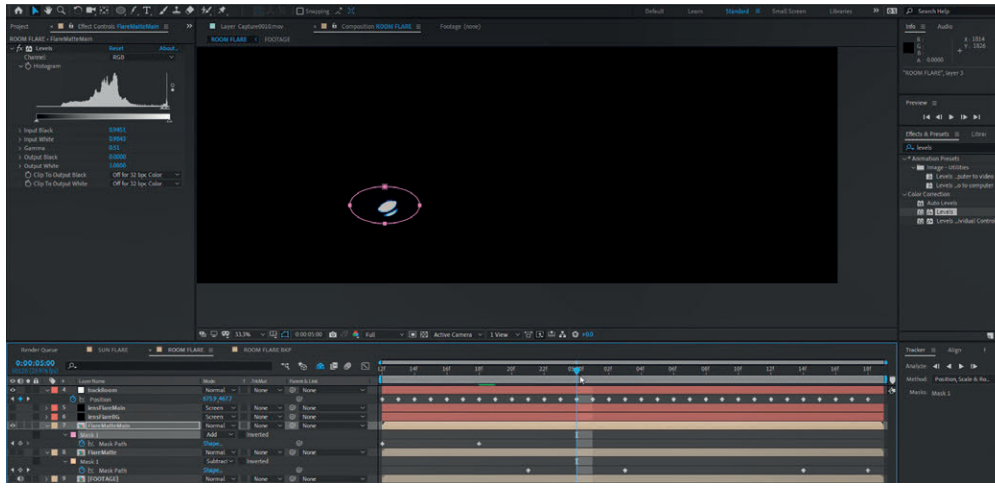
Some of the highlights should be peeking out. Still using the Pen tool, adjust the mask to cover them again. This is going to automatically create another keyframe. Scrub through the timeline to make sure nothing of the flashlight and stuffed animal are showing again.

Try to do as few keyframes as possible and take into consideration that After Effects will interpolate the shape of the mask between each keyframe.

Once that is finished you can disable the visibility for the “flareMatte” and enable back the footage layer and “lensFlareBG”. This wraps up the work for the background flares. You can tweak its “Brightness” and “Scale” values if desired, but I am moving on to the flashlight flare.

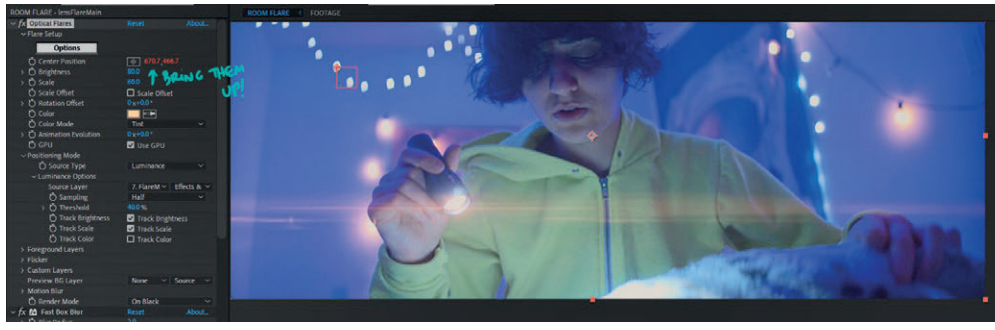
A good way to save time here and have consistent flares is to just duplicate “lensFlareBG”, but rename the copy to “lensFlareMain”. It should keep the same settings as the background, except we need a new “Source Layer” for the flares instead of “flareMatte”.

Go ahead and duplicate “flareMatte” too and rename the copy to “flareMatteMain”. The mask we made to hide the flashlight and stuffed animal is not very useful anymore, so delete it and make a new one just around the flashlight.



Add a keyframe to the mask's shape and scrub through the timeline making sure you are getting only the flashlight inside the mask. No need to change the mask's mode to "Subtract" this time.

Once the flashlight is tracked and ready, enable/disable the layers we have been working with and switch the "Source Layer" in "lensFlareMain" to "flareMatteMain" and bump up "Brightness" and "Scale" for the flares.



After all that fiddling, it is good to see something come together with just a few clicks!

Add in some color correction to the whole thing and you are more than qualified to work with Luminance-based flares using Optical Flares. The key is to have control over the elements and to separate them in different layers if necessary, like what we did here for the background and the flashlight.

CONCLUSION

With this guide I aim at bringing back the “budget” aspect of my original project - Anamorphic on a Budget - showing many different and cheap ways of achieving the anamorphic look without necessarily relying on anamorphic glass. Techniques range from post production through lens mods and all the way through post production, so you can pick and choose the ones that work best for you!

Sigma 18-35mm with Cinemorph filter!

Take this as an introduction. Experiment with the techniques, see if this look is really what you are searching for. If you like what you get, consider looking into actual anamorphics but beware: if you feel the steps for some of the techniques in this guide are too challenging, shooting with anamorphic adapters has about the same

level of complexity, if not higher and quirkier. Plus the price of admittance is really high right now and the limitations imposed by adapters are severe.

Decide if shooting with anamorphic glass is something that is important to you, to the project, or to your audience. I find myself anamorfaking more and more these days in an attempt to keep the budgets down and still be able to deliver a unique look. Yes, it is unique. Technically speaking, it is something in between both worlds, spherical and anamorphic, capturing the best of each one. The simplicity and practice of shooting spherical and the added production value of the anamorphic look.

Rokinon 50mm anamorfake on Sony A7s II - by Marcus Hann

Lastly, I would love to see what kind of inventions and films you are coming up with, so please send me emails! All feedback on this guide is appreciated and remember to use the tag #anamorfake, encouraging others to try it out. The anamorphic look should not be kept a secret and we can only get better looking pictures by spreading it around!

Thank you,

Tito Ferradans

ANAMORFAKE IT UNTIL YOU MAKE IT!
HOW TO ACHIEVE THE ANAMORPHIC LOOK ON THE CHEAP